

Onconeurology: Kidney Disease and Cancer

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CHAPTER OUTLINE

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KEY POINTS

- Acute and chronic kidney disease are common complications in patients with cancer and are primarily due to the underlying malignancy or its treatment. Study of the growing connection between the two diseases is often referred to as “onconeurology”
- Acute kidney injury (AKI) is a relatively common complication of cancer and its treatment. Both hematologic malignancies and solid cancers are associated with AKI. The causes can be classified as prerenal (e.g., volume depletion, hypercalcemia, capillary leak, renal sodium wasting), intrarenal (e.g., direct cancer invasion, paraneoplastic effects, drug toxicities), and postrenal (e.g., retroperitoneal cancer or fibrosis, bladder cancer).
- Cancer is frequently complicated by chronic kidney disease (CKD). A number of causes lead to CKD including the underlying malignancy, drugs used to treat cancer, shared risk factors (e.g., hypertension, diabetes mellitus), nephrectomy for renal cell cancer, and therapies such as hematopoietic stem cell transplantation.
- Diseases involving the glomeruli also complicate various malignancies. Membranous nephropathy, minimal change disease, and proliferative glomerulonephritides are noted paraneoplastic effects of cancer. Paraproteinemias with monoclonal immunoglobulin or light- or heavy-chain synthesis also cause glomerular injury (e.g., with primary “AL” amyloidosis and monoclonal immunoglobulin deposition disease).
- Electrolyte and acid–base disorders frequently complicate cancer and its therapy. Hyponatremia, hypernatremia, hypokalemia, and hyperkalemia, as well as imbalances in calcium, magnesium, and phosphorus, are the result of direct or paraneoplastic effects or adverse effects of drugs on the gastrointestinal tract (e.g., vomiting, diarrhea) and/or kidneys.
- Hematopoietic stem cell transplantation is a life-saving procedure for patients with certain malignancies. However, this procedure is complicated by AKI, glomerular disease, hypertension, electrolyte abnormalities, and CKD, depending on the patient’s underlying risk and the conditioning regimen and type of stem cell transplant used.
- Medications used to treat cancer are associated with adverse renal effects. Conventional chemotherapeutic drugs, immunotherapies, and targeted agents are associated with AKI, glomerulopathies, and electrolyte and acid–base disturbances, and CKD.

Onconeurology refers to the intersection of the disciplines of nephrology, hematology, and oncology. Kidney disease in multiple forms, ranging from glomerular diseases to electrolyte abnormalities, is common in patients with cancer. In many cases, these kidney disorders are not unique to patients with cancer but can affect the care of the patient, such as affecting the dosage of appropriate chemotherapy in patients with chronic kidney disease (CKD). In other circumstances, either the malignancy itself or its associated therapy may lead to unique kidney diseases, such as acute kidney injury (AKI), glomerulonephritis, or electrolyte disorders.

ACUTE KIDNEY INJURY IN PATIENTS WITH MALIGNANCIES

EPIDEMIOLOGY

AKI is a common occurrence in hospitalized patients with cancer and, when it occurs, it is associated with higher costs, increased hospital length of stay, and increased morbidity and mortality. In some cases, AKI may lead to changes in chemotherapy regimens that may lessen the chances of disease remission, or AKI may exclude patients from clinical trials. In the largest study on this topic, 37,267 incident patients with cancer in Denmark had follow-up over a 7-year period in the early 2000s.¹ The 1-year risk of AKI, as defined by RIFLE risk (>50% increase in serum creatinine), was 17.5%. The 1-year risk for more severe AKI, the RIFLE injury (>100% increase in serum creatinine), and failure (>200% increase in serum creatinine of >4 mg/dL and requiring dialysis) risk categories was 8.8% and 4.5%, respectively. The 5-year risk for the RIFLE risk, injury, and failure AKI categories was even higher, at 27%, 14.6%, and 7.6%, respectively. Critically ill patients with cancer have a higher incidence of AKI and AKI requiring dialysis than critically ill patients without cancer.^{2,3} In those patients with cancer who suffer an episode of AKI, the mortality is elevated, such that the mortality was 13.6% in those without AKI and progressively increased with higher RIFLE stage AKI (risk, 49%; injury, 62.3%; failure, 86.8%).² In patients with hematologic malignancies who were undergoing induction therapy, the mortality in those with RIFLE risk AKI was 13.6%, whereas the mortality in those with no AKI was 3.8% over an 8-week period.⁴ Importantly, those patients who required dialysis during the period of induction therapy had a mortality of 61.7%.⁴ It is important to note that more contemporary studies may demonstrate that outcomes in patients with cancer and AKI are not universally poor, with one study showing that 82% of critically ill patients with cancer with AKI completely recovered kidney function, whereas partial recovery was observed in 12%, and chronic renal replacement therapy (RRT) was required in only 6% of patients.⁵ The prognosis of AKI greatly depends on the underlying functional and premorbid conditions of the patient, as well as the overall state of the malignancy. Thus individualized decisions regarding the appropriateness of RRT in these patients is warranted.

A specific group of patients with a high incidence of AKI is those undergoing hematopoietic stem cell transplantation (HSCT). Various studies have shown an incidence of AKI

Table 57.1 Risk Factors for Acute Kidney Injury in Patients With Cancer

Modifiable Risks	Nonmodifiable Risks
Nephrotoxic medications (nonchemotherapeutic), such as aminoglycosides, amphotericin B, calcineurin inhibitors	Age >65 years
Nephrotoxic medications (chemotherapeutic), such as cisplatin, checkpoint inhibitors, ifosfamide, interferon, tyrosine kinase inhibitors, BRAF (serine-threonine protein kinase) inhibitors	Underlying chronic kidney disease, especially diabetic kidney disease
Hypovolemia • True volume depletion—nausea, vomiting, diarrhea • Effective volume depletion—cirrhosis, heart failure, nephrotic syndrome, hypoalbuminemia	Specific cancer types—multiple myeloma, hepatocellular cancer, renal cell carcinoma, pelvic malignancies
Intravenous radiographic contrast	Hematopoietic stem cell transplantation
High tumor bulk and risk for tumor lysis syndrome	Female gender

ranging from 23% to 73% according to the different cutoff points used to define AKI.^{6,8} Furthermore, the incidence of AKI varies according to the type of HSCT, with most data supporting the finding that myeloablative allogeneic transplantation is associated with a higher incidence of AKI.⁸

The risk factors for AKI in patients with cancer are multiple and depend on the specific type of cancer. These can be broadly divided into those that are modifiable and those that are not (Table 57.1). Certain cancers appear to carry a higher AKI risk than others, with renal cell carcinoma, hepatocellular carcinoma, multiple myeloma, and lymphoma having higher rates of AKI.¹ As an example, patients undergoing total (radical) nephrectomy for renal cell carcinoma have an incidence of AKI as high as 33.7%.⁹ Even partial nephrectomy to spare nephron loss is associated with AKI, albeit less commonly.¹⁰ Patients with acute lymphoma or leukemia undergoing induction chemotherapy are also at especially high risk to develop AKI, primarily due to tumor lysis syndrome and drug nephrotoxicity. In a series of 537 patients undergoing induction therapy for acute myelogenous leukemia or high-risk myelodysplastic syndrome, 36% developed AKI.⁴ Whenever possible, modifiable risk factors should be addressed in an attempt to lower the incidence of AKI.

CAUSES OF ACUTE KIDNEY INJURY IN PATIENTS WITH CANCER

The causes of AKI in patients with solid organ cancers are similar to those of the general population, with an overrepresentation of obstructive nephropathy (especially with prostate, ovarian, and cervical cancer), chemotherapy-associated nephrotoxicity, sepsis-associated ischemic acute tubular necrosis in the setting of neutropenia, and AKI due to hypercalcemia.¹¹⁻¹³ In patients with hematologic

malignancies, the causes of AKI are also similar to those of the general population, with some notable unique causes, such as AKI associated with multiple myeloma (e.g., cast nephropathy, light- or heavy-chain-associated glomerulonephritis or associated with hypercalcemia), tumor lysis syndrome, and tumor infiltration of the kidneys associated with lymphoma and leukemia.¹¹⁻¹³ In those patients receiving a stem cell transplant, there are other situation-specific causes of AKI including hepatic venoocclusive disease and cytokine release syndrome.¹⁴

Prerenal causes of AKI are common in patients with malignancies and often result from poor oral intake, as well as chemotherapy-induced nausea, vomiting, and diarrhea. Hypercalcemia and its natriuretic effect may also lead to volume depletion and prerenal azotemia. In addition, patients with cancer may be prescribed common medications, such as diuretics, angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, or nonsteroidal anti-inflammatory drugs that can affect renal autoregulation and exacerbate prerenal AKI. Prerenal causes are so common in this patient population that judicious trials of intravenous (IV) fluids are reasonable in most patients presenting with AKI.

Many of the causes of AKI seen in patients with cancer are not unique and are covered in other chapters. However, certain intrarenal and postrenal causes are seen almost exclusively in the setting of malignancy and are covered here and shown in [Table 57.2](#). AKI associated with glomerular disorders, chemotherapeutic agents, HSCT, and hypercalcemia and seen in the setting of renal cell carcinoma are covered in their respective sections in this chapter.

TUMOR LYSIS SYNDROME

Tumor lysis syndrome (TLS) is encountered in patients with bulky, rapidly growing and chemosensitive malignancies, such as high-grade lymphomas (e.g., Burkitt lymphoma), leukemia, or other cancers with large cellular burdens.^{15,16} Typically, TLS is seen after the administration of chemotherapy but can occur spontaneously as well. TLS has been defined by the Cairo-Bishop criteria ([Table 57.3](#)) and is biochemically typified by the findings of numerous electrolyte disorders (e.g., hyperkalemia, hyperphosphatemia, hypocalcemia, hyperuricemia) that result from the release of intracellular substances after cell lysis.^{15,16} Importantly, TLS may lead to sudden death and seizures due to the electrolyte disturbances encountered in these patients. The mechanism of AKI in these patients is at least partially due to uric acid nephropathy but may also be influenced by cytokine release and nephrocalcinosis, the latter due to elevated serum concentrations of phosphate and calcium.¹⁷

Uric acid nephropathy and AKI are the result of the precipitation of insoluble uric acid in the renal tubules that occurs when purine nucleotides are released from dying cancer cells. These nucleotides are metabolized by xanthine oxidase into uric acid, which is subsequently filtered in the glomerulus and concentrated in the renal tubules. This forms an insoluble precipitate, leading to a combination of intratubular obstruction, vasoconstriction, and inflammation, culminating in a fall of the glomerular filtration rate and AKI.¹⁸ Understanding this pathophysiology leads to rational protocols for the prevention of TLS ([Fig.](#)

Table 57.2 Intrarenal and Postrenal Causes of Acute Kidney Injury in the Patient With Cancer

Cause	Representative Example
Vascular—thrombotic microangiopathy	Gemcitabine, post stem cell transplantation
Tubular injury—acute tubular necrosis	Ischemia due to sepsis, nephrotoxicity due to chemotherapeutic agents such as cisplatin
Tubular injury—intratubular precipitation of crystals	Methotrexate
Tubular injury—cast nephropathy	Multiple myeloma
Tubular injury—lysozymuria	Acute promyelocytic, acute monocytic leukemia, chronic myelomonocytic leukemia
Interstitial injury—interstitial nephritis	Immune checkpoint inhibitors
Interstitial injury—tumor infiltration of the kidneys	Lymphoma
Glomerular injury	Various types of amyloidosis Various paraneoplastic or drug-induced glomerulonephritis Paraneoplastic membranous nephropathy
Obstruction	Retropertoneal lymphadenopathy (lymphoma) Tumor bulk obstructing urine flow (pelvic malignancies)

Table 57.3 Cairo-Bishop Criteria for Tumor Lysis Syndrome

Type of Criteria	Features
Laboratory criteria for tumor lysis syndrome	Requires 2 or more of the following criteria within 3 days to or 7 days after initiation of chemotherapy: - Uric acid level ≥ 8 mg/dL or 25% increase from baseline - Potassium level ≥ 6.0 mEq/L or 25% increase from baseline - Phosphorus level ≥ 6.5 mg/dL for children - Phosphorus level ≥ 4.5 mg/dL for adults or 25% increase from baseline - Calcium level ≤ 7 mg/dL or 25% decrease from baseline
Clinical criteria for tumor lysis syndrome	Laboratory tumor lysis syndrome plus one or more of the following criteria: - Creatinine >1.5 times upper limit of age-adjusted reference range - Cardiac dysrhythmia or sudden death - Seizure

From Cairo MS, Bishop M. Tumor lysis syndrome: new therapeutic strategies and classification. *Br J Hematol*. 2004;127(1):3–11.

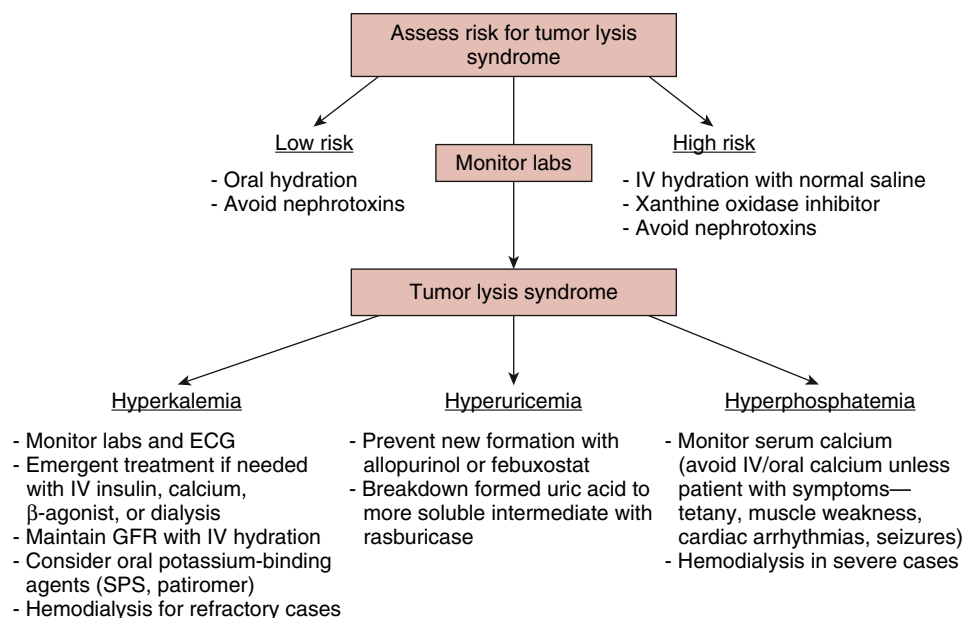


Fig. 57.1 Approach to the management of tumor lysis syndrome. ECG, Electrocardiogram; GFR, glomerular filtration rate; IV, intravenous; SPS, sodium polystyrene sulfonate. (From Rosner MH, Perazella MA. Acute kidney injury in patients with cancer. *N Engl J Med.* 2017;376(18):1770–1781.)

57.1). While most of the literature supports crystalopathy-related AKI as the primary mechanism of injury, one study has postulated a novel pathophysiologic model through which AKI is associated with endothelial dysfunction from high levels of extracellular histones.¹⁹ This animal study proved that extracellular histones are released in vast amounts during TLS, causing a profound endothelial injury in the mouse model. The mechanisms of histone-mediated damage implicate endothelial cell activation mediated by Toll-like receptor 4. In addition, TLS has now been associated with novel agents, such as the B cell lymphoma 2 inhibitor venetoclax, the monoclonal antibody obinutuzumab, and even immune checkpoint inhibitors, but data on such associations are limited to case reports and case series.^{20–22}

The first step in a preventive strategy rests on identifying patients at risk for TLS, which should include any patient with a large, rapidly growing and chemosensitive tumor or those patients who, before treatment, may already be manifesting signs of tumor lysis with an associated electrolyte disturbance. Low-risk patients should be treated with oral fluids and avoidance of other nephrotoxins; high-risk patients should receive IV 0.9% saline to maintain adequate glomerular filtration and tubular flow rates that facilitate rapid clearance and dilution of uric acid, as well as the clearance of potassium and phosphate. The prophylactic use of a xanthine oxidase inhibitor (allopurinol or febuxostat) is recommended because these drugs will block de novo formation of uric acid and limit further increases in the levels of this compound. For patients who already have elevated uric acid levels, treatment with recombinant urate oxidase (rasburicase) is beneficial for rapidly lowering uric acid levels and is recommended.²³ Rasburicase converts uric acid to a soluble and readily excreted compound, allantoin.²³ It should be cautioned that rasburicase also leads to the production of hydrogen peroxide, which can cause methemoglobinemia and hemolytic anemia in patients with glucose-6-phosphate dehydrogenase (G6PD)

deficiency, and at-risk patients should be tested for this condition before administration.²⁴

For those patients manifesting signs of TLS, close monitoring of serum electrolyte levels and clinical status is required (see Fig. 57.1) and electrolyte disorders require rapid therapy. Patients should receive aggressive IV hydration, a xanthine oxidase inhibitor, and rasburicase. For patients presenting with established AKI, emergent hemodialysis to correct electrolyte disorders may be required. Urine alkalization, which increases uric acid solubility in the renal tubules, is not recommended because it can increase the likelihood of calcium phosphate precipitation.¹⁸

LYSOZYMURIA

Lysozyme can be released from certain hematologic malignancies (e.g., acute promyelocytic, monocytic, chronic myelomonocytic leukemia-CMML) and is filtered by the glomerulus and then reabsorbed by proximal tubular cells, where it leads to tubular damage and AKI.²⁵ The condition is rare, and suspected cases can be confirmed by urinary protein electrophoresis, which can identify lysozyme in the urine. Kudose and colleagues²⁶ published the largest case series comprising 37 patients with lysozyme-associated nephropathy, among whom 15 were diagnosed with CMML. The kidney manifestations commonly observed with lysozymuria included AKI, CKD, and subnephrotic range proteinuria, with a median estimated glomerular filtration rate (eGFR) of 21.7 mL/min per 1.73 m² and median proteinuria of 1.7 g at the time of biopsy. Some patients also exhibited microscopic hematuria and partial Fanconi syndrome, consistent with previous reports. The hallmark finding on pathology was the presence of abundant periodic acid-Schiff–positive hypereosinophilic intracytoplasmic inclusions, exhibiting a characteristic staining pattern with lysozyme immunohistochemical stain. Notably, these proximal tubular inclusions tested negative on silver

methenamine Jones (JMS) stain, appeared fuchsinophilic on trichrome, and exhibited a pale pink hue on Congo Red staining. Further characterization through electron microscopy revealed these intracytoplasmic inclusions as membrane-bound vacuoles containing homogenous or granular electron-dense material, which likely represent autophagolysosomes. These degenerating cellular organellar debris were diffusely distributed in proximal tubular epithelia.²⁷

TUMOR INFILTRATION OF THE KIDNEYS

For unclear reasons, leukemia and lymphoma cells have a predilection for infiltrating the kidney and, on autopsy, up to 60% of these patients will have tumor cell infiltrates in the kidney.²⁸ In most cases, these infiltrates have little clinical significance but, in approximately 1% of cases, they may be so massive that they result in AKI.^{28,29} AKI in these cases likely results from compression of the renal parenchyma by tumor cells, with distortion of the microvasculature and tubular architecture. Patients may present with hypertension (HTN), flank pain, and hematuria or, more often, are asymptomatic. Renal imaging with ultrasonography or computed tomography shows bilaterally enlarged kidneys with a heterogeneous texture. A kidney biopsy is diagnostic and reveals tumor cell infiltration. Appropriate chemotherapy with tumor response can lead to rapid improvement in kidney function.

THROMBOTIC MICROANGIOPATHY

Thrombotic microangiopathy (TMA) can be a consequence from the tumor itself or, more likely, from associated chemotherapy, such as gemcitabine; vascular endothelial growth factor (VEGF) inhibitors, such as bevacizumab; and other tyrosine kinase inhibitors.³⁰⁻³³ TMA has been associated with mucin-producing gastric, lung, and breast cancers. TMA may also be encountered in patients undergoing HSCT, in the setting of graft-versus-host disease (GVHD, associated with cyclosporine use), and with radiation nephropathy.³³ The presentation of TMA in patients with cancer may be subtle and can be acute (after a clearly defined insult) or delayed (months after starting a chemotherapy regimen, such as gemcitabine). In some patients, new-onset and relatively severe HTN and proteinuria may be an early clue to the diagnosis of TMA.³² Unfortunately, the classic laboratory signs of TMA—falling hemoglobin, falling platelet count, and elevated lactate dehydrogenase level, along with falling haptoglobin and schistocytes on a peripheral blood smear—may not be present or may be mild. In some cases, kidney biopsy should be considered for a definitive diagnosis, where pathologic findings of edematous intimal expansion of arteries, fibrinoid necrosis of arterioles, ischemic collapse of capillaries, and focal thrombosis of injured vessels can be seen. Kidney-limited TMA may not be uncommon in patients with cancer who receive novel targeted therapies. In VigiBase, a large global pharmacovigilance database, TMA was most commonly reported with gemcitabine/mitomycin, proteasome inhibitors, and antiangiogenic agents. Time to TMA was shortest in patients receiving proteasome inhibitors, while immune checkpoint inhibitor-associated TMA had the highest mortality. Authors used the information component (IC), a

disproportionate Bayesian analysis comparing the number of observed versus expected adverse drug reactions, to determine the potential association between antineoplastic agents and TMA. The IC025 indicates the lower end of 95% of the information component, in which a value >0 suggests a disproportionality signal between the drug of interest and the adverse drug reaction. Carfilzomib had the highest IC025 for TMA among all studied chemotherapies, followed by gemcitabine, mitomycin, bevacizumab, and bortezomib.²⁷ There are no trials that guide therapy in these cases, and treatment relies on removing any insults, such as medications, and consideration of using plasmapheresis, eculizumab, and rituximab in more refractory cases.^{32,33} Malignancy-associated TMA usually responds poorly to plasmapheresis and carries a poor prognosis overall.

LIGHT-CHAIN CAST NEPHROPATHY AND ACUTE KIDNEY INJURY ASSOCIATED WITH MYELOMA

Patients with myeloma have an incidence of AKI that can be as high as 50%.³⁴ The most common cause of AKI in these patients is cast nephropathy, followed by light chain–related proximal tubulopathy, various glomerular diseases (including light-chain deposition disease and primary [AL] amyloidosis), hypercalcemia, hyperuricemia, and other contributing factors (Fig. 57.2).

Cast nephropathy is due to the filtration of large amounts of myeloma-produced free light chains into the renal tubules, which bind to Tamm-Horsfall protein (also known as uromodulin) to form insoluble aggregates.³⁵ These intratubular precipitates lead to intrarenal obstruction and local inflammation, which may be further exacerbated by the proximal tubular uptake of pathogenic light chains, which also leads to activation of inflammatory pathways.³⁶ Sensitive assays for serum free light chains, serum and urine protein electrophoresis, and immunofixation facilitate the diagnosis of myeloma-related AKI. Typically, patients with myeloma-related glomerular diseases will present with high levels of albuminuria (>2 g/day) indicative of the glomerular injury. Patients with cast nephropathy will demonstrate high levels of urine and serum light chains and lower levels of albuminuria.³⁷ Of note, 15% of patients with AKI in the setting of myeloma may have a cause of AKI completely unrelated to myeloma; thus kidney biopsy should be performed in unclear cases.³⁷

Therapy for cast nephropathy focuses on a combination of hydration and ensuring sufficient renal tubular flow, correction of hypercalcemia, treatment of other precipitating or aggravating factors, and chemotherapy to lower the myeloma production of pathogenic light chains. Chemotherapeutic regimens for myeloma generally include a proteasome inhibitor, such as bortezomib, which does not require dosing adjustments for renal function.³⁸ Regimens including bortezomib (usually in combination with dexamethasone and another agent, such as thalidomide or melphalan) have led to renal response rates as high as 72% and dialysis discontinuation in 57% of patients.³⁹ Other effective agents include thalidomide, lenalidomide, pomalidomide, and carfilzomib.⁴⁰ A great deal of controversy remains surrounding the benefit of removing light chains by therapeutic plasma exchange or high-flux hemodialysis; currently, these methods cannot be routinely recommended.⁴¹

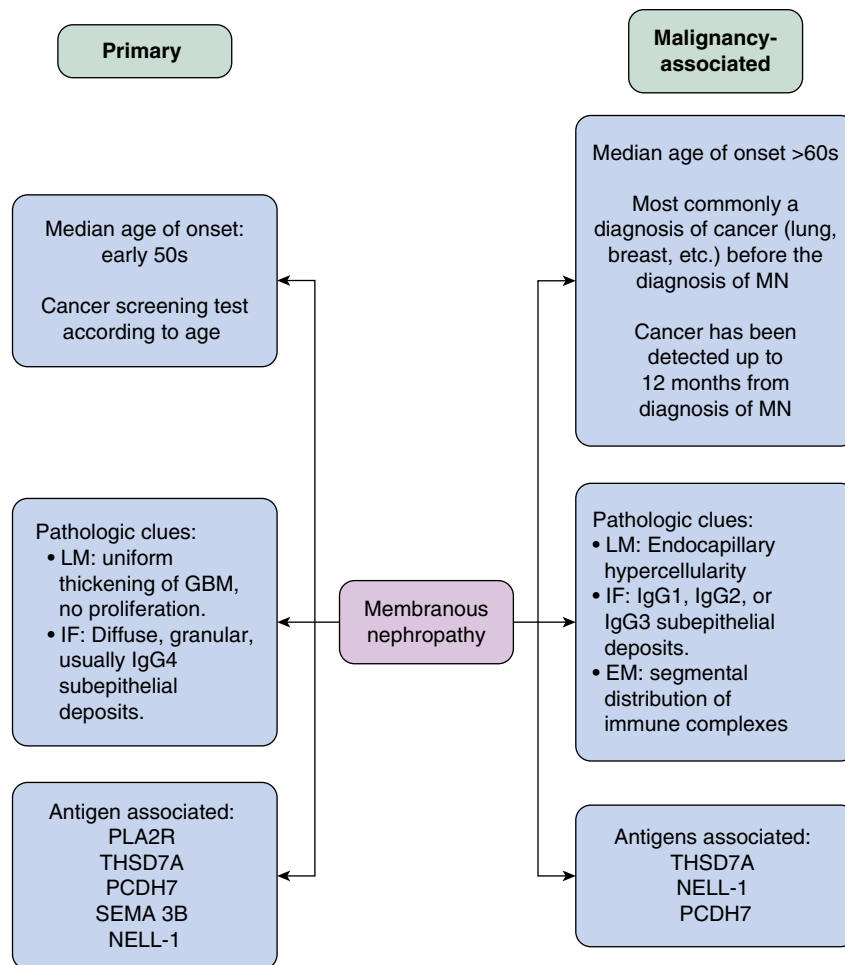


Fig. 57.2 Approach to distinguish primary membranous nephropathy from cancer-associated membranous nephropathy. EM, electron microscopy; GBM, glomerular basement membrane; IF, immunofluorescence; Ig, immunoglobulin; LM, light microscopy; MN, membranous nephropathy; NELL-1, neural epidermal growth factor-like 1 protein; PCDH7, protocadherin 7; PLA2R, phospholipase A2 receptor; SEMA 3B, semaphorin-3B; THSD7A, thrombospondin type I domain-containing 7A.

URINARY TRACT OBSTRUCTION

Obstruction of the urinary tract is a common cause of AKI in the patient with cancer. Obstruction can be intratubular due to precipitation of uric acid, as in TLS, or from medications such as high-dose methotrexate or acyclovir.⁴² In all cases, knowledge of the risk for intratubular obstruction should lead to the administration of prophylactic IV fluids to maintain urinary dilution of the drug or uric acid and maintenance of tubular fluid flow.

More commonly, urinary tract obstruction will be extrarenal and due to extrinsic compression of the ureters by a tumor mass or due to prostatic disease compressing the urethra. The clinical spectrum of patients presenting with malignant ureteral obstruction was illustrated by a case series of 102 patients.⁴³ Obstruction was bilateral in 68% of patients, and initial management with a percutaneous nephrostomy or ureteral stent was successful in 95% of cases. Despite successful decompression, 53% of patients developed complications, mostly urinary tract infection and obstruction of nephrostomy tubes or stents, and overall survival was poor (median, 7 months), reflecting the advanced stage of malignancy in such patients. Although renal ultrasonography will usually demonstrate hydronephrosis in patients with

obstruction, patients with retroperitoneal fibrosis may not have this finding and, if clinical suspicion is high for this condition, other imaging techniques such as nuclear medicine scans should be pursued.⁴⁴

DECISION MAKING IN THE PATIENT WITH CANCER AND ACUTE KIDNEY INJURY

As discussed earlier, the prognosis of critically ill patients with cancer and AKI is poor, with mortality rates approaching 60% to 70% if RRT is needed. Thus clinicians may be faced with difficult decisions when offering invasive intensive therapy for such a patient population and may appropriately opt for more conservative and palliative approaches. However, it is important to individualize these decisions because select patients (e.g., those with fewer comorbid conditions, higher functional status, and cancers that are responsive to therapy) may benefit from more intensive therapy. Therefore decisions regarding the initiation of dialysis therapy require input from the entire care team, as well as the patient and family. These decisions should consider the reversibility of the AKI, long-term cancer prognosis, quality of life before critical illness, and the patient's preferences.

CHRONIC KIDNEY DISEASE IN PATIENTS WITH CANCER

EPIDEMIOLOGY

As has been observed with AKI, CKD is also noted to be a possible complication of numerous cancers and their associated therapy. There are several explanations for this observation. An important driver of this untoward complication is the fact that preexisting and often unrecognized CKD is common in patients with various types of malignancy. In addition to underlying kidney disease, incomplete recovery of kidney function after multiple episodes of AKI, nephrectomy for kidney cancer with a reduction in nephron mass, and exposure to several courses of nephrotoxic medications can ultimately lead to the development of glomerulosclerosis and tubulointerstitial fibrosis, all of which either exacerbate underlying kidney disease or promote the development of new CKD.

In regard to preexisting CKD in patients with cancer, the Renal Insufficiency and Anticancer Medications (IRMA)-1 and -2 studies clearly illustrate the extent of this problem.^{45,46} In the two IRMA studies, 52.9% (IRMA-1) and 50.2% (IRMA-2) of patients with an active malignancy had an estimated glomerular filtration rate (eGFR) <90 mL/min/1.73 m², respectively.^{45,46} In this group of patients, the prevalence of stage G3 CKD was 12% (IRMA-1) and 11.8% (IRMA-2), respectively. Stage G4 CKD was rare in these two studies (0.9% and 0.7%, respectively). In an Australian cohort with various cancers (*n* = 4077) using a prospective population cohort design, eGFR was <60 mL/min/1.73 m² in 30% of patients and <45 mL/min/1.73 m² in 8.3%.⁴⁷ In a cohort of Chinese patients with underlying cancer observed from 2000 to 2004, stage G3 or more severe CKD (defined as an eGFR < 60 mL/min/1.73 m²) was noted in 12.8% (1051 of 8223).⁴⁸ The relatively high prevalence of CKD in patients with cancer has been confirmed in other studies, irrespective of the type of malignancy.

OUTCOMES AND PROGNOSIS

Similar to patients with malignancy who develop AKI, CKD in patients with malignancy experience an increased risk of death compared with those without kidney disease.⁴⁷⁻⁴⁹ Higher mortality is not uniformly noted; however, the association of CKD with mortality appears to be more pronounced with certain cancer types and definitively related to CKD severity. Patients with cancer and stage G3 CKD had an adjusted hazard ratio (HR) for cancer-specific mortality of 1.12 (95% confidence interval [CI], 1.01–1.26).⁴⁸ In contrast, patients with stage G4 CKD had an adjusted HR for cancer-specific mortality of 1.75 (95% CI, 1.32–2.32). As described in a population-based cohort study, stage G3 CKD (eGFR <30 to <60 mL/min/1.73 m²) was a significant risk factor for death from cancer.⁴⁷ In this study, the adjusted HR for cancer-specific mortality was 1.18 (95% CI, 1.08–1.29) for each 10 mL/min/1.73 m² decrease in eGFR. It is notable that mortality risk was highest for patients with breast and urinary tract cancers.⁴⁷ Patients with kidney failure receiving hemodialysis who underwent radiofrequency ablation for hepatocellular cancer (HCC) were identified using the Japanese Diagnosis Procedure Combination database.⁵⁰

Four patients without kidney failure were matched for each patient with kidney failure based on age, sex, treatment hospital, and treatment year. In-hospital mortality was more than 7-fold higher in the 437 patients with kidney failure compared with the 1345 matched controls (1.1% vs. 0.15%, respectively; odds ratio [OR] = 7.77; *P* < .001).

In addition to CKD identified by eGFR measurement, the presence of proteinuria was also associated with mortality in patients with cancer. A retrospective study of 9465 patients with newly diagnosed cancer from January to December 2010 observed an increase in all-cause mortality in patients with eGFR <60 mL/min/1.73 m² and proteinuria.⁴⁹ As seen in other studies, CKD was not associated with mortality in all type of cancers. For example, an eGFR <60 mL/min/1.73 m² was independently associated with death from hematologic cancers (adjusted HR, 2.93; 95% CI, 1.36–6.31) and gynecologic cancers (adjusted HR, 2.82; 95% CI, 1.19–6.70), but not for other cancers. Furthermore, a study in patients with colorectal cancer (*n* = 3379) noted that eGFR <60 mL/min/1.73 m² was not associated with mortality.⁵¹ In addition, a prospective cohort study of patients with cancer that examined the risk of death as a secondary endpoint did not show an association between CKD and mortality.⁵² Thus it appears that underlying CKD in patients with cancer is associated with mortality in some, but not all types of malignancy.

CANCER AND CHRONIC KIDNEY DISEASE: BIDIRECTIONAL RELATIONSHIP

As has been previously reviewed, CKD is a well-recognized complication of various malignancies and their therapies. However, in addition to the observed increase in CKD prevalence in patients with cancer, both CKD and ESKD appear to be risk factors for the development of a number of malignancies.^{53,54} Thus the relationship between the kidney and cancer can be viewed as bidirectional.⁵³ The association between severity of kidney disease and the risk of incident cancer was examined over 6,000,420 person-years of follow-up. During this time, 76,809 incident cancers were identified in 72,875 persons. After adjustment for time-updated confounders, eGFR was inversely associated with the risk of kidney cancer. The adjusted HR was 2.28 (95% CI, 1.78–2.92) for an eGFR <30 mL/min/1.73 m².⁵⁵ The authors also observed an increased risk of urothelial cancer at an eGFR <30 mL/min/1.73 m² but no significant associations between eGFR and other types of cancer. Several observational studies have suggested an increased cancer risk in patients with kidney failure receiving maintenance dialysis RRT⁵⁵⁻⁶¹ (Table 57.4). A study from Taiwan published in 2015 demonstrated higher risks of cancer in patients with ESKD (HR, 3.43) with higher risks specifically observed for oral, colorectal, liver, blood, breast, renal, upper urinary tract, and bladder cancer than those seen in the general population.⁵⁹ In one study, the observed increased risk for renal parenchymal cancer in patients with ESKD was related to the development of acquired renal cystic disease, which increases with time on dialysis.⁵⁶ A U.S. study that examined data from 1996 to 2009 observed a higher incidence of cancer in patients with ESKD. The cumulative 5-year incidence of any cancer was 9.48% compared with a 5-year cumulative incidence of cancer in U.S. transplant recipients of 4.4%.⁶⁰

Table 57.4 Standardized Incidence Rates of Cancer in Hemodialysis Patients^a

Type of Cancer	Standardized Incidence Rate
All cancer types	1.42
Kidney parenchyma/pelvis	4.03
Bladder	1.57
Breast (female)	1.42
Non-Hodgkin lymphoma	1.37
Pulmonary, lung	1.28
Colonic, rectal	1.27
Pancreatic	1.08
Prostatic	1.06

^aSee references.⁵⁵⁻⁶¹

POTENTIAL EFFECTS OF CHRONIC KIDNEY DISEASE ON CANCER

The increased mortality observed as a complication in certain patients with cancer with CKD compared with those without kidney disease has several plausible explanations. The presence of preexisting CKD may limit the use of effective agents or promote underdosing of anticancer regimens that might otherwise be curative.^{53,62} In addition, underlying CKD may alter the bioavailability and/or safety profile of some anticancer drugs, which can potentially lead to different and suboptimal treatment choices. The development of AKI superimposed on CKD, which is a common complication, can lead to the cessation of effective chemotherapeutic regimens, allowing unhindered and potentially rapid tumor growth.^{53,62} Furthermore, CKD and ESKD may also be associated with impaired immune surveillance, which would allow cancers to grow and metastasize, especially when combined with insufficient drug dosing. This risk is further increased when these patients receive a kidney transplant, where immunosuppressive regimens (and other factors) enhance the risk for recurrent or de novo cancers that have been associated with mortality.^{63,64} It is also possible that the adverse renal effects of the anticancer drug used leads to acute and/or progressive kidney injury or to worsening of preexisting CKD, which leads to increased all-cause mortality unrelated to cancer.^{53,62} Finally, a nihilistic view of patients with CKD or ESKD who develop cancer may lead to undertreatment of potentially curable or modifiable malignancies.

ASSESSMENT OF KIDNEY FUNCTION

Patients with cancer require frequent assessment of kidney function to ensure proper dosing of chemotherapeutic agents.^{53,62,65} In addition, close monitoring during ongoing anticancer therapy is critical for timely recognition of drug-induced nephrotoxicity (either AKI or progression of underlying CKD).^{62,65} The Kidney Disease: Improving Global Outcomes (KDIGO) guidelines have recommended that the glomerular filtration rate (GFR) should be the standard measure to evaluate kidney function using the most accurate method available in the individual cancer patient.⁶⁶ This is of particular importance because the incidence of

CKD in patients with cancer is as high as 50%.⁴⁵ Lack of appreciation for the reduced GFR in patients with CKD can lead to erroneous, harmful, and potentially life-threatening dosing of medications that are cleared predominantly by the kidneys. However, the optimal method for assessing GFR in patients with cancer is currently unknown. Several small studies have examined the accuracy and precision of the traditional eGFR equation, including the four-variable Modification of Diet in Renal Disease, Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI), and two equations derived specifically in patients with cancer (the Wright and Martin formulas).^{67,68} The previously described bedside eGFR equations all performed similarly well when compared with GFR determined by technetium-99m diethylenetriamine pentaacetic acid (^{99m}TcDTPA) clearance and when used for selecting the appropriate drug dosage for carboplatin.⁶⁹ It is worth noting that all GFR estimating equations significantly underdosed carboplatin, highlighting the need for continued research to determine the most accurate methodology for assessing GFR in this growing patient population.

The Onco-GFR study by Silva and colleagues⁷⁰ used measured GFR using chromium-51-labeled ethylenediamine tetraacetic acid (⁵¹Cr-EDTA) clearance in 1200 patients with solid tumors to test six GFR estimating equations. They reported both the bias (median of the differences between measured GFR and estimated GFR) and accuracy (1 minus the percentage of GFR estimates within 30% of measured GFR in mL/min/1.73 m² [1–P30]) of each equation. The 2012 CKD-EPI equation, using both serum creatinine and cystatin C, performed the best among all equations. Among the GFR estimating equations that used serum creatinine alone, Cockcroft-Gault and 2009 CKD-EPI had the largest bias and Cockcroft-Gault had the least accuracy. In subgroup analyses, the authors showed that several patient-specific factors strongly influenced the accuracy of GFR estimation. Creatinine-based equations were much more likely to overestimate GFR in women and in those with low body mass index (BMI [<25 kg/m²]). In these populations, the CKD-EPI 2012 equation that uses cystatin C alone was most accurate. This suggests that GFR estimation could be personalized on the basis of patient-specific factors such as BMI and sex. The authors also demonstrate that in patients with measured GFR <60 mL/min/1.73 m², all equations overestimated GFR. In another study, data from 2011 to 2023 on adult patients with cancer who underwent measured GFR (mGFR) at Mayo Clinic were analyzed. GFR estimating equations were compared using performance metrics. Among 1145 patients with both serum creatinine and cystatin C concentrations available, the CKD-EPI creatinine–cystatin C equation showed the best performance, with low bias and highest precision across both solid and hematologic cancers. The European Kidney Function Consortium equation performed similarly, while the Cockcroft–Gault equation had the poorest precision. Findings support the use of combined creatinine–cystatin C equations over creatinine-only or cystatin C-only equations in patients with cancer.⁷¹

It is noteworthy that the new 2021 CKD-EPI race-free equation that uses both serum creatinine and cystatin C concentrations was not used in this study.

Currently, we recommend use of the CKD-EPI creatinine–cystatin C formula for determination of eGFR in most

patients with cancer. Note that the eGFR is not reliable in the setting of AKI, and the KDIGO guidelines have therefore recommended that timed clearances of urea and creatinine may be of great value for persons with AKI.⁶⁶ See [Chapter 23](#) for a detailed discussion of methods for measuring and estimating GFR.

DRUG DOSING IN CHRONIC KIDNEY DISEASE AND END-STAGE KIDNEY DISEASE

Anticancer drugs are cleared from the body predominantly by the kidneys, extrarenal pathways, or a combination of the two. One of the most important drug-related problems in patients with cancer and impaired kidney function is inappropriate medication use and dosing errors. For example, many cytotoxic drugs and their active and potentially toxic metabolites are eliminated by the kidneys via some combination of glomerular filtration, tubular secretion, and/or tubular reabsorption. Furthermore, patients with CKD receiving chemotherapeutic agents often have alterations in one or more of their pharmacokinetic parameters.⁷² Alterations in medication absorption, volume of distribution and drug distribution in the body, protein binding, biotransformation, and kidney excretion may result in the accumulation of potentially toxic components and overdosage.^{73,74} It is therefore critical that clinicians adjust doses of drugs that are excreted primarily by the kidneys appropriately. Drug dosing will use either the calculated or measured creatinine clearance or GFR estimating equations (as discussed earlier), which should allow for safer and more effective use of chemotherapy in patients with cancer and underlying CKD. Drug dosing and appropriate adjustments for the most commonly used chemotherapeutic agents, which may require dose modification in the setting of underlying kidney disease, are noted in [Table 57.5](#).^{69,75,76}

It is important to recognize that CKD also alters the pharmacokinetics of drugs that are cleared by nonrenal pathways. It has been demonstrated that uremia (in the setting of both AKI and CKD) affects hepatic drug metabolism and coupled transport.⁷⁷ Selected uremic toxins interfere with transcriptional activation, cause downregulation of gene expression mediated by proinflammatory cytokines, and directly inhibit the activity of the cytochrome P450 enzymes and drug transporters.⁷⁸ These additional nuances in pharmacokinetics will make prescribing of some drugs to patients with CKD that much more complicated. As a result, clinicians will need to watch closely for symptoms and signs of drug-related overdose and toxicity even when the patient has received what appears to be appropriate renal dosing of an anticancer drug regimen. Thus clinicians should work closely with clinical pharmacists when these drugs are prescribed to CKD patients, and special considerations should be taken to optimize exposure to cytotoxic drugs and reduce the risk of adverse effects.⁷⁹ Precautions should include consideration of current medication pharmacokinetic and pharmacodynamic information, use of less nephrotoxic agents, and documented preventive measures, which are well integrated into oncology practice and vigilant surveillance.^{80,81}

The management of cancer in the ESKD population is particularly challenging. Little is known about anticancer

drug management in ESKD patients and even less about the optimal timing and necessary dosage adjustments, depending on all forms of dialysis. In most patients receiving maintenance hemodialysis, kidney function no longer contributes a significant proportion of drug clearance. As a result, these patients may require a reduction in drug dose to avoid excessive drug exposure and associated toxicity. This is clearly needed with drugs that are normally eliminated by the kidneys and also when extrarenal (hepatic primarily) metabolic and excretory pathways are impaired due to uremia. Because most cytotoxic drugs used are excreted predominantly in the urine as unchanged drug or active or toxic metabolite(s), any reduction in kidney clearance (as in CKD and ESKD) may result in the accumulation of potentially toxic components and overdosage.⁷⁵ Careful dosage adjustment is mandatory to optimize exposure to cytotoxic drugs and reduce the risk of adverse effects.

The efficiency (or lack thereof) of drug clearance by the modality of dialysis must also be considered to allow for appropriate timing of chemotherapy administration. Therefore it is necessary to determine what fraction of drug(s) is removed by hemodialysis to allow chemotherapy dosing after hemodialysis sessions (for medications with excellent dialytic clearance) to avoid drug removal, which may result in a loss of anticancer efficacy. Drugs that are not removed efficiently by hemodialysis can be administered at any time, either before or after the hemodialysis session. Hemodialysis influences drug pharmacokinetics primarily by the following three properties—drug clearance by dialysis, extraction coefficient, and dialysis extraction factor.⁷⁵ Hemodialysis clearance is the removal rate relative to blood concentration when entering the dialyzer. It is calculated by the amount removed (in mg/min) relative to the rate of presentation (mg/mL). Extraction coefficient, also called the “extraction ratio,” is the percentage of drug removed from blood across the dialyzer. It is calculated by the rate of removal (mL/min) relative to the rate of presentation (mL/min).^{82,75} Hemodialysis clearance and the extraction coefficient measure the ability of a hemodialysis system to remove a drug from the blood but do not indicate how readily the drug is removed from the body. Because hemodialysis clearance and extraction ratio cannot be extrapolated to the clinical setting, the hemodialysis extraction factor of the drug (when available) should be used to determine the clinical impact of hemodialysis session on drug removal. This extraction factor, which is derived from total body clearance and the dialysis clearance of a drug, represents the actual influence of dialysis on drug pharmacokinetics. Hemodialysis is considered clinically relevant if the dialysis extraction factor exceeds 25% of the overall drug elimination.^{83,84}

Ultimately, it is important to measure the pharmacokinetic parameters in patients receiving maintenance hemodialysis and to compare these data with those of patients with normal kidney function. The current lack of pharmacokinetic information of anticancer drug therapy in patients receiving maintenance hemodialysis underlines the need for prospective studies in these patients. Data on dosage adjustments in peritoneal dialysis, hemodiafiltration, and continued forms of renal replacement therapy are limited in most anticancer agents.

Table 57.5 Dose Modifications for Commonly Used Chemotherapeutic Agents in Chronic Kidney Disease Patients^{69,75,76}

Agent	Creatinine Clearance (mL/min)			
	90-60	60-30	30-15	Hemodialysis
Bleomycin	100%	75%	75%	50% ^a
Capecitabine	100%	75%	Avoid	Avoid
Carboplatin	Dosing based on AUC Calvert formula—total carboplatin dose (mg) = target AUC × (eGFR + 25); AUC varies between 5 (treated pts) and 7 (untreated pts; mg/mL × min)			In dialysis patients, consider GFR = 0 and target dose = 125–175 mg; E, 84% ± 3% ^b
Carmustine	100%	80% if CrCl ≥45; 75% if CrCl <45	Consider alternative	Consider alternative
Cisplatin	100%	50%	50%	25%–50% ^b
Crizotinib	100%	100%	50%	NA
Cyclophosphamide	100%	75%–100%	75%	75% ^a ; E, 40%–90%
Cytarabine (high dose)	100%	60% if CrCl ≥45; 50% if CrCl <45	30% or consider alternative	30% consider alternative ^a
Dacarbazine	100%	80%–70%	NA	100 mg daily for 5 days per cycle
Eribulin	100%	≥40 mL/min: 100%; <40 mL/min — NA	NA	NA
Etoposide	100%	75%	75%	50% ^c
Fludarabine	100%	80%, United States; 50%, United Kingdom	Avoid	Avoid
Hydroxyurea	100%	50%	50%	20% ^b
Ifosfamide	100%	75%–100%	75%–100%	75% ^b ; E, 87%
Irinotecan	100%	NA	NA	30%–40%
Lenalidomide	25 mg daily	10 mg daily	15 mg every 2 days	15 mg every 2 days; E, 31% ^b
Lomustine	100%	75%	75%	25%–50%
Melphalan	100%	75%	75%	50%
Methotrexate	100%	80%	50%	Avoid; HD Cl, 92 ± 10 mL/min ^b
Mitomycin	100%	100%	100%	75% ^a
Oxaliplatin	100%	100%	Avoid if CrCl <30 (Canada) or <20 (United States)	Avoid
Pentazocine	100%	66%	66%	50%
Pentostatin	100%	50%–75%	50%	50% (1–2 h before dialysis)
Pemetrexed	100%	100%	Avoid	Avoid
Regorafenib	100%	100%	NA	NA
Sorafenib	100%	100%	100%	25% of the dose; increase to 100% according to clinical safety and efficacy ^c
Sunitinib	100%	100%	100%	25%; then increase to 100%, according to clinical safety and efficacy; E, 0 ^c
Topotecan	100%	50%	50%	25% ^b
Vandetanib	100%	25%	NA	NA

^aIn the absence of data on its removal during dialysis, the drug will be administered after the session.

^bThe drug is dialyzable. Therefore it will be administered after the session, on days with HD.

^cThe drug may be administered indifferently, before or after the HD session.

CANCER SCREENING IN END-STAGE KIDNEY DISEASE

Although cancer screening recommendations for the general population focus on early detection and treatment to improve outcomes and survival,⁸⁵ it is not clear that these same principles apply to patients with ESKD.⁸⁶ Reasons to screen or not screen for malignancy in ESKD include a foreshortened lifespan, occurrence rates (common vs. rare) of specific cancers, variable accuracy of screening tools, and safety and efficacy of interventions when a cancer is identified.

Although maintenance dialysis prevents death from uremia, mortality among patients with ESKD remains high.⁸⁰ Generally, patients with ESKD younger than 80 years of age live less than one-third as long as their non-ESKD counterparts, whereas those aged older than 80 years are projected to live less than half as long as those without ESKD.⁸⁷ The most common cause for mortality, accounting for approximately 50% of deaths, is cardiovascular disease; death due to malignancy is relatively uncommon.⁸⁷ Individual characteristics should be taken into account because groups

include those with higher and lower mortality risks. For example, the 50-year-old man with uncontrolled type 2 diabetes mellitus, heart failure, and peripheral arterial disease has a much higher mortality than the 50-year-old man with polycystic kidney disease and no other significant comorbid conditions. Thus a patient-specific understanding of life expectancy is required. Due to a likely shortened life expectancy in ESKD patients primarily due to cardiovascular disease and infection, cancer screening strategies developed in the general population may not apply to them.

The goal of cancer screening is early cancer detection to enhance patient longevity. This appears to be an effective strategy for many cancers (e.g., using flexible sigmoidoscopy for colon cancer screening) in the general population when a patient's expected lifespan exceeds 10 years.⁸⁸ Thus colon cancer screening is unlikely to be effective in many ESKD patients with multiple comorbidities whose lifespan is 3 to 5 years. In addition, screening requires patients to undergo additional procedures and tests, which have the potential for harm. Thus in the absence of a survival benefit or if the survival benefit of screening is only realized in those with an expected lifespan of more than 10 years, cancer screening in most patients with ESKD is unlikely to improve outcomes or be cost-effective. It is noteworthy, however, that certain ESKD patients enjoy life expectancies more than 10 years and may benefit from individualized cancer screening. While life expectancy plays a key role in determining potential benefits of screening, underlying disease prevalence also plays a role. Both CKD and ESKD have been associated with an increased incidence of certain cancers,⁵⁵⁻⁶¹ some of which may be amenable to screening strategies. The degree to which incidence estimates reflect a true increase versus a detection bias remains unclear.

SCREENING TESTS FOR CANCER IN THE PATIENT WITH END-STAGE KIDNEY DISEASE

Cancer screening in ESRD patients is also highly dependent on the sensitivity and specificity of screening tests. Unfortunately, studies examining the accuracy of screening tests are generally performed on highly selected patient populations with minimal comorbidities.⁸⁹ Thus the extrapolation of these test findings to those with ESKD is highly questionable. For most commonly used screening tests, large-scale clinical studies assessing their predictive value in detecting cancer in the ESKD population have not been performed. In addition, there are a number of important caveats for several screening tests in the ESKD population. False-positive screening tests are observed in patients with ESKD due to underlying issues such as uremic platelets (positive fecal occult blood), mammary microcalcifications, dense breast tissue (positive mammography), and elevations in several tumor markers in the absence of cancer.⁹⁰ Importantly, false-positive screening tests may lead to inappropriate and potentially harmful diagnostic tests and workups, unnecessary costs, and minimal benefits.

OUTCOMES OF CANCER THERAPY IN PATIENTS WITH END-STAGE KIDNEY DISEASE

In view of the apparent increased incidence of selected malignancies in ESKD, the most important question is whether a malignancy identified by screening is amenable to effective and safe therapy. Unfortunately, robust data on

outcomes of cancer therapy in patients with ESKD are not available. Although cancer in this group is generally treated in the same manner as in the general population, there are important differences due to difference in pharmacokinetics and pharmacodynamics in ESKD (previously reviewed). In patients with ESKD, underlying drug dosing modifications and the challenges related to the highly variable dialyzability of drugs limit the applicability of outcome data generated in the general population.⁹¹ Cancer response rates may differ greatly with adjusted dosing and dialytic removal, which likely affect drug levels and overall drug exposure.

Although efficacy concerns abound for anticancer drug therapy, adverse drug effects are generally more frequent in patients with ESKD and can lead to morbidity and mortality. Clearly, more data are required on the efficacy of chemotherapeutic regimens in patients with ESKD, and an individualized approach including input from nephrologists, pharmacists, and oncologists is needed to optimize outcomes and mitigate complications of drug therapy. In addition to drug toxicity, studies have observed more frequent adverse outcomes in patients with ESKD undergoing surgical treatment for lung and colorectal cancers.⁹²⁻⁹⁴ Patients with colorectal cancer undergoing tumor resection suffered a higher risk of reintubation and a need for longer ventilator support, as well as higher rates of sepsis, deep surgical site infections, and pneumonia.⁹² Most importantly, the patients also had a much higher risk of death.⁹² Similar findings were found in patients undergoing surgical resection for lung cancer.^{93,94} These findings highlight the importance of considering the risk-benefit ratio when considering screening for malignancy in ESKD.

RECOMMENDATIONS FOR CANCER SCREENING

Data from routine cancer screening studies in ESKD patients have not shown this approach significantly increases life expectancy or is cost-effective.⁹⁵⁻⁹⁸ For example, the anticipated increase in life expectancy for a 60-year-old woman with ESKD and type 2 diabetes mellitus undergoing breast cancer screening ranges from 1 to 16 days.⁹⁵ Furthermore, a cost analysis demonstrated that the costs per unit of survival benefit from cancer screening range from 1.6 to 19.3 times higher in patients with ESKD, with the gain-in-life expectancy only 5 days or less, even considering high performance testing and curative therapy.⁹⁷ A Canadian study investigating the utility of breast and cervical cancer screening in patients with ESKD demonstrated similar results.⁹⁶ These studies were performed more than 10 years ago; since then, survival rates associated with kidney failure have improved somewhat, although we can anticipate that benefits of screening and subsequent therapy are likely to be modest. A cohort study in patients with ESKD aged older than 50 years noted that colon cancer screening was performed more often among healthier patients and those with the highest likelihood of transplantation; the overall screening rate was eight times higher than the rate in patients without ESKD but similar limited life expectancies, perhaps related to the requirement for colorectal cancer screening as part of kidney transplant evaluation.⁹⁹

Despite the fact that the general ESKD population may not benefit from universal cancer screening, an individualized approach is critical to identify patients who might benefit. The patient should be assessed for the following: 1. life

expectancy; 2. risk of cancer; 3. transplantation candidacy; and 4. his or her perspective and preferences. Gauging patients' prognosis and life expectancy help in the decision making process. A poor prognosis is likely when two or more of the following factors are present: 1. age older than 75 years; 2. high comorbidity score on a tool such as the Charlson score or a negative response to the surprise question (i.e., "Would I be surprised if this patient were alive in 1 year?"); and 3. poor functional status or poor nutritional state.¹⁰⁰ A number of prognostic scoring tools can be used to estimate life expectancy and prognosis.¹⁰⁰⁻¹⁰³ In contrast, patients with an acceptable life expectancy may engender benefit from age-appropriate cancer screening, as do patients with a strong family history of cancer or another specific cancer risk (e.g., acquired cystic kidney disease). In young patients and those considered transplantation candidates, screening for acquired cystic kidney disease is appropriate. Cancer screening is reasonable for patients who are transplantation candidates determined to have an acceptable longevity. Finally, it should be emphasized that even if cancer screening in patients without signs or symptoms is not warranted, testing should be strongly considered in patients with signs and/or symptoms that could represent malignancy (e.g., cough and hemoptysis in a patient with a history of tobacco use or weight loss and gastrointestinal bleeding).

GLOMERULAR DISEASES ASSOCIATED WITH MALIGNANCY

Patients with both solid and liquid cancers can present with a host of glomerular diseases either related to the primary disease or to its therapy. Glomerular diseases associated with chemotherapy are discussed in the section on chemotherapeutic agents. Glomerular diseases associated with HSCT are discussed in the section on HSCT. Typically, patients with glomerular diseases present with hematuria, proteinuria, nephrotic syndrome, and variable degrees of renal insufficiency. For those patients with a known malignancy who develop proteinuria and/or hematuria, a malignancy-related glomerular disease should be suspected. Table 57.6 lists the commonly seen glomerular diseases associated with malignancies and their treatment. Many of these associations are based on isolated cases and case series and are likely rare associations, yet they should be considered, particularly if there are no other obvious causes. Some glomerular disorders in patients with cancer represent paraneoplastic syndromes in which the clinical manifestation is not directly related to the tumor burden, invasion, or metastases but instead is caused by the action of a secreted tumor product, such as a cytokine, growth factor, or tumor cell antigen.¹⁰⁴ There is little information about the prevalence of paraneoplastic glomerular diseases.¹⁰⁵ For example, in one study of 120 patients with glomerulonephritis, 14.1% had a concomitant cancer but causality could not be determined.¹⁰⁶ Furthermore, it is important to realize that the treatment of certain glomerular diseases with selective immunosuppressive therapies can increase the risk for subsequent malignancies (e.g., cyclophosphamide, which can lead to acute myelogenous leukemia, and mycophenolate, which can lead to non-Hodgkin lymphoma).

The diagnosis of paraneoplastic glomerular diseases is difficult; determining causation between the glomerular lesion and the cancer can be facilitated by carefully reviewing the sequence of clinical events and the timing of the cancer diagnosis in relation to the diagnosis of the glomerular lesion. In one study, most cancer diagnoses that were thought to be related to the glomerular lesion were found either at the time of kidney biopsy or within the first year afterward.¹⁰⁶ The importance of determining the link between the cancer and glomerular disease is that if this link is present, treatment of the cancer may result in remission of the glomerular lesion.

MEMBRANOUS NEPHROPATHY

Membranous nephropathy is the most commonly reported glomerular disease in patients with cancer.¹⁰⁷⁻¹¹⁰ The evidence base for this assertion is modest and, in the largest study of 240 patients with membranous nephropathy, 10% of patients were found to have cancer that was more commonly found in the lung, prostate, and stomach.¹⁰⁹ In this case series, 21 cases of cancer were identified at the time of kidney biopsy, and the remaining 3 cases were identified within 1 year post biopsy. Overall, the standardized incidence ratio of cancer in patients with membranous nephropathy has been estimated to be 2.25, with 80% of cancers detected before or at the time of kidney biopsy.¹¹¹ From case-series data, the cancers that have been most often associated with membranous nephropathy include lung, gastric, renal cell, prostate cancers, and thymomas.^{107,108} Other cancers that have been reported with membranous nephropathy include breast, colorectal, pancreatic, esophageal, and hepatic cancers.^{107,108} In patients older than 65 years or with a significant smoking history, the incidence of malignancy with membranous nephropathy may be higher.¹⁰⁹

Malignancy-associated membranous nephropathy may be due to one of several mechanisms that lead to subepithelial deposits forming in the glomerular basement membrane.¹¹¹ Tumor cells may shed antigens that can do the following: 1. form circulating immune complexes that become trapped in the capillary wall and elicit an immune reaction in a subepithelial location; 2. form immune complexes that initially deposit in a subendothelial location but dissociate and reform in a subepithelial location; and 3. deposit in a subepithelial location (depending on the size and charge of the antigen) where they can react with a circulating antibody. A fourth mechanism may be related to an infection with an oncogenic virus or altered immune function, which can simultaneously lead to both malignancy and membranous nephropathy.

Distinguishing primary (idiopathic) versus secondary (malignancy-related) membranous nephropathy is challenging but perhaps made easier by the recognition that a high percentage of patients with primary membranous nephropathy have an elevated circulating level of autoantibodies to the podocyte transmembrane glycoprotein M-type phospholipase A2 receptor (anti-PLA2R) or increased expression of anti-PLA2R on kidney biopsy.¹¹²⁻¹¹⁴ Fig. 57.2 lists features of malignancy-associated membranous nephropathy compared with primary membranous nephropathy that can be helpful in diagnosis.

Table 57.6 Glomerular Diseases Associated With Cancer and Chemotherapeutic Agents

Glomerular Lesion	Malignancy	Drug
Membranoproliferative glomerulonephritis	Lung, renal cell, breast, esophageal, gastric, Wilms, melanoma, thymoma, Hodgkin and non-Hodgkin lymphoma, acute and chronic leukemia, monoclonal gammopathy, myeloma	Gemcitabine, sirolimus, immune checkpoint inhibitor
Minimal change disease	Lung, colon, pancreas, bladder, renal cell, ovarian, mesothelioma, thymoma, Hodgkin and non-Hodgkin lymphoma, acute and chronic leukemia	Interferon- α , β , γ ; pamidronate, doxorubicin, daunorubicin, sirolimus, immune checkpoint inhibitors, tyrosine kinase inhibitors
Focal segmental glomerulosclerosis (FSGS)	Lung, renal cell, breast, esophageal, thymoma, Hodgkin and non-Hodgkin lymphoma, acute and chronic leukemia, T cell leukemia	Sirolimus, temsirolimus, everolimus, doxorubicin, daunorubicin, bisphosphonates, tyrosine kinase inhibitors
Collapsing FSGS	Unlikely, but causes of FSGS listed above could be considered	Interferon- α , β , γ ; pamidronate, gefitinib, sirolimus, doxorubicin, daunorubicin, clofarabine, bisphosphonates
Membranous nephropathy	Lung, colon, gastric, pancreas, prostate, breast, head and neck, Wilms, teratoma, ovarian, cervical, endometrial, melanoma, skin, pheochromocytoma, Hodgkin and non-Hodgkin lymphoma, acute and chronic leukemia	Immune checkpoint inhibitors
Lupus-like glomerulonephritis	None	Immune checkpoint inhibitors
IgA nephropathy	Lung, pancreas, renal cell, head and neck, Hodgkin and non-Hodgkin lymphoma	Sirolimus, immune checkpoint inhibitors

There should be a high index of suspicion for an underlying malignancy in patients with a new diagnosis of membranous nephropathy, especially if there are other risk factors for cancer. However, the most efficient and cost-effective method for screening these patients has not been identified.¹¹⁵ It is recommended that age- and sex-appropriate cancer screening be performed on patients with a new diagnosis of membranous nephropathy. In those patients in whom no malignancy is found on initial screening, follow-up vigilance for cancer should be maintained because there are case reports of cancers being detected up to 5 years after the initial diagnosis of membranous nephropathy.¹¹⁶ Whether these cancers are directly related to the occurrence of glomerular disease is unknown.

Malignancy-associated membranous nephropathy may resolve with surgical removal of the tumor or with effective chemotherapy.¹¹¹ In the absence of curative therapy, standard therapy for membranous nephropathy can be attempted but may have a lower likelihood of inducing remission when compared with patients with primary membranous nephropathy.

OTHER GLOMERULAR DISEASES ASSOCIATED WITH SOLID TUMORS

There are reported associations among solid-organ malignancies and glomerular diseases for minimal change disease (MCD; e.g., lung, colorectal, thymoma, and renal cell cancers); focal segmental glomerulosclerosis (e.g., renal cell carcinoma, thymoma); membranoproliferative glomerulonephritis (e.g., lung, renal cell, and gastric carcinomas); IgA nephropathy (e.g., renal cell carcinoma, upper respiratory tract tumors, nasopharyngeal cancers); Henoch-Schönlein purpura (e.g., lung, upper respiratory tract tumors, digestive tract tumors); crescentic glomerulonephritis (e.g., renal cell, gastric, and lung

carcinomas); and AA-type amyloidosis (e.g., renal cell carcinoma).¹¹¹ These potential associations highlight the fact that in patients presenting with glomerular diseases, clinicians should remain vigilant for the possibilities of an underlying associated malignancy. Age- and sex-appropriate cancer screening should be undertaken in these patients. The association is made stronger if effective therapy for the cancer leads to concomitant remission of the glomerular disease.

HEMATOLOGIC MALIGNANCY–ASSOCIATED GLOMERULAR DISEASES

Lymphomas and leukemias have been associated with an increased risk of various forms of glomerular diseases.¹¹⁷ The associations are strongest between Hodgkin disease and MCD and chronic lymphocytic leukemia (CLL) and membranoproliferative glomerulonephritis (MPGN).

HODGKIN LYMPHOMA

Minimal change disease is the most frequent glomerular disease seen in association with Hodgkin lymphoma but is uncommon, with an incidence of 0.4% in two large studies.¹¹⁸ Typically, the nephrotic syndrome occurs early in the disease, with up to 40% of cases presenting at the time of or just before diagnosis.¹¹⁹ Minimal change disease in this setting is typically both steroid and cyclosporine-resistant, and effective therapy of the lymphoma usually induces remission of the nephrotic syndrome.¹¹¹ Importantly, recurrence of the nephrotic syndrome in patients with previously treated Hodgkin disease can signal the recurrence of lymphoma.^{118,119} The pathogenesis of MCD in Hodgkin disease may be related to the induction of *c-maf*-inducing protein (c-mif), which can lead to cytoskeletal disorganization and foot process effacement of the podocyte.¹²⁰

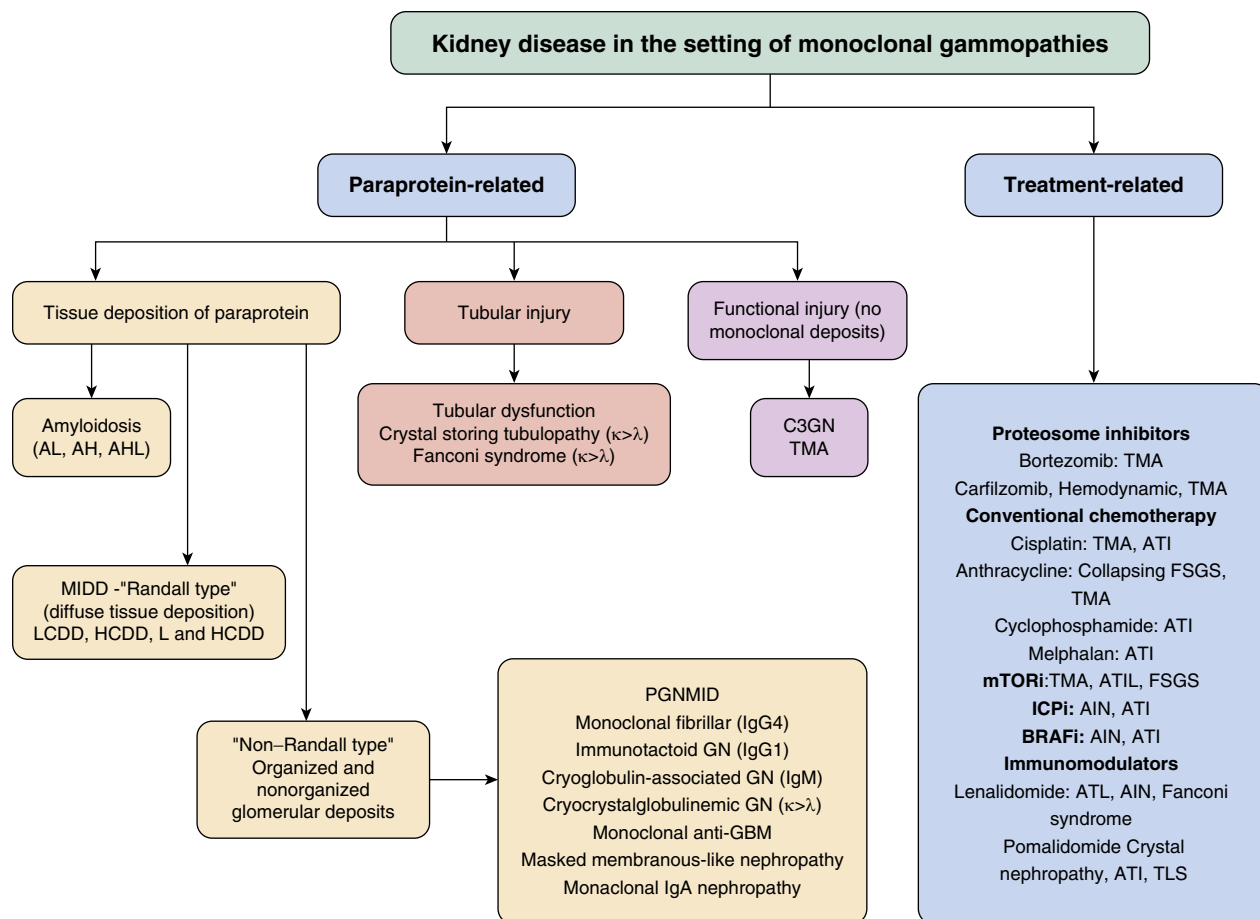


Fig. 57.3 An approach to the differential diagnosis of kidney disease associated with paraproteinemias. AH, Amyloid heavy chain; AHL, amyloid light and heavy chain; AIN, acute interstitial nephritis; AL, amyloid light chain; ATI, acute tubular injury; BRAFi, v-raf murine sarcoma viral oncogene homolog B1 inhibitor; C3, complement 3; FSGS, focal segmental glomerulosclerosis; HCDD, heavy chain deposition disease; ICPI, immune checkpoint inhibitor; Ig, immunoglobulin; IgG 1&4, immunoglobulin G types 1 and 4; IgM, immunoglobulin M; LCDD, light chain deposition disease; LHCDD, light-heavy chain deposition disease; MIDD, monoclonal immunoglobulin deposition disease; mTORi, mammalian target of rapamycin inhibitor; PGMIDD, proliferative N with monoclonal IgG deposits; TLS, tumor lysis syndrome; TMA, thrombotic microangiopathy; TNF, tumor necrosis factor.

CHRONIC LYMPHOCYTIC LEUKEMIA AND B-CELL LYMPHOMAS

The nephrotic syndrome is seen in approximately 1% to 2% of patients with CLL.¹²¹ The most common glomerular lesion seen in patients with CLL is MPGN, followed by membranous nephropathy.¹²² The pathogenesis of MPGN in these cases may involve cryoglobulin production by the abnormal B cell clone. In addition, B cell lymphomas may produce a pathogenic immunoglobulin that can lead to a host of deposition diseases including monoclonal immunoglobulin deposition disease and immunotactoid glomerulopathy, in which the deposits are organized in a microtubular array.¹¹¹ A clear link between the lymphoma and glomerular disease is solidified when effective chemotherapy leads to a remission of both the lymphoma or CLL and nephrotic syndrome.

PLASMA CELL DYSCRASIA

Plasma cell dyscrasias are associated with a host of glomerular diseases and nephrotic syndrome, the most common of which is AL amyloidosis; AL amyloidosis has been found in

5% to 11% of patients with myeloma at autopsy.¹²³ Other glomerular diseases include light-chain deposition disease (up to 50% of these patients are found to have myeloma) and heavy-chain deposition disease (up to 25% of these patients have myeloma).¹²⁴ In AL amyloidosis, the predominant light-chain isotope is lambda (λ), whereas in light-chain deposition disease, it is kappa (κ).¹¹¹ The diagnosis of glomerular diseases in a patient with suspected myeloma is facilitated by finding urine albumin excretion rates >2 g/day, and a kidney biopsy is indicated in these patients (see Fig. 57.3). Several rarer forms of glomerular disease can be found in the presence of a plasma cell dyscrasia and paraproteinemia. In all cases, effective chemotherapy or HSCT can lead to remission of the glomerular disease.¹¹¹

FLUID AND ELECTROLYTE ABNORMALITIES ASSOCIATED WITH MALIGNANCIES

It is not uncommon to encounter electrolyte imbalances in patients with malignancies. In general, these abnormalities may be directly linked to the malignancy as a paraneoplastic

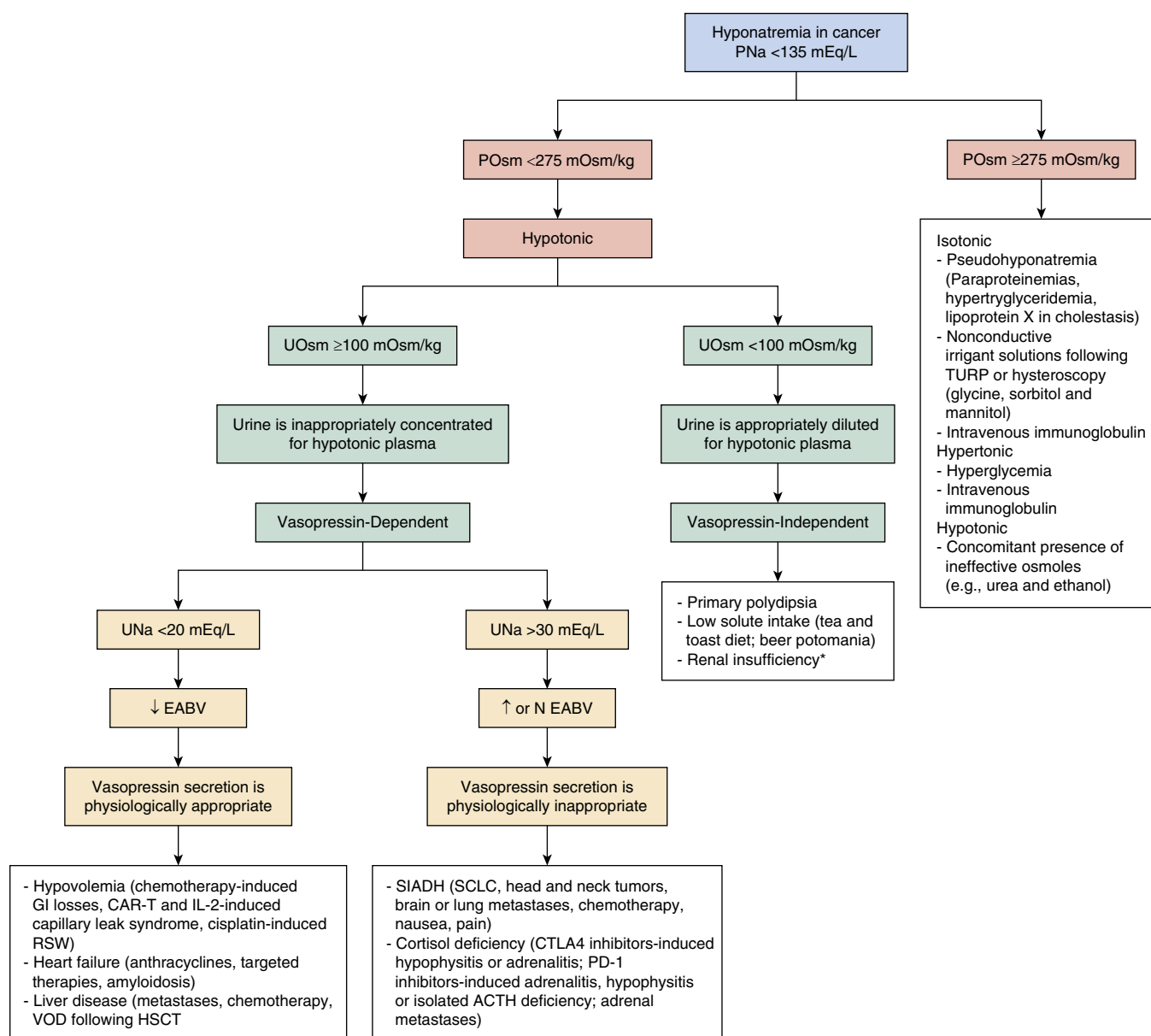


Fig. 57.4 Diagnostic approach to hyponatremia in the patient with cancer.

process or as a result of chemotherapeutic regimens. Conversely, they may be unrelated to the malignancy and may be similar to electrolyte disorders seen in patients without cancer (e.g., diuretic-induced hypokalemia).

HYPONATREMIA

Disorders of water homeostasis leading to hyponatremia are among the most common electrolyte disorders encountered in patients with cancer, with a wide range of prevalence reported in the literature (4%–47%).^{125,126} In patients presenting to the hospital with severe hyponatremia, approximately 14% had an underlying malignancy.¹²⁷ Perhaps most importantly, hyponatremia in patients with cancer is associated with poor outcomes; the hospital length of stay is doubled, and mortality (within 90 days) is increased 3- to 5-fold.^{125,128-130} Whether hyponatremia has a direct link to these poor outcomes is debatable; it is more likely

that hyponatremia is a marker of severity of illness, disease progression, and overall debility.

The symptoms attributable to hyponatremia are generally nonspecific (e.g., lethargy, nausea) or neurologic (e.g., headache, confusion, seizures) in origin, and the diagnosis relies on correlating laboratory studies with these symptoms. Elucidating the cause of hyponatremia requires careful history taking, physical examination, and judicious laboratory evaluation. An approach to the patient with hyponatremia and cancer is shown in Fig. 57.4. Clearly understanding the cause of a patient's hyponatremia is critical in tailoring appropriate therapy.

The syndrome of inappropriate antidiuretic hormone secretion (SIADH) is a common cause of hyponatremia in the patient with cancer and has numerous causes, ranging from paraneoplastic to medication induced.¹³¹ Paraneoplastic causes include small cell lung cancer (most common and seen in 10%–15% of patients) and head and neck tumors. Case

reports have described SIADH with numerous other solid-organ malignancies, from brain to sarcoma.¹³¹⁻¹³³ Hyponatremia in these patients typically develops slowly and insidiously, and patients rarely have symptoms unless their serum sodium level is <125 mmol/L.¹³⁴ Serial measurements of ADH, at least for small cell lung cancer, reflect the state of the cancer, and levels fall with remission and rise with recurrence.¹³⁵ Many chemotherapeutic agents can also lead to SIADH. These include cyclophosphamide, cisplatin, vinblastine, and vincristine.¹³⁶ Many of the associations between chemotherapeutic agents and hyponatremia are derived from case reports, and the causal linkage between the drug and SIADH is not conclusive. Of note, in many cases, chemotherapeutic agents (including cyclophosphamide) are administered with substantial volumes of IV hydration, which, depending on the levels of antidiuretic hormone (ADH) and urine osmolality, may lower serum sodium concentration further.

Therapies for hyponatremia in the patient with cancer are similar to those for other causes of hyponatremia and should be guided by three factors: 1. the presence of hyponatremia-related symptoms; 2. the duration of hyponatremia; and 3. the volume status of the patient. When possible, removal or correction of the underlying cause should always be pursued. In the vast majority of cases, hyponatremia associated with cancer is chronic in nature; thus, correction rates should be slow (no more than 6–8 mmol/L/day). In the presence of severe neurologic symptoms (e.g., seizures, obtundation, coma), 3% hypertonic saline can be administered as small boluses or continuous infusion to raise the serum sodium level by 4 to 6 mmol/L, which usually improves these symptoms dramatically.¹³⁷ Fluid restriction (typically defined as oral intake 500 mL less than urine output) is a reasonable treatment for patients with mild and transient hyponatremia due to SIADH. However, given the need for hydration protocols with several chemotherapies, as well as the need to maintain adequate nutrition, fluid restriction can be extremely challenging. In these cases, as well as those cases of SIADH in which the urine osmolality is high (>600 mOsm/kg), the use of vasopressin type 2 receptor antagonists (e.g., conivaptan [IV only] or tolvaptan [oral only]) can be considered.¹³⁸ These drugs block the action of antidiuretic hormone (ADH) in the distal tubule and allow for the excretion of more dilute urine with increased free water excretion.¹³⁸ Tolvaptan is contraindicated in patients with hypovolemic hyponatremia, and volume depletion and/or in patients who cannot perceive or respond to thirst. In addition, given concerns for hepatotoxicity with tolvaptan, the current recommendation is to limit its use to 1 month or less. Finally, oral urea administration can also induce a free water diuresis and can be considered as a therapy for SIADH refractory to fluid restriction.¹³⁷ Relatively poor palatability of commercially available urea formulations can limit their utility in patients with cancer, many of whom have impaired taste and a general propensity toward nausea and vomiting.

HYPOKALEMIA

Hypokalemia is commonly encountered in patients with cancer.¹³⁹ The causes of hypokalemia are often overlapping and multifactorial in a given patient (Table 57.7). Unique causes in the patient with cancer include the following: 1. tubular damage associated with chemotherapeutic agents

Table 57.7 Causes of Hypokalemia in the Patient With Cancer

Type	Cause
Tumor-related	Lysozymuria with acute leukemia
	Mineralocorticoid excess syndromes
	• Primary hyperaldosteronism (adrenal carcinoma) • Renin-producing tumors (extremely rare) • Ectopic adrenocorticotropin syndrome
Chemotherapy-related	Intracellular shifts (pseudohypokalemia) in the setting of high white blood cell counts
	Chemotherapeutic agents (cisplatin, ifosfamide)
	Antimicrobials and antifungal agents (aminoglycosides, amphotericin B)
Other related	Excessive gastrointestinal losses (vomiting, diarrhea)
	• Hypercalcemia
	• Hypomagnesemia
	• Postobstructive diuresis
	• Use of growth factors

and antimicrobials (e.g., cisplatin, ifosfamide, amphotericin B, aminoglycosides); 2. tubular damage associated with a light-chain-induced proximal tubulopathy, such as in multiple myeloma; 3. ectopic adrenocorticotropin hormone (ACTH) syndrome; 4. lysozymuria associated with hematologic malignancies; and 5. chemotherapy-induced gastrointestinal losses (vomiting and diarrhea).

Transcellular shifts that occur post phlebotomy, while the blood is in a sample tube, can lead to spurious hypokalemia, especially when the patient has marked leukocytosis (>100,000/ μ L) or when the blood is kept at room temperature for prolonged periods.¹⁴⁰ When spurious hypokalemia is suspected, rapid separation of the plasma and storage of blood at 4°C will usually confirm a normal serum potassium level.

The syndrome of paraneoplastic ACTH production can be encountered with several tumors, with the most common causes being bronchial carcinoid tumors, small cell lung carcinoma, lung adenocarcinoma, thymic cancers, and pancreatic tumors.¹⁴¹ Patients typically present with signs of severe hypercortisolism—diabetes, bone loss, hyperlipidemia, HTN, and, depending on the length of symptoms, Cushingoid habitus—along with severe hypokalemia.¹⁴¹ The hypokalemia results from excessive cortisol production, which overwhelms the distal tubular mechanism that normally limits mineralocorticoid receptor activation by cortisol (11 β -hydroxysteroid dehydrogenase). Excessive production of cortisol essentially creates a syndrome similar to that of excessive aldosterone secretion, with resulting HTN and hypokalemia. Paraneoplastic ACTH production can be diagnosed through measurement of excessive ACTH, hypercortisolism, and localization of the primary tumor with radiographic studies. Optimal

management requires control of the primary tumor, which may or may not be possible, so drugs that antagonize glucocorticoid synthesis (metyrapone and ketoconazole) are often required.¹⁴¹ Rarely, adrenalectomy may be required.

A rare form of hypokalemia associated with acute myelogenous leukemia (subtypes M4 and M5) has been reported. It is usually associated with other electrolyte and acid-base disorders including hyponatremia, hypocalcemia, hypophosphatemia, hypomagnesemia, and a non-anion gap acidosis.^{142,143} The magnitude and breadth of these electrolyte issues suggest a global tubular defect and inappropriate renal potassium losses. This defect has been postulated to be secondary to increased tumor-derived lysozyme secretion and lysozymuria-induced tubular damage.¹⁴³

HYPERKALEMIA

Hyperkalemia in patients with cancer is often seen in association with AKI, rhabdomyolysis, or tumor lysis syndrome. Less common causes include adrenal insufficiency (secondary to adrenal metastases) and drugs (e.g., heparin, ketoconazole, trimethoprim, calcineurin inhibitors, nonsteroidal anti-inflammatory drugs). An important consideration in patients with marked leukocytosis or thrombocytosis is pseudohyperkalemia, which results from a shift of potassium out of platelets or white blood cells post phlebotomy and after clotting of the sample.¹⁴⁴ If pseudohyperkalemia is suspected, plasma rather than serum samples should be used for laboratory measurement.

HYPERCALCEMIA

Hypercalcemia associated with malignancy occurs in up to 30% of patients with cancer. Hypercalcemia portends a poor prognosis and limited lifespan.^{145,146} Symptoms attributable to hypercalcemia are generally nonspecific (e.g., lethargy, nausea, vomiting, confusion) and are related to both the absolute level of serum calcium and rapidity at which the disorder develops. Patients presenting with serum calcium concentrations above 13 mg/dL are generally symptomatic.

There are three mechanisms that describe the pathophysiology in most cases of malignancy-associated hypercalcemia (Fig. 57.5). The most common cause is paraneoplastic production of parathyroid hormone related peptide (PTHrP).¹⁴⁷ PTHrP is similar to parathyroid hormone (PTH) and acts to increase bone resorption (increases bone calcium release) via activation of osteoclasts and increase reabsorption of calcium in the renal tubules.¹⁴⁷ However, PTHrP is less likely than PTH to stimulate 1,25-dihydroxyvitamin D production. Therefore PTHrP does not increase intestinal calcium absorption. Hypercalcemia is typically accompanied by hypophosphatemia, reduced tubular reabsorption of phosphorus, enhanced tubular reabsorption of calcium, and increased excretion of nephrogenous cyclic adenosine monophosphate (cAMP), reflecting the PTH-like actions of PTHrP. The second most common mechanism of malignancy-associated hypercalcemia is local osteolytic effects of the cancer and is seen in patients with myeloma, breast cancer, lymphomas, and other cancers. For example, myeloma has a predilection to grow in bone environments, facilitated by the secretion

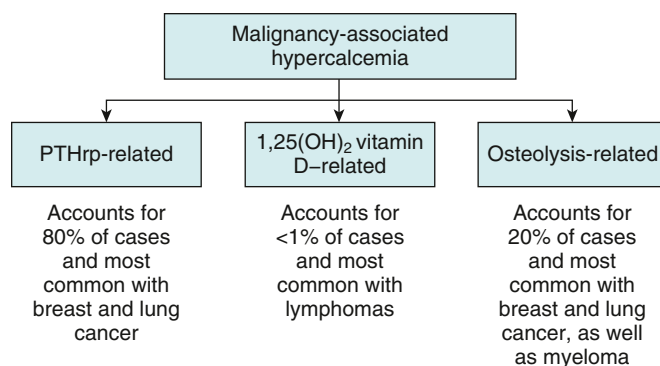


Fig. 57.5 Causes of malignancy-associated hypercalcemia. PTHrP, Parathyroid hormone related peptide.

of bone reabsorbing factors such as interleukin-1 (IL-1), IL-6, and tumor necrosis factor, leading to extensive bone destruction, lytic bone lesions, and the release of calcium.¹⁴⁸ Typically, these patients will have advanced cancers, with skeletal surveys demonstrating lytic bone lesions. The third mechanism of malignancy-associated hypercalcemia is due to activation of vitamin D by the tumor itself, which can be seen in lymphomas and, rarely, with myeloma. These patients will have elevated serum 1,25 dihydroxyvitamin D and suppressed PTH concentrations.¹⁴⁹ Elevated activated vitamin D concentrations facilitate increased gastrointestinal calcium absorption and increased bone calcium release, both leading to hypercalcemia.

The diagnostic approach to the patient with hypercalcemia is presented in Fig. 57.6. Once hypercalcemia is confirmed with measurement of an elevated serum ionized calcium, the diagnostic algorithm centers on measurement of PTH. Serum PTH concentrations are suppressed in patients with malignancy-associated hypercalcemia, with the rare exceptions of tumors such as parathyroid adenocarcinomas, which produce excessive PTH. In patients with suppressed PTH, measurement of PTHrP and 25-hydroxy vitamin D is the next diagnostic step and allows for accurate diagnosis in most cases.¹⁵⁰

Patients with hypercalcemia are usually severely volume depleted, as hypercalcemia frequently leads to nephrogenic diabetes insipidus, as well as nausea and vomiting. In addition, hypercalcemia, via activation of the calcium-sensing receptor in the thick ascending limb of the loop of Henle, leads to natriuresis and worsening of the volume depletion.¹⁵¹ Thus the cornerstone of the treatment of hypercalcemia is aggressive restoration of the extracellular volume, as well as induction of natriuresis with IV fluids (typically, 0.9% saline), facilitating excretion of excess calcium. Loop diuretics should only be added in the presence of volume overload and are not a routine part of management.¹⁵² Data support the use of bisphosphonates as highly effective agents in the management of patients with malignancy-associated hypercalcemia, largely due to their effect to inhibit osteoclast activity.¹⁴⁵ Both pamidronate and zoledronic acid are U.S. Food and Drug Administration (FDA) approved for this indication; dosing guidelines are shown in Table 57.8. These agents are generally well tolerated, with the most common side effect being infusion-related fever, but serious nephrotoxicity of these agents has been described including glomerular lesions and tubular toxicity.¹⁵³ Thus the serum creatinine should be monitored before each dose of bisphosphonate and the drug withheld in patients with

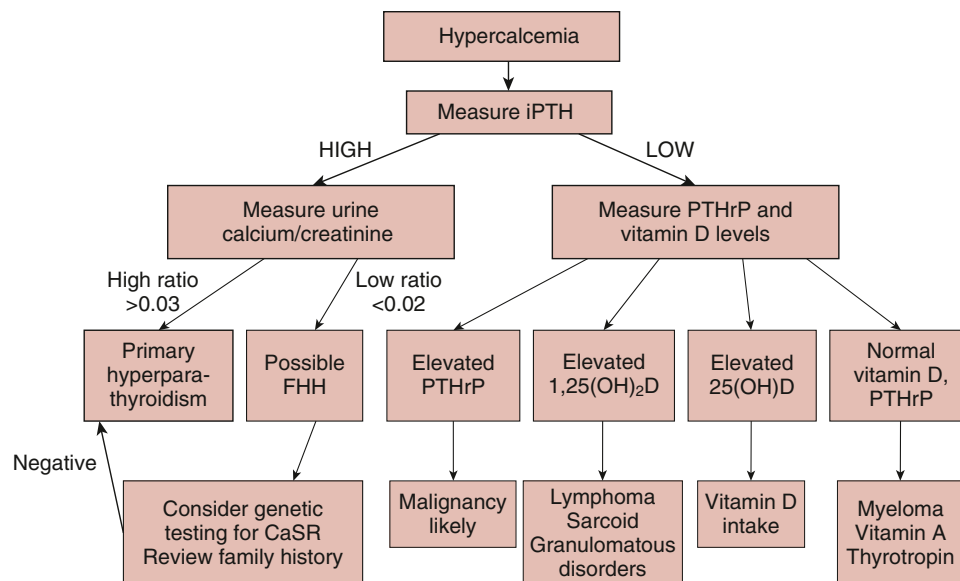


Fig. 57.6 Diagnostic algorithm for the patient presenting with hypercalcemia. Measurement of PTH is the first step in the diagnostic approach. Patients with malignancy-associated hypercalcemia typically have low levels of PTH due to feedback inhibition of the high blood calcium levels on the parathyroid gland. In patients with low levels of PTH, measurement of PTHrP and 1,25-dihydroxyvitamin D (1,25(OH)₂D) levels will allow classification of patients into several diagnostic groupings. In patients with elevated PTH levels, a 24-hour urine collection for calcium and creatinine levels should be obtained. *CaSR*, Calcium-sensing receptor; *FHH*, familial hypocalciuric hypercalcemia; *iPTH*, intact parathyroid hormone; *PTH*, parathyroid hormone; *PTHrP*, parathyroid-related peptide; *25(OH)D*, 25-hydroxy vitamin D. (From Reagan P, Pani A, Rosner MH. Approach to diagnosis and treatment of hypercalcemia in a patient with malignancy. *Am J Kidney Dis*. 2013;63(1):141–147.)

declining kidney function. Urine albumin excretion should be monitored at 3- to 6-month intervals, and the drug should be held if albuminuria develops. Patients with PTHrP-mediated hypercalcemia may remain mildly hypercalcemic, even after bisphosphonate administration, due to the effects of PTHrP on renal calcium excretion. An alternative antiresorptive agent is denosumab, a monoclonal antibody directed against the receptor activator of nuclear factor kappa B ligand (RANKL).¹⁵⁴ Denosumab does not require dose adjustment for impaired kidney function and has demonstrated benefit in reducing skeletal-related events in patients with metastatic cancer.¹⁵⁵ For patients with hypercalcemia secondary to tumors producing 1,25-dihydroxyvitamin D, corticosteroid therapy may be of benefit.¹⁵⁶ Finally, in patients with kidney failure and hypercalcemia, hemodialysis with a low calcium dialysate can be beneficial.¹⁴⁵

HYPOPHOSPHATEMIA

Cancer can lead to disorders of phosphate homeostasis via a number of mechanisms. Hypophosphatemia can result in the setting of malnutrition and cachexia and with vitamin D deficiency; these patients typically have low normal serum calcium, low vitamin D, and elevated PTH concentrations. In addition, patients with hypercalcemia due to elevated PTH or PTHrP levels may have low-normal serum phosphate concentrations due to the effects of these hormones on renal phosphorus handling. In some cases, proximal tubular injury (either through chemotherapy such as cisplatin or ifosfamide or due to light-chain-induced tubular injury from myeloma) can lead to phosphaturia and hypophosphatemia.

A rare and unique form of hypophosphatemia is also encountered in patients with cancer, the syndrome of tumor-induced osteomalacia (TIO), or oncogenic osteomalacia.¹⁵⁷

Table 57.8 Dosing Guidelines for Use of Bisphosphonates in Treatment of Malignancy-Associated Hypercalcemia

Bisphosphonate	Dosing Guidelines	Interval Between Doses
Pamidronate	CrCl >60 mL/min: 90 mg IV administered over 2-3 hours	Every 3-4 weeks
	CrCl, 30-60 mL/min: 60-90 mg IV administered over 2-4 hours	
	CrCl < 30 mL/min: 60-90 mg IV administered over 4-6 hours	
Zoledronate	CrCl >60 mL/min: 4 mg IV over 15 minutes	Every 3-4 weeks
	CrCl, 50-60 mL/min: 3.5 mg IV over 15 minutes	
	CrCl, 40-49 mL/min: 3.3 mg IV over 15 minutes	
	CrCl 30-49 mL/min: 3 mg IV over 15 minutes	
	CrCl <30 mL/min: not recommended	

In this syndrome, tumor production of phosphaturic factors such as fibroblast growth factor-23 (FGF-23) results in renal phosphate wasting, hypophosphatemia, and osteomalacia. Tumor types producing these factors include chondrosarcoma, osteoblastoma, and hemangiopericytomas. These tumors may be small and indolent and may require extensive and

specialized imaging for localization. Patients with TIO typically have normal serum calcium concentrations, and the diagnosis is facilitated by measuring the percentage tubular reabsorption of phosphate or the tubular maximum for phosphate, corrected for the GFR.¹⁵⁸ Once renal phosphate wasting is confirmed, measurement of elevated serum concentrations of FGF-23 can be diagnostic of TIO. Importantly, 1,25-dihydroxyvitamin D levels are low because FGF-23 inhibits the activity of the 1 α -hydroxylase. Therapy rests on surgical removal of the tumor, which is usually curative.

HYPOMAGNESEMIA

Hypomagnesemia is a relatively common electrolyte disorder in patients with cancer; the unique causes in this patient population relate to chemotherapeutic agents such as cisplatin and cetuximab. Otherwise, patients usually develop hypomagnesemia due to gastrointestinal losses, especially with prolonged diarrhea, or renal losses due to diuretic therapy. Other drugs associated with hypomagnesemia in the patient with cancer may include proton pump inhibitors, aminoglycoside antibiotics, calcineurin inhibitors (in patients with stem cell transplants), and amphotericin B.¹⁵⁹

Cisplatin causes magnesium wasting in more than 50% of treated patients, and the incidence increases with the cumulative dose.¹⁶⁰ Renal magnesium wasting continues after cessation of the drug for several months but can persist for years. The occurrence of magnesium wasting does not correlate with cisplatin-induced AKI. Cetuximab, a monoclonal antibody against the epithelial growth factor receptor (EGFR), has been associated with hypomagnesemia due to renal losses in up to 30% of patients receiving this drug.¹⁶¹ The mechanism of magnesium wasting during treatment with cetuximab is associated with the blockage of the EGFR-dependent transient receptor potential channel 6 (TRPM6) in the nephron, which results in insufficient activation of the TRPM6 epithelial ion channel and magnesium wasting in the distal convoluted tubule. Cessation of therapy with cetuximab leads to reversal of the magnesium wasting and return of the serum magnesium concentration to normal.

KIDNEY DISEASE ASSOCIATED WITH HEMATOPOIETIC STEM CELL TRANSPLANTATION

Hematopoietic stem cell transplantation is an important therapy for many patients with cancer. However, it is complicated by acute and chronic kidney disease, often associated with increased morbidity and mortality. Causes of HSCT-associated kidney injury are often multifactorial, including the conditioning chemotherapy regimen, radiation exposure, nephrotoxic medications, sepsis, sinusoidal obstruction syndrome (SOS), HSCT-associated TMA, and GVHD.^{162,163} Short- and long-term morbidity and mortality are both increased in the setting of HSCT.¹⁶⁴ Other complications include fluid disturbances, electrolyte abnormalities, glomerular disease, and HTN.

HSCT involves administering near-lethal doses of chemotherapy and radiotherapy, followed by bone marrow rescue by engraftment of stem or progenitor cells, which are harvested from bone marrow, peripheral blood, or umbilical

cord blood. Stem or progenitor cells can be autologous (patient) or allogeneic (donor). In myeloablative HSCT, high-dose chemotherapy and radiation are administered to eradicate the underlying malignancy and bone marrow, followed by marrow reconstitution with the infusion of stem or progenitor cells. However, because myeloablative HSCT is extremely toxic, older patients and those with numerous comorbidities are often excluded from transplantation candidacy. To allow this group of higher-risk patients to benefit from HSCT, less toxic nonmyeloablative regimens have been created, using a so-called graft-versus-tumor effect for therapeutic efficacy.

HEMATOPOIETIC STEM CELL TRANSPLANTATION–ASSOCIATED ACUTE KIDNEY INJURY

The incidence of AKI with HSCT is quite varied (ranging 15%–73%) due to the AKI definition used, the type of HSCT performed (allogeneic vs. autologous), and the chemotherapeutic conditioning regimen used (high dose vs. reduced intensity).¹⁶² The incidence and clinical course of AKI following allogeneic HSCT are well documented. In one study, AKI developed in 53% of patients, of whom half required RRT.¹⁶⁵ Subsequent studies have noted that the incidence of moderate to severe AKI (defined as a doubling of the baseline serum creatinine concentration) ranges from 36% to 78% with allogeneic HSCT.^{6,7,166} Dialysis was required in 21% to 33% of these patients, who suffered a mortality of 78% to 90%. The incidence of AKI with autologous HSCT is less common.⁶ In patients with breast cancer, autologous HSCT was associated with the development of moderate to severe AKI in 21% of patients and a mortality rate of 18%. The lower AKI incidence in this group may be due to calcineurin inhibitor avoidance.¹⁶⁷

Nonmyeloablative HSCT is also associated with a lower incidence of AKI, which is attributed to the milder conditioning regimen used.¹⁶⁸ One study in nonmyeloablative HSCT patients demonstrated a cumulative AKI incidence of 40.4% at 4 months, and dialysis was required in 4.4% of patients. In contrast to myeloablative HSCT, AKI in these patients was usually associated with calcineurin inhibitor exposure. In addition, the development of AKI during nonmyeloablative HSCT occurs over the first 3 months, which is in contrast to myeloablative HSCT, in which AKI typically develops in the first 3 weeks.

As with AKI incidence, the risk for requiring KRT after HSCT varies widely. Overall, approximately 5% of HSCT recipients with AKI require dialysis, which is associated with increased mortality risk.^{168–174} However, the need for dialysis ranges from 0% to 30%,^{9–15} and the risk is higher with myeloablative therapy.¹⁷⁵ Outcomes in patients requiring acute KRT are difficult to quantify accurately. Most studies have reported that acute dialysis in this group is associated with an extremely high mortality rate, ranging from 80% to 100%.^{163,168–171,173,176,177}

Common risk factors for and causes of AKI after HSCT include volume depletion, sepsis, nephrotoxic medication exposure, SOS, and GVHD (Table 57.9).^{163,174,178,179} Because of a propensity for increased gastrointestinal fluid losses and poor oral intake, HSCT patients are highly susceptible to volume depletion and prerenal AKI. GVHD is unique to HSCT and causes tissue and endothelial damage via

Table 57.9 Risk Factors and Causes of Acute and Chronic Kidney Lesions With Hematopoietic Stem Cell Transplantation

Acute Kidney Injury	Chronic Kidney Disease
Prerenal—nausea, vomiting, and diarrhea associated with acute gastrointestinal GVHD, drug-induced nausea and vomiting	Thrombotic microangiopathy—calcineurin inhibitors, chronic GVHD
Prerenal/acute tubular injury—sepsis, sinusoidal obstruction syndrome, acute GVHD, marrow infusion syndrome	Arteriolonephrosclerosis—hypertension, radiation, TBI
Thrombotic microangiopathy—acute GVHD, calcineurin inhibitors, TBI	Viral nephropathy—BK virus, cytomegalovirus
Acute tubular injury, crystalline nephropathy—amphotericin, vancomycin, aminoglycosides, acyclovir, other nephrotoxins	Membranous nephropathy, minimal change disease—chronic GVHD, drug-induced glomerular injury

GVHD, Graft-versus-host disease; TBI, total body irradiation.

T cell and cytokine-mediated injury.¹⁶³ The gastrointestinal mucosa is a common site of injury by GVHD, contributing to inadequate fluid intake and increased fluid losses. SOS is also an independent risk factor for AKI and results in a picture similar to hepatorenal syndrome, with sodium avidity and edema formation. Medications commonly prescribed that cause AKI include vancomycin, aminoglycosides, acyclovir, and amphotericin. These agents cause AKI by direct tubular toxicity, acute interstitial nephritis, and crystal-related tubular injury (acyclovir). Calcineurin inhibitors can lead to renal arteriolar vasoconstriction and have been associated with the development of TMA.

MARROW INFUSION SYNDROME

AKI associated with symptoms of nausea, vomiting, and abdominal pain may develop within 24 to 48 hours of HSCT from hemoglobin pigment nephropathy caused by the infusion of lysed red blood cells. The process of stem cell preservation is associated with exposure to various cryoprecipitants, such as dimethyl sulfoxide, which can cause red cell lysis, with hemoglobinuria in the recipient resulting in the precipitation of heme proteins in the distal tubule.¹⁸⁰ Renal vasoconstriction, direct cytotoxicity of hemoglobin, and intratubular hemoglobin cast formation are the mechanisms whereby AKI develops with hemoglobinuria. Management of this now rare complication includes alkalization of the urine to increase heme solubility and mannitol-induced diuresis to prevent intratubular heme trapping.^{181,182}

HEPATIC SINUSOIDAL OBSTRUCTION SYNDROME

AKI following HSCT can also be due to acute liver injury. Hepatic SOS, formerly known as venoocclusive disease, is

a commonly encountered complication after HSCT and has been implicated as an independent risk factor for the development of AKI.¹⁸⁰ It typically occurs within 30 days of HSCT, and the incidence varies widely, ranging from 0% to 62.3% (mean, 13.7%).¹⁸⁰ Although the pathophysiology of AKI in the SOS setting is poorly defined, it appears to be a variant of hepatorenal syndrome (hemodynamic AKI). Hepatic sinusoidal obstruction occurs due to iatrogenic damage to sinusoidal endothelial cells and hepatocytes, which results in fibrous narrowing of small hepatic venules and sinusoids. Injury appears to be triggered by the pretransplantation cytoreductive regimen and is more common after allogeneic than autologous HSCT. Furthermore, cytokine release and glutathione depletion in this setting cause hepatocellular necrosis and fibrosis.^{183,184} The development of SOS is usually associated with pretreatment with busulfan, cyclophosphamide, and/or total body irradiation.¹⁸⁵

Criteria have been established for the diagnosis of hepatic SOS. The Seattle criteria require the presence of at least two of three manifestations—jaundice, painful hepatomegaly, and fluid retention or weight gain within 20 days of HSCT.^{184,185} The Baltimore criteria require a bilirubin level of >2 mg/dL, plus two or more of the following: painful hepatomegaly, weight gain >5%, or ascites, within 21 days of HSCT.¹⁸⁶ Hepatic SOS severity is graded as mild (the illness fulfills SOS diagnostic criteria but is self-limiting), moderate (SOS subsides with diuretic or analgesic treatment), and severe (SOS persists for more than 100 days, or death).¹⁸⁴

Hepatic SOS is characterized by painful hepatomegaly, jaundice, fluid retention, and ascites. Patients have hyperdynamic vital signs along with hyponatremia, oliguria, and low fractional excretion of sodium. Urinalysis reveals minimal proteinuria and muddy brown granular casts as a result of the toxic tubular effects of bile acids and bilirubin in the urine. Hypervolemia is usually resistant to diuretics, and spontaneous recovery is rare.

Risk factors that predispose to the development of AKI include a baseline serum creatinine >0.7 mg/dL, weight gain, hyperbilirubinemia, and exposure to amphotericin B, vancomycin, or acyclovir. AKI in this setting is associated with mortality. In patients who require KRT, the mortality rate approaches 80%.

The management of hepatic SOS has included a number of agents. Small trials using infusions of prostaglandin E, pentoxifylline, and low-dose heparin have shown promise in the prevention and treatment of SOS.¹⁸⁷⁻¹⁸⁹ Smaller trials with defibrotide, an antithrombotic and fibrinolytic agent, have also shown benefit in patients with SOS, especially when used within the first 2 days of HSCT.^{190,191} Beneficial effects are thought to be due to the prevention of platelet aggregation and antiinflammatory properties. However, bleeding risk is a concern and must be closely monitored.

ACUTE THROMBOTIC MICROANGIOPATHY

Acute TMA is characterized by endothelial damage that leads to thickened glomerular and arteriolar vessels, fibrin thrombi in capillary loops and arterioles (Fig. 57.7), the presence of fragmented red blood cells, thrombosis, and endothelial cell swelling.^{192,193} In the setting of HSCT, patients may develop subclinical kidney disease, AKI, or CKD.^{194,195} Given the challenges of obtaining kidney tissue

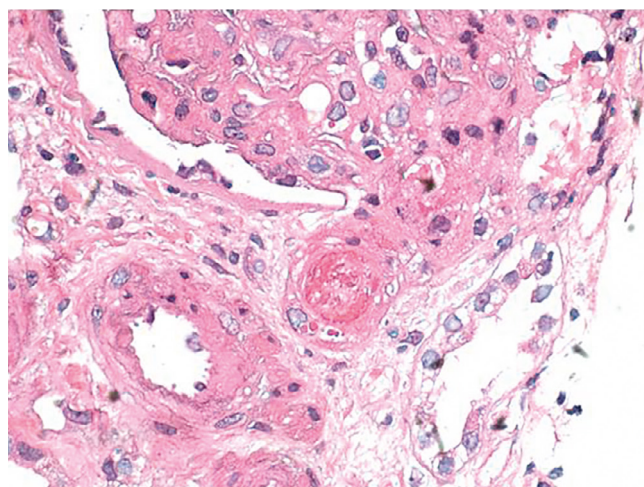


Fig. 57.7 Kidney biopsy from a patient with thrombotic microangiopathy in the setting of hematopoietic stem cell transplantation exhibits a fresh fibrin thrombus in a preglomerular arteriole. (Hematoxylin-eosin, $\times 600$.)

in patients with HSCT, two consensus guidelines have been published outlining clinical criteria for the diagnosis of TMA.^{196,197} To meet criteria, the guidelines require the following: 1. schistocytes on a peripheral smear and 2. an elevated lactate dehydrogenase concentration. The Blood and Marrow Transplant Clinical Trials Network also includes AKI (doubling of serum creatinine), unexplained central nervous system dysfunction, and a negative Coombs test result as part of diagnostic criteria.¹⁹⁶ The International Guidelines from the European Group for Blood and Marrow Transplantation include thrombocytopenia, anemia, and a decreased haptoglobin concentration.¹⁹⁷ Despite these criteria, making a diagnosis of TMA remains challenging and often requires a high index of suspicion, as supported by validation studies and autopsy studies in which clinical criteria often do not correlate with histologic findings.¹⁹⁸

It is debated whether TMA is a distinct complication of HSCT or a manifestation of several other post-HSCT complications, such as GVHD, infection, and drug toxicity.¹⁹⁹⁻²⁰² Despite histologic similarities, endothelial cell damage, primarily in the renal vasculature, underlies the pathogenesis of TMA post-HSCT, which differs from thrombotic thrombocytopenic purpura, which is characterized by low levels of von Willebrand factor–cleaving protease ADAMTS13 (genetic or acquired).²⁰³⁻²⁰⁷ At times, HSCT may unmask a previously undiagnosed genetic mutation in the alternate complement pathway, resulting in atypical hemolytic-uremic syndrome,²⁰⁸ which may respond to eculizumab.²⁰⁹

In addition to the conditioning regimens for and other exposures during HSCT, some studies have suggested that renal endothelial injury with subsequent TMA may also reflect direct endothelial cell injury by GVHD.^{210,211} In fact, the risk of TMA diagnosed on renal pathology at autopsy was fourfold higher in patients with acute GVHD after HSCT. Renal endothelial injury may occur due to circulating inflammatory cytokines or may reflect direct injury to renal endothelial cells by GVHD. Furthermore, plasma markers of endothelial injury and coagulation activation are elevated in patients with acute GVHD after HSCT, suggesting an

association among endothelial injury, acute GVHD, and the subsequent development of TMA.^{210,211}

Treatment remains challenging for clinicians. Currently available studies are limited by retrospective study designs and the inclusion of heterogeneous patient populations. Options that are used include adjustment of GVHD prophylaxis (either stopping or reducing the calcineurin inhibitor dose), therapeutic plasma exchange (TPE), rituximab, and defibrotide.^{196,198,212-214} In uncontrolled studies, reported response rates for TPE have ranged between 27% and 80%.²¹⁵⁻²²¹ In a prospective study of 112 HSCT patients, TMA developed in 11, and 64% of these patients responded with treatment consisting of TPE and cyclosporine withdrawal.²¹⁷ Case reports have described a benefit with rituximab treatment of TMA post HSCT.^{197,222-225} More data are required to understand the true utility of rituximab for treatment of this form of TMA.

GRAFT-VERSUS-HOST DISEASE

The use of donor stem cells requires that all allogeneic HSCT recipients receive prophylaxis against GVHD. Prophylactic regimens include cyclosporine A, mycophenolate mofetil, tacrolimus, and sometimes short-term methotrexate.^{226,227} GVHD can be classified into acute and chronic categories based on the duration of onset after HSCT and associated clinical findings. Acute GVHD has great significance because it can independently serve as a risk factor for the development of AKI in HSCT recipients.²²⁸ GVHD may lead directly to AKI through cytokine-mediated inflammation affecting the glomeruli and/or renal tubules or indirectly from exposure to certain medications, such as cyclosporine, which itself can predispose to nephrotoxicity. Furthermore, GVHD also promotes viral reactivation, such as cytomegalovirus, which can contribute to AKI. Gastrointestinal involvement by GVHD is associated with extrarenal fluid losses from vomiting and diarrhea, which can cause prerenal AKI. Treatment options for GVHD include prednisone, antithymocyte globulin, sirolimus, and mycophenolate mofetil.²²⁹ Supportive measures such as IV fluids and avoiding other nephrotoxins are also important in management.

CKD also develops in many patients following HSCT. It can be due to chronic TMA, various viral infections, or a sequelae of acute or chronic GVHD, particularly in patients whose pre-HSCT conditioning regimen did not include total body irradiation (TBI).^{162,230} A retrospective study of 1635 HSCT recipients has noted that neither TBI nor cyclosporine use was associated with the development of CKD (eGFR < 60 mL/min/1.73 m²) at 1 year post transplantation. Subgroup analyses of patients receiving cyclosporine have supported the theory that GVHD, independent of calcineurin inhibitor therapy, increases the risk for the development of CKD.¹⁶²

HEMATOPOIETIC STEM CELL TRANSPLANTATION–ASSOCIATED CHRONIC KIDNEY DISEASE

HSCT also appears to be associated with the development of CKD. However, studies reporting the risk of CKD after HSCT are difficult to compare because the definitions used are not consistent, the populations studied are often heterogeneous, and follow-up times vary.²³¹ In general, the prevalence of CKD after HSCT ranges from 20% to 30%.²³²

Microalbuminuria and macroalbuminuria, HTN, and various types of renal tubular dysfunction also complicate the course of HSCT. A meta-analysis has reviewed the published data on CKD after HSCT through 2006.²³¹ Kidney function was assessed with creatinine-based, radioisotope, or inulin measurements of the GFR. The overall prevalence of CKD was 16.6% (range, 3.6%–89%) and was similar in those receiving autologous and allogeneic transplants, although the allogeneic transplant recipients experienced a more pronounced decline in kidney function. Regardless of underlying cause, CKD can progress to ESRD, which increases the mortality risk post HSCT.²³³

Causes of CKD following HSCT are quite varied (see [Table 57.9](#)). The incidence of post-HSCT TMA ranges from 2% to 21%¹⁹⁶ and this complication often follows an indolent course, resulting in CKD and, possibly, progression to ESRD.²³⁴ Although several disease processes, such as TMA, radiation nephritis, nephrotic syndrome, chronic GVHD, and BK virus nephropathy, have been associated with CKD after HSCT, much of the long-term kidney injury in patients undergoing HCT remains unexplained.²³⁵ Multiple bouts of AKI in the setting of HSCT, especially when severe and prolonged, may lead to CKD. Risk factors considered to be potentially important for the development of CKD include acute and chronic TMA post HSCT, calcineurin inhibitor use, TBI, and GVHD.²³⁶ CKD occurring from GVHD-associated T cell and cytokine-mediated damage kidney tissue is thought to be due to a chronic inflammatory state.²³⁷ Glomerular lesions related to HSCT occur primarily in the setting of chronic GVHD. Membranous nephropathy (67%) and MCD (33%) are the two most common pathologies^{238,239} and occur within several months of GVHD, approximately 8 to 14 months after HSCT. Hypertension commonly complicates HSCT and likely contributes to CKD and its overall progression. BK virus–related genitourinary and renal lesions also complicate the course of HSCT and are associated with CKD.²⁴⁰

ANTICANCER DRUGS AND KIDNEY DISEASE

The number of anticancer agents available to treat various malignancies has grown significantly in the past 10 to 20 years. Chemotherapy-related nephrotoxicity has always been an important area for the nephrologist; however, the number of agents causing kidney injury and the assortment of electrolyte and acid-base disturbances has increased dramatically.^{79,241,246}

Furthermore, understanding renal metabolism and excretion of these drugs is critical in ensuring efficacy while also avoiding or at least reducing toxicity, especially in the setting of acute or chronic kidney disease. Nephrotoxicity associated with the more common chemotherapeutic drugs includes AKI primarily from acute tubular injury, acute interstitial nephritis, and a variety of glomerular injuries; a number of tubulopathies (alone or combined with kidney injury) may develop and result in metabolic perturbations.^{79,241,242} In addition to conventional agents, the emergence of targeted therapies and immunotherapies has further increased the amount of drug-induced nephrotoxicity seen in patients with cancer.^{32,244-249} ([Table 57.10](#)).

CONVENTIONAL CHEMOTHERAPEUTIC AGENTS

GEMCITABINE

Gemcitabine is a cell cycle–specific pyrimidine antagonist used to treat a number of malignancies.^{32,250} Following gemcitabine approval, TMA was described as a complication, with an incidence ranging from 0.015% to 2.4%.^{11-14,32,250-252} The risk for TMA is associated with a higher cumulative dose and prior exposure to other chemotherapeutic agents that cause TMA.^{32,250-252} Endothelial injury from direct drug toxicity is the most likely mechanism. Disruption of endothelial cell integrity releases von Willebrand factor (vWF) multimers and plasminogen activator inhibitor, reduces the synthesis of prostacyclin and tissue plasminogen activator, and exposes a denuded endothelial surface to fibrin and platelet, which deposit within the renal microvasculature.^{32,250-252}

Gemcitabine-induced TMA usually develops weeks to months after initiating therapy, and AKI develops in most cases.^{32,250-252} A case series of 29 patients with gemcitabine-associated TMA noted microangiopathic hemolytic anemia (MAHA), thrombocytopenia, and AKI in all patients, whereas new or worsening HTN, proteinuria, and hematuria were noted in 26 of 29 patients.²⁵³ Gemcitabine discontinuation or future avoidance resulted in full ($n = 8$) or partial ($n = 11$) recovery, with CKD noted in three and ESRD in seven patients (25%). Plasma exchange and plasmapheresis had minimal or no effect on renal recovery. Reported mortality was wide ranging (0%–90%), with TMA-specific death estimated at 15%.^{32,250-253} Complement inhibition has been successful in small case series.

CISPLATIN

Cisplatin is an effective and widely used chemotherapeutic agent. However, nephrotoxicity is the most common dose-limiting side effect. This relates to its renal site of elimination—in particular, transport through the proximal tubule.^{79,241,242} Following drug entry into the tubule cell via the organic cation transporter, direct tubular toxicity occurs in part due to a low chloride environment. In the intracellular compartment, chloride molecules are replaced with water molecules in the *cis* position of cisplatin, forming hydroxyl radicals that injure the neutrophilic binding sites on DNA.^{79,241,242,254,255} Nonoliguric AKI, primarily from renal tubular injury (apoptosis and necrosis), typically develops after approximately 3 to 5 days of drug exposure. Furthermore, tubular injury from high-dose cisplatin is often associated with a so-called tubulopathy, which is characterized by various electrolyte (e.g., hypokalemia, hypomagnesemia, hypocalcemia) and acid-base disturbances (e.g., renal tubular acidosis).^{2,4,79,241,242} Hypomagnesemia, which occurs with prolonged exposure, is a challenging management issue because it can be permanent and require IV supplementation.^{79,241,242,254,255}

Although kidney injury is generally reversible, repeated cisplatin doses (>100 mg/m²) may cause permanent kidney injury (GFR reduction of 15%). Isotonic or hypertonic saline administration and avoidance of concomitant nephrotoxins are the most effective ways to prevent cisplatin-induced nephrotoxicity. Amifostine may reduce cisplatin nephrotoxicity through the promotion of improved DNA repair and elimination of free radicals.^{2,3,18,79,241,242,256} Newer platinum agents (carboplatin and oxaliplatin) are less nephrotoxic than

Table 57.10 Anticancer Drug Nephrotoxicity

Drug Class	Drug Name	Mechanism of Injury	Kidney Effect
Conventional Chemotherapy			
Platinum based	Cisplatin, Carboplatin, Oxaliplatin.	Platinum-DNA adducts mediate arrest of cell cycle, initiate apoptosis. ATP depletion	ATI, renal magnesium wasting, proximal tubulopathy, NDI
Antimetabolite	Methotrexate	Intratubular crystal formation. Afferent arteriolar constriction	Crystal nephropathy, ATI
	Gemcitabine	Endothelial injury	TMA, HTN
	Pemetrexed	Unknown	ATI, Chronic interstitial fibrosis, proximal tubulopathy, NDI
Alkylating agent	Cyclophosphamide	Toxic metabolite, acrolein	hemorrhagic cystitis
	Ifosfamide	Toxic metabolite, chloroacetaldehyde	ATI, proximal tubulopathy, NDI
	Melphalan	Increase ADH release	SIADH
	Nitrosoureas	Alkylation of tubular cell proteins	Chronic interstitial nephritis
Antitumor antibiotics	Mitomycin C	Endothelial injury	TMA, HTN
Targeted Therapy			
VEGF inhibitors	Bevacizumab	Endothelial injury	TMA, HTN, ATI
Tyrosine kinase inhibitors	Sorafenib, sunitinib, axitinib, pazopanib, lenvatinib.	Endothelial injury, podocyte injury.	ATIN, ATI, TMA, FSGS
BCR-ABL tyrosine kinase inhibitor	Imatinib, dasatinib.	Tubular injury, endothelial injury	ATI, TMA
BRAF inhibitors	Vemurafenib, dabrafenib	Tubular injury, ERK activation.	ATIN, ATI
BCL-2 inhibitors	Venetoclax	Tubular injury	AKI, TLS
ALK inhibitors	Crizotinib, lorlatinib, alectinib	Inhibition of creatinine secretion, renal arteriolar myocyte vacuolization	Pseudo-AKI, ATIN, ATI, kidney cyst, podocytopathies
CDK4/6 inhibitors	Palbociclib, ribociclib	Inhibit MATE1 and MATE2 transporters	Pseudo-AKI, ATI
PARP inhibitors	Olaparib, talazoparib	Inhibition of creatinine secretion	Pseudo-AKI
MET tyrosine kinase	Capmatinib, tepotinib	Inhibition of creatinine	Pseudo-AKI
EGFR monoclonal antibodies	Cetuximab, panitumumab	Inhibition of EGFR signaling at the DCT	Renal magnesium wasting
mTOR inhibitors	Everolimus	Decrease cubilin and megalin, VEGF inhibition	ATI, podocytopathies
Protease inhibitors	Bortezomib, carfilzomib	Endothelial injury, autoantibody formation	TMA, HTN
Bruton tyrosine kinase inhibitors	Ibrutinib	Endothelial injury	ATI, HTN
XPO inhibitor	Selinexor	Volume depletion	Hemodynamic AKI, hyponatremia
Immunotherapies			
CAR-T therapy	Axicabtagene ciloleucel, Idecabtagene vicleucel, brexucabtagene autoleucel, Tisagenlecleucel.	Systemic hyperinflammatory state. Ischemic injury	ATI, hemodynamic mediated AKI
CTLA-4 inhibitor	Ipilimumab, tremelimumab	Tubular injury, endothelial injury, podocyte injury	ATIN, ATI, MCD, lupus-like GN, necrotizing GN, TMA
PD-1 inhibitors	Pembrolizumab, cemiplimab, nivolumab	Tubular injury, endothelial injury, podocyte injury	ATIN, ATI, MCD, IgA nephropathy, FSGS, Necrotizing GN, Amyloidosis, Immune-complex mediated GN, C3 glomerulonephropathy
PDL-1 inhibitor	atezolizumab, avelumab, durvalumab	Tubular injury	ATIN, ATI

ALK, Anaplastic lymphoma; ATI, acute tubular injury; ATIN, acute tubulointerstitial nephritis; ATP, adenosine 5'-triphosphate; BCL-2, B cell lymphoma 2 gene; BCR-ABL, breakpoint cluster region-tyrosine protein kinase ABL-1 gene; BRAF, proto-oncogene B-Raf; BTK, Bruton tyrosine kinase; CAR-T, chimeric antigen receptor T cells; CDK4/6, cyclin-dependent kinase 4/6; CTLA-4, cytotoxic T-lymphocyte-associated protein 4; DCT, distal convoluted tubule; EGFR, epidermal growth factor receptor; ERK, extracellular signal-regulated kinase; FSGS, focal segmental glomerulosclerosis; HTN, hypertension; MATE, multidrug and toxin extrusion; MCD, minimal change disease; MET, mesenchymal-epithelial transition; mTOR, mammalian target of rapamycin; NDI, nephrogenic diabetes insipidus; PARP, poly-ADP-ribose polymerase; PD-1, programmed cell death protein 1; PDL-1, programmed death-ligand 1; SIADH, syndrome of inappropriate antidiuretic hormone secretion; TLS, tumor lysis syndrome; TMA, thrombotic microangiopathy; VEGF, vascular endothelial growth factor; XPO, export protein exportin 1.

cisplatin and are available for patients with underlying kidney disease and for those who previously suffered from cisplatin nephrotoxicity.²⁵⁷⁻²⁵⁹ Clinicians must recognize that these agents can still cause AKI and electrolyte disorders.

A large multicenter cohort study developed and externally validated a prediction model for severe cisplatin-associated AKI in adults receiving their first dose of intravenous cisplatin across six major U.S. cancer centers.²⁶⁰ Using data from >24,000 patients between 2006 and 2022, researchers created a multivariable logistic regression model and a simplified risk score using nine clinical variables including age, comorbidities, laboratory values, and cisplatin dose. The model demonstrated strong predictive ability (C statistic 0.75) and outperformed previously published models. In both derivation and validation cohorts, patients in the highest risk category had more than 17- to 24-fold greater odds of developing cisplatin-induced AKI than those in the lowest category. Additionally, severe cisplatin-related AKI was significantly associated with reduced 90-day survival. Each of the following factors were independently associated with cisplatin-associated AKI in the derivation cohort: age, HTN, diabetes mellitus, serum creatinine level, hemoglobin level, white blood cell count, platelet count, serum albumin level, serum magnesium level, and cisplatin dose. The study concludes that this straightforward risk score can help clinicians identify high-risk patients before treatment, potentially improving outcomes through targeted monitoring or intervention. While mannitol was evaluated and was harmful, intravenous magnesium seems to be preventive for cisplatin-associated AKI.^{261,262}

Newer platinum agents (carboplatin and oxaliplatin) are less nephrotoxic than cisplatin and are available for patients with underlying kidney disease and for those who previously suffered from cisplatin nephrotoxicity.²⁵⁰⁻²⁵² Clinicians must recognize that these agents can still cause AKI and electrolyte disorders.

IFOSFAMIDE

Ifosfamide is an effective alkylating agent that is associated with nephrotoxicity, predominantly injuring the renal tubules. Like cisplatin, this drug enters proximal tubular cells via the organic cation transporter.^{79,241,242} Once inside the cell, the drug or its metabolite chloroacetaldehyde causes direct tubular epithelial cell damage.^{79,241,242} Ifosfamide-induced tubular injury is associated clinically with AKI and a tubulopathy, which can be associated with severe hypokalemia, hypophosphatemia, hypomagnesemia, and hyperchloremia. Patients can experience Fanconi syndrome, with hypophosphatemic rickets and osteomalacia, as well as nephrogenic diabetes insipidus.^{79,241,242,263}

Risk factors for ifosfamide nephrotoxicity include previous exposure to cisplatin, CKD, and a high cumulative dose (>84 g/m²).^{79,241,242,264,265} Amifostine may offer some protection, whereas mesna, which prevents hemorrhagic cystitis, is ineffective in the kidney. Most patients recover from ifosfamide-induced tubular injury; however, long-term complications such as CKD and tubulopathies may develop. Chronic renal fibrosis, with a progressive decline in the GFR, can rarely lead to ESKD.^{79,241,242,266}

METHOTREXATE

Methotrexate (MTX) is an antifolate drug that is an effective antineoplastic agent when administered in a

high dose (>1 g/m²).^{79,241,242,267-271} Nephrotoxicity occurs primarily because of the precipitation of an insoluble parent drug and metabolites (7-hydroxymethotrexate) within the tubular lumens, a phenomenon known as “crystalline nephropathy.”^{79,241,242,267-271} True or effective volume depletion and acidic urine are two major risk factors for the development of AKI; direct tubular toxicity may also contribute to the development of kidney injury. The overall incidence rate of AKI is approximately 1.8% (range, 0%–12%) but can be as high as 50% with high doses in at-risk patients.^{79,241,242,267-271} In general, kidney injury is reversible.

Preventive measures include volume repletion to achieve high urine flow rates. Isotonic saline infusion and furosemide may be necessary to maintain a high urine output. An increase in the clearance rate of MTX (and reduction in crystal precipitation) is seen when the urine pH is increased from 5.5 to 8.4, which can be accomplished with an isotonic solution containing bicarbonate. Once AKI develops, the excretion of MTX is reduced, and systemic toxicity is increased. In this case, therapy is primarily supportive. However, it may be necessary to remove the drug with hemodialysis. Hemodialysis, using high blood flow rates with a high-flux dialyzer, is an effective method of removing MTX.²⁷² High-dose leucovorin therapy can reduce the systemic toxicity associated with MTX and AKI.

Glucarpidase (carboxypeptidase G) may be considered when severe systemic toxicity and AKI are present, and dialysis access is risky. This costly drug metabolizes MTX to benign metabolites.^{273,274} A large study evaluated the impact of glucarpidase on outcomes in 708 adults with MTX-AKI across 28 U.S. cancer centers. Using a target trial emulation framework, outcomes of 209 patients who received glucarpidase within 4 days of MTX initiation were compared with those who did not. Glucarpidase was linked to a 2.7-fold higher odds of kidney recovery, faster recovery time, and reduced risks of neutropenia and liver toxicity. However, it did not affect mortality.

IMMUNOTHERAPIES

The immune system represents a potent tool to destroy cancer cells. Enhancing the immune system as a therapy for cancer has been used for several decades. Initial therapies included interferon and IL-2. More recently, immune checkpoint inhibitors and chimeric antigen receptor T cells have been used to treat certain forms of malignancy.

INTERFERON

Interferon is used to treat various malignancies (e.g., chronic myelogenous leukemia [CML], renal cell carcinoma) and infectious agents (e.g., hepatitis C). Treatment with interferon results in two renal lesions: 1. podocytopathy with MCD and focal segmental glomerulosclerosis (FSGS) and 2. thrombotic microangiopathy.^{275,276} MCD was first described, but more recent reports have described collapsing and noncollapsing FSGS. Nephrotic-range proteinuria and AKI are observed within weeks to months of commencing interferon therapy. Although proteinuria declines with interferon discontinuation, complete resolution is more common with MCD than with FSGS.^{275,276} In particular, collapsing FSGS is associated with CKD. Treatment with corticosteroids leads to complete remission

in less than one-third of patients with FSGS. Interferon-related glomerular injury may be due to the direct binding of interferon to podocyte receptors and an alteration of normal cellular proliferation. Macrophage activation and skewing of the cytokine profile toward IL-6 and IL-13, which are possible permeability factors in MCD and FSGS, are also possible mechanisms.^{275,276}

INTERLEUKIN-2

High-dose IL-2 has been used to treat renal cell carcinoma, malignant melanoma, and other cancers. High-dose IL-2 is associated with a cytokine release syndrome and capillary leak; this causes a prerenal form of AKI, which typically develops within 24 to 48 hours of infusion.²⁷⁷ This hemodynamic disturbance is further exacerbated by concurrent nonsteroidal anti-inflammatory drug therapy, which is used to reduce fever and myalgias associated with IL-2.²⁷⁷ Kidney dysfunction generally reverses with drug discontinuation, although ischemic/nephrotoxic acute tubular injury can be associated with a more prolonged course of AKI.²⁷⁷ This is particularly the case for patients with underlying risk factors, such as CKD, diabetes mellitus, and HTN. To reduce the likelihood and severity of AKI during IL-2 therapy, antihypertensive agents should be withdrawn and fluid resuscitation initiated. Low-dose dopamine (2 µg/kg/min) may prevent some of the nephrotoxicity of IL-2, reversing oliguria and hastening time to recovery of kidney function.²⁷⁸

IMMUNE CHECKPOINT INHIBITORS

Another method to activate the immune system against the tumor microenvironment is the strategy of immune checkpoint inhibitor (ICI) therapy.^{248,249} Activation by an immunologic event such as infection signals cytotoxic T cells to undergo damping of their response to maintain immune homeostasis by expressing several receptors, including cytotoxic T lymphocyte-associated antigen 4 (CTLA-4), programmed death-1 protein (PD-1), and programmed death ligand-1 (PD-L1) that downregulate T cell function.^{248,249,279} These checkpoints normally prevent the development of unwanted autoimmunity. Tumor survival is enhanced in malignancies that overexpress ligands that bind inhibitory T cell receptors (PD-1), thereby decreasing activated T cell infiltration into the tumor microenvironment and inhibiting antitumor T cell responses. To combat this, monoclonal antibody drugs that block ligand binding to PD-1 and CTLA-4 receptors were designed to facilitate T cell rescue and restore antitumor immunity. Ipilimumab, a fully human, immunoglobulin G1 (IgG1) monoclonal antibody blocking CTLA-4, nivolumab, a fully human IgG4 antibody blocking the PD-1 receptor, and pembrolizumab, a humanized monoclonal IgG4-kappa isotype antibody against PD-1, are the ICIs available in clinical practice. However, blocking immune checkpoints risks the development of pathologic autoimmunity and end-organ injury, such as acute tubulointerstitial nephritis.⁷⁰ A large multicenter retrospective study collected >400 cases of patients with AKI from immune checkpoint inhibitors.²⁸⁰ These were compared with controls who received ICIs but did not develop AKI. ICI-AKI occurred at a median 16 weeks (IQR 8–32). The use of proton pump inhibitor therapy (PPI), underlying low estimated glomerular filtration rate

(eGFR), and presence of extrarenal immune-related adverse events were risk factors. Acute interstitial nephritis was the most common kidney biopsy finding in 83% of patients. Treatment of ICI-AKI with corticosteroids within 14 days following the diagnosis was associated with a higher odds of recovery, with early initiation favoring renal recovery. Of the 121 patients who were rechallenged, 20 (16.5%) developed recurrent ICI-AKI. There was no difference in survival among patients rechallenged versus those who were not, following ICI-AKI. A meta-analysis examined the effects of AKI on outcomes in patients being treated with ICIs. This meta-analysis included seven studies with a total of 3767 patients demonstrating that ICI-associated AKI is of mild to moderate severity in most cases. The results revealed high rates of AKI, with incidence rates up to 43% for stage 2 and up to 57% for stage 3 AKI. Kidney biopsy was performed in 43% of patients, the most frequent finding was of tubulointerstitial disease. The presence and severity of AKI were associated with all-cause mortality, as was the absence of recovery of kidney function. Even though these results are not definitive, they emphasize the need for clear management guidelines for AKI during the course of ICI therapy. Early recognition and proper management have the potential to improve the clinical outcome. The duration of corticosteroid therapy has not been well defined. Single- and multicenter data suggest that a shorter course of corticosteroids with a quick taper is as effective as a longer course for the treatment of ICI-induced AIN.

The addition of ICIs to standard chemotherapy has improved progression-free and overall survival in patients with metastatic nonsquamous non-small cell lung cancer regardless of PD-L1 expression. This raised initial concern for an increased rate of AKI in patients receiving this triple combination therapy (pembrolizumab, pemetrexed, and platinum-based chemotherapy). Real-world data have shown that both groups (monotherapy and triple therapy) had a similar distribution of AKI severity. However, AKI occurred earlier in the triple therapy group (105 days vs. 202 days). The kidney biopsy findings were more suggestive of tubular and interstitial damage than AIN alone.⁸ Studies have now confirmed the safety of ICI in patients with kidney failure on maintenance dialysis, as they had similar incidence of immune-related adverse events when compared with the general population, likely explained by the fact that the main mechanism of ICI clearance is through proteolytic catabolism and receptor-mediated endocytosis, without major influence of kidney function.

Ideally, a noninvasive test to predict a kidney-related adverse immune event would be an invaluable resource in patients treated with ICIs. A study by Isik and colleagues reviewed the role of commonly available laboratory tests, such as C-reactive protein and urine retinol binding protein, in predicting ICI related-AKI.^{280a} Elevated C-reactive protein with an elevated urine retinol binding protein/creatinine was helpful in identifying patients with ICI-related AKI. A newer study showed that soluble interleukin-2 receptor (sIL-2R) level in peripheral blood was significantly higher in patients with ICI-related nephritis compared with ICI-treated controls and hemodynamic AKI controls. A sIL-2R cutoff point of 1.75-fold the upper limit of normal was highly diagnostic of ICI nephritis (AUC >96%) when compared with either ICI-treated or hemodynamic AKI controls. By

peripheral blood flow cytometry analysis, lower absolute CD8⁺ T cells, CD45RA⁺CD8⁺ T cells, memory CD27⁺ B cells, and expansion of plasmablasts were prominent features of ICI-nephritis compared with ICI-treated controls. Another emerging biomarker is urine TNF- α as its level was significantly elevated in ICI nephritis when compared with other causes of AKI. TNF- α in conjunction with specific T-cell responses contribute to AKI-ICI injury and could potentially serve as targets for therapeutic intervention, as well as potential biomarkers. Urine proteomics has revealed that urine was more sensitive and specific than plasma in distinguishing ICI-acute interstitial nephritis from nonacute interstitial nephritis cases. Pathway analyses highlighted the involvement of JAK-STAT and TNF signaling in ICI-associated AIN. IL-5+Fas-in one study achieved a statistical advantage for diagnosing ICI associated acute interstitial nephritis.²⁸¹

Imaging biomarkers using computed tomography (CT) and positron emission tomography (PET) have also been investigated in larger, single-center and multicenter cohorts. Bilateral increase in kidney size, new/increasing perinephric stranding, and bilateral wedge-shaped hypoenhancing cortical foci can occur in ICI-related nephritis. On PET-CT, a diffuse increase in radiotracer uptake throughout the renal cortex and a decrease in radiotracer activity in the renal pelvis can be seen. In the large multicenter study using PET-CT, patients with ICI-AKI had a considerable increase in SUV_{mean} on F¹⁸-FDG PET-CT from baseline to the time of AKI. In contrast, patients with AKI from non-ICI causes, as well as ICI treated patients without AKI at follow-up imaging, did not exhibit this increase. These findings suggest that, when a baseline F¹⁸-FDG PET-CT is available, these scans have diagnostic utility in differentiating ICI-AKI from AKI caused by other etiologies and could offer a noninvasive alternative to kidney biopsy.

New glomerular diseases have now been reported in association with ICI as well. In an analysis of all published case reports and case series of glomerular pathology findings associated with immune checkpoint inhibitors, several lesion types were observed, with the most frequent being pauci-immune glomerulonephritis (GN) and renal vasculitis (27%), podocytopathies (24%), and complement 3 GN (C3GN; 11%).

CHIMERIC ANTIGEN RECEPTOR T CELL THERAPY

Chimeric antigen receptor T cell (CAR-T) therapy is a genetically modified immunotherapy used to treat certain blood cancers including non-Hodgkin lymphoma and leukemia. Patient's T cells are collected, genetically engineered to express a CAR targeting tumor antigen, expanded in the laboratory, and then infused back into the patient. Several CAR-T therapies, such as tisagenlecleucel and axicabtagene ciloleucel, have been approved to treat specific B cell hematologic malignancies, including multiple myeloma (Fig. 57.8). However, CAR-T therapy can trigger cytokine release syndrome, occurring in >40% of patients, potentially leading to multiorgan dysfunction, including AKI. Cytokine release syndrome-induced AKI results from high cytokine levels causing vascular issues and reduced kidney perfusion. A systematic review found the incidence of cytokine release syndrome at 75%, AKI at 19%, and AKI needing kidney replacement therapy at

4%, with higher AKI rates in children. Different CAR-T agents, like tisagenlecleucel, may have lower AKI risk. AKI risk factors include lower baseline kidney function, IV contrast exposure, and certain medications. AKI is generally reversible, with most patients recovering within 30 days. Electrolyte imbalances including hyperphosphatemia, hyperkalemia, hypokalemia, and hypophosphatemia are also associated with CAR-T therapy, some related to tumor lysis syndrome.²⁸²⁻²⁸⁵

TARGETED THERAPIES

ANTIANGIOGENESIS DRUGS

The discovery that VEGF is an important mediator of tumor growth and angiogenesis led to the development of drugs targeting this pathway for cancer therapy. However, VEGF is also an essential growth factor for maintaining glomerular and microvascular endothelial health.^{244,286} In the glomerulus, VEGF is secreted by podocytes and traverses the basement membrane, where it binds to VEGF receptors on endothelial cells and maintains glomerular endothelial cell integrity and filtration barrier function.^{244,286} Anti-VEGF drugs, such as the monoclonal antibody bevacizumab, as well as drugs that inhibit tyrosine kinase, such as sunitinib, block VEGF function. Blocking the action of VEGF leads to glomerular endothelial cell dysfunction and filtration barrier disruption, resulting in proteinuria, thrombotic microangiopathy, HTN, and AKI. Mild and asymptomatic proteinuria is common, although heavy proteinuria occurs in <10% of subjects.^{244,286} New-onset HTN or worsening of preexisting HTN is common. The reported incidence varies between 17% and 80%. The effects of anti-VEGF therapy on the development of HTN have been reviewed in a meta-analysis of seven trials, which noted that the relative risk of HTN was threefold to sevenfold higher, depending on dose.²⁸⁷ Hypertension reflects the efficacy of VEGF blockade, which may underlie the observed association between bevacizumab-induced HTN and improved clinical outcomes in certain malignancies. However, while several studies have supported the correlation, others have failed to replicate these findings, suggesting variability in the association. Hypertension is managed with standard antihypertensive medications, although the use of angiotensin-converting enzyme inhibitors or angiotensin receptor blockers (ARBs) is preferable in patients with proteinuria. Adverse kidney effects with anti-VEGF drugs may require dose reduction or drug discontinuation, especially with TMA. Table 57.11 summarizes the anti-VEGF therapies and kidney effects.

EPIDERMAL GROWTH FACTOR RECEPTOR INHIBITORS

The EGF pathway has multiple functions in several organs including the kidneys.^{79,242} Several novel agents that target the EGF pathway are used to treat various cancers; these include receptor antagonists, tyrosine kinase inhibitors, and antisense oligodeoxynucleotides. Cetuximab is a monoclonal antibody against the EGF receptor and was first approved for use in metastatic colon cancer. The primary abnormality observed following EGF receptor antagonism is hypomagnesemia due to renal magnesium wasting.^{79,242,288} Hypocalcemia and hypokalemia may also develop as a result of the effects of hypomagnesemia on parathyroid

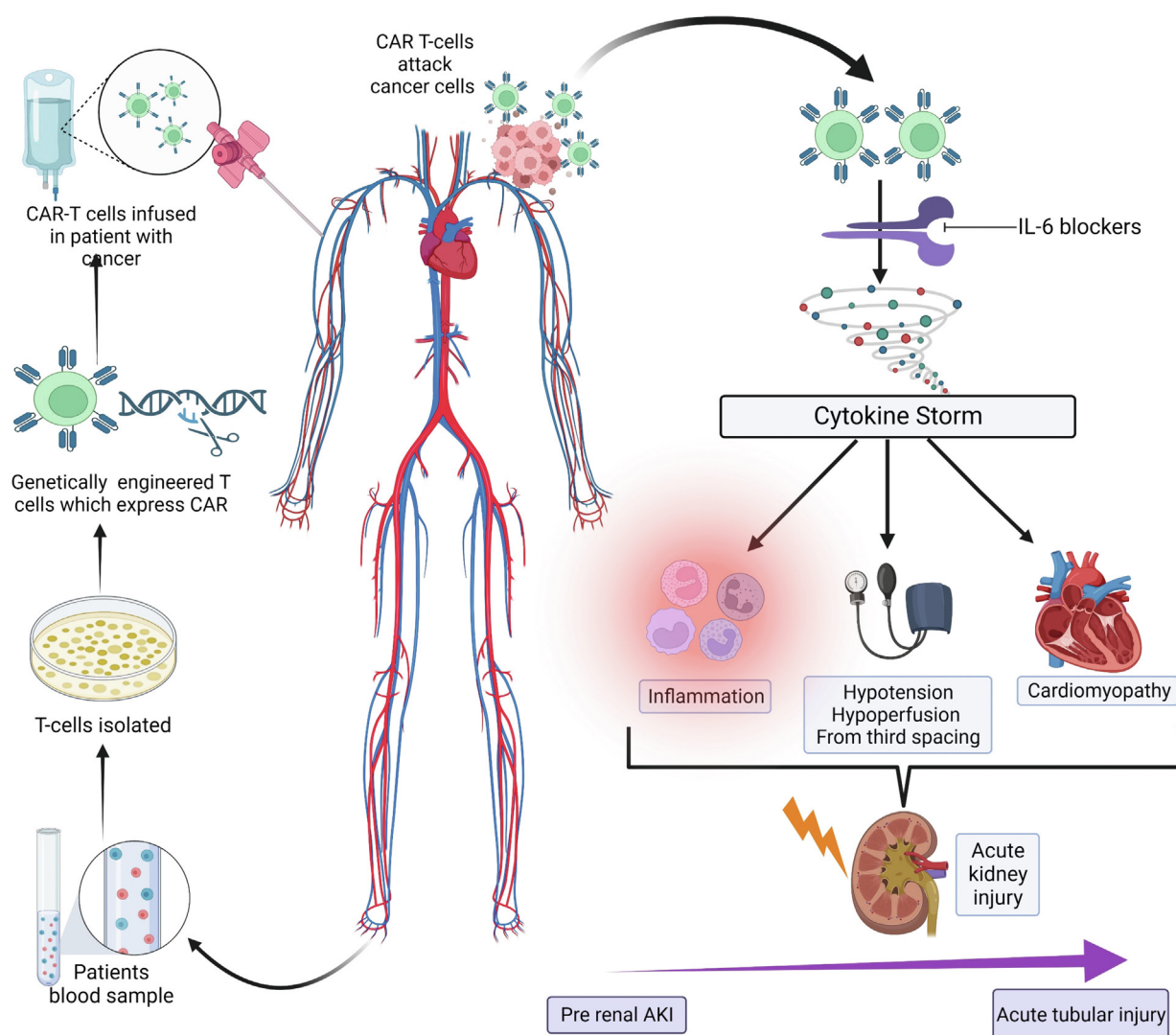


Fig 57.8 Mechanism of how chimeric antigen receptor T cell therapy works and may lead to cytokine release storm and acute kidney injury.

hormone.^{79,242} EGF normally activates the renal magnesium channel (TRP-M6) in the apical membrane of the distal convoluted tubule to stimulate magnesium absorption. The incidence of severe hypomagnesemia is 10% to 15% but is reversible with drug discontinuation.²⁸⁹ Many patients require IV magnesium repletion to continue the drug. Small molecule tyrosine kinase inhibitor of EGF receptor, such as Afatinib, erlotinib, gefitinib, dacomitinib, osimertinib, and mobocertinib, are utilized in non-small cell lung cancer treatment. These drugs have been associated with electrolyte imbalances including hypomagnesemia, hypokalemia, and hypophosphatemia, though the incidence appears lower compared to EGF receptor monoclonal antibodies. Additionally, there have been rare instances of nephrotic syndrome in patients receiving gefitinib, with kidney biopsy results indicative of MCD and membranous nephropathy.

FGF INHIBITORS

Collectively, all fibroblast growth factor receptor (FGFR) inhibitors including erdafitinib, infigratinib, and pemigatinib induce hyperphosphatemia due to

their class effect. This is because the FGFR pathway plays a vital role in regulating phosphate levels through feedback mechanisms involving fibroblast growth factor (FGF) 23, 1,25-dihydroxyvitamin D, and parathyroid hormone. Consequently, hyperphosphatemia represents a pharmacodynamic consequence of these drugs. Although >60% of treated individuals experience hyperphosphatemia, it is typically mild (grade 1 or 2). It usually occurs shortly after starting treatment (around 15–20 days) and can be managed through dietary phosphate restriction, concurrent use of phosphate binders, dose adjustments, or treatment interruptions. Notably, there have been reports of hypophosphatemia as well, potentially linked to the use of low-phosphate diets or phosphate binders during periods off treatment or feedback mechanisms affecting phosphate balance.^{290,291} Furthermore, AKI has been observed in patients receiving FGFR inhibitors, with incidences ranging from 2% to 6%, and kidney biopsy in one case revealed acute tubular necrosis. AKI improved upon discontinuation of treatment, and reintroduction at a lower dose did not result in kidney injury recurrence.²⁹⁰

Table 57.11 Anti-VEGF Signaling Pathway Agents Kidney Adverse Events

Anti-VEGF Agents Subclasses	Drugs	Indications (Approved and Potential)	Proteinuria (Risk Factors: CKD, HTN, RCC)	Hypertension (Risk Factors: Age >60, BMI >25 kg/m ² , >1 Anti-VEGF Agent)
VEGF-Trap (fusion protein)	Alficercept	Melanoma, prostate cancer, pancreatic cancer, non-small lung cancer	Incidence of 40%-60%	Incidence about 40%
VEGF-A monoclonal antibody	Bevacizumab	Glioblastoma, metastatic colorectal cancer, non-small cell lung cancer, renal cell carcinoma	Incidence of 20%-60%	Incidence about 20%
VEGFR2 monoclonal antibody	Ramucirumab	Metastatic colorectal cancer, breast cancer, non-small lung cancer	Incidence about 10%	Incidence about 42%
Tyrosine kinase inhibitors	Axitinib	Renal cell carcinoma, thyroid cancer, hepatocellular cancer, gastrointestinal stromal tumors, pancreatic neuroendocrine tumors, soft tissue sarcomas, refractory chronic myelogenous leukemia, and refractory metastatic colorectal cancer	Incidence about 20%	Incidence about 40%
	Cabozantinib		Incidence about 22%	Incidence about 30%-60%
	Lenvatinib		Incidence about 11%	Incidence about 70%
	Nintedanib		n/a	n/a
	Pazopanib		Incidence about 13%	Incidence about 35%
	Regorafenib		Incidence about 7%	Incidence about 28%
	Sorafenib		Incidence about 11%	Incidence about 19%
	Cabozantinib		Incidence about 8%	Incidence about 20%
	Vandetanib		Incidence about 10%	Incidence about 30%
	Apatinib		Incidence about 15%	Incidence about 9%

BMI, Body mass index; CKD, chronic kidney disease; HTN, hypertension; RCC, renal cell carcinoma; VEGF, vascular endothelial growth factor.

HER2 INHIBITORS

Trastuzumab, ado-trastuzumab emtansine, and pertuzumab are used to treat HER2-positive breast cancer. There are no published cases of nephrotoxicity associated with trastuzumab or pertuzumab. Some kidney-related adverse events were reported, but their causal relationship to the drug remains uncertain and it is unclear whether these events were directly attributable to the treatment. Rare cases of fetal nephrotoxicity have been linked to trastuzumab during pregnancy, but improvements occurred upon discontinuation. Ado-trastuzumab emtansine may lead to hypokalemia in some patients. While trastuzumab is minimally excreted by the kidneys, preexisting kidney impairment has been associated with a higher risk of cardiotoxicity. Pertuzumab does not require dose adjustment for mild to moderate impairment in kidney function. Lapatinib, a dual tyrosine kinase inhibitor, has been linked to grade 3 toxicities including hyponatremia, but no cases of nephrotoxicity have been reported. Some cases of hypokalemia, AKI, and electrolyte imbalances were reported in association with lapatinib.²⁹²

B-RAF AND MEK INHIBITORS

Malignant melanoma frequently has a B-RAF V600 mutation that can be effectively targeted by selective B-RAF inhibition by vemurafenib and dabrafenib. As with other effective cancer agents, these drugs have also been observed to cause nephrotoxicity. Impaired kidney function, defined as an eGFR decline at 1 and 3 months of therapy, occurred in 15 of 16 patients, which was complicated by persistent kidney injury after 8 months of follow-up.²⁹³ In addition, eight

patients developed AKI (one with ATN on biopsy) following vemurafenib therapy.²⁹³ AKI that was clinically suggestive of acute interstitial nephritis was observed in four patients treated with vemurafenib, with three of four patients recovering kidney function after drug discontinuation, although no biopsy data were available. Of 74 patients treated with vemurafenib, about 60% developed AKI, primarily KDIGO stage 1, within 3 months of drug exposure.²⁴⁵ Biopsy in two patients revealed tubulointerstitial injury. Kidney function recovered within 3 months of B-RAF discontinuation. Acute kidney injury and metabolic disturbances from B-RAF inhibitors have been described in the FDA Adverse Events Reporting System.²⁹³ Although the mechanism of AKI is unknown, these drugs may interfere with the downstream MAPK pathway, increasing susceptibility to ischemic tubular injury. Trametinib and cobimetinib are potent and highly specific inhibitors of the mitogen-activated extracellular kinases 1 and 2 (MEK1/MEK2). They are both used for the treatment of advanced melanoma. There have been no published reports of nephrotoxicity associated with trametinib monotherapy, although monotherapy with MEK inhibitors can lead to HTN. B-RAF and MEK inhibitors are frequently combined for the treatment of several tumors including advanced melanoma. Impaired kidney function, hyponatremia, and rare cases of glomerulonephritis have been described in patients treated with the combination of BRAF and MEK inhibitors.²⁹⁴

ANAPLASTIC LYMPHOMA KINASE INHIBITORS

Anaplastic lymphoma kinase 1 (ALK-1) is a member of the insulin receptor tyrosine kinase family. Small-molecule

inhibitors of ALK-1 include crizotinib and ceritinib.^{247,295} Crizotinib is an effective agent for the treatment of advanced ALK-positive non-small cell lung cancer; however, it is complicated by two major adverse renal effects. These include AKI and an increased risk for the development and progression of renal cysts.^{247,295} In addition to these complications, crizotinib is also associated with the development of peripheral edema and electrolyte disorders (relatively rare). Regarding AKI, it appears that crizotinib is associated with both true AKI (tubulointerstitial injury) and pseudo-AKI, which is likely due to a drug-induced reduction in renal tubular creatinine secretion. Four other ALK inhibitors are available: ceritinib, alectinib, brigatinib, and lorlatinib. Increases in creatinine have been seen in 11% to 28% of patients treated with alectinib, and proteinuria has been reported with lorlatinib.^{296,297}

BCL-2 INHIBITORS

Although many targeted treatments used for treatment of hematologic malignancies are associated with tumor lysis syndrome (TLS; e.g., obinutuzumab, ofatumumab), the B cell lymphoma-2 (BCL-2) inhibitor venetoclax, which is used in the treatment of refractory chronic lymphocytic leukemia, is associated with a particularly high incidence of TLS, which can cause AKI and severe electrolyte abnormalities. A gradual, stepwise dose-escalation strategy has been introduced in an effort to reduce this risk.²⁹⁸

PROTEASOME INHIBITORS

Proteasome inhibitors used in multiple myeloma treatment include bortezomib, carfilzomib, and ixazomib. Bortezomib rarely causes nephrotoxicity, with some reported cases TMA and acute interstitial nephritis. Dose adjustments are not necessary for patients with impaired kidney function, including those on dialysis. Carfilzomib has been associated with serum creatinine elevations in up to 25% of patients, potentially leading to AKI and biopsy-proven TMA. Ixazomib, another proteasome inhibitor, can cause rare instances of TMA as well.²⁹⁹

BCR::ABL AND KIT INHIBITORS

Small molecule inhibitors play a crucial role in targeting BCR::ABL, a tyrosine kinase responsible for the persistent activation of the Philadelphia chromosome in CML. Some of these inhibitors also extend their reach to the tyrosine kinase receptor KIT (CD117) and the platelet-derived growth factor receptor (PDGFR), both constitutively active in gastrointestinal stromal tumors. Bosutinib, a dual tyrosine kinase inhibitor, hinders both ABL and SRC pathways, but not KIT or PDGFR. It is FDA approved for refractory CML, though long-term use may lead to hypophosphatemia and a potentially reversible decrease in GFR, necessitating more intensive monitoring of kidney function. Dasatinib, a second-generation TKI primarily for imatinib-resistant CML, affects BCR::ABL, PDGFR, and KIT. Rare instances of AKI have been documented, along with reports of proteinuria and nephrotic syndrome, which resolved upon discontinuation. Unlike other BCR-ABL-targeting TKIs, dasatinib may influence glomerular podocytes, potentially independent of VEGF inhibition. Imatinib, a first-generation TKI for BCR::ABL and KIT, used in CML and gastrointestinal stromal tumors, has

been associated with acute and chronic kidney disease in prolonged treatments. These injuries may stem from various mechanisms, including tumor lysis syndrome, tubular injury, and rhabdomyolysis. Imatinib use can also lead to hypophosphatemia, possibly due to altered renal tubular phosphorus reabsorption. Careful monitoring and dose adjustments are essential for managing potential kidney-related complications in TKI therapy.³⁰⁰⁻³¹⁰

BTK INHIBITORS

Ibrutinib is an oral covalent inhibitor of Bruton tyrosine kinase (BTK), a critical player in the B cell receptor signaling pathway that suppresses malignant B cell survival. Although ibrutinib is commonly employed in the management of chronic lymphocytic leukemia, it has been linked to severe and potentially fatal episodes of AKI. In a study involving 111 patients with mantle cell lymphoma receiving ibrutinib, AKI occurred in 3 (2.7%) cases. Moreover, two-thirds of patients exhibited elevated serum creatinine concentrations up to 1.5 times the upper limit of normal, while roughly 1% to 10% percent displayed levels 1.5 to 3 times the upper limit of the population reference range. The precise mechanism behind kidney disease associated with ibrutinib remains uncertain, but tumor lysis syndrome could be a contributing factor. Pirtobrutinib, in contrast, is an orally administered noncovalent BTK inhibitor, targeting the same B cell receptor signaling pathway to hinder malignant B cell survival. To our knowledge, this agent has no adverse kidney effects.³¹¹

MISCELLANEOUS AGENTS

BISPHOSPHONATES

Bisphosphonates are administered primarily to treat patients with the hypercalcemia of malignancy and to minimize skeletal complications in the setting of bone metastases and multiple myeloma. These agents undergo renal excretion and have been associated with two major types of kidney disease—AKI from acute tubular injury and the podocytopathies, MCD and FSGS.³¹² Biopsy-proven ATN has been observed in patients treated with zoledronate, which is often associated with various levels of impaired kidney function, despite drug discontinuation.³¹² Pamidronate has also been shown to cause AKI, as well as the nephrotic syndrome, from a collapsing variant of FSGS.³¹³ The mechanism of bisphosphonate-induced kidney injury is unknown, but direct toxicity to the glomerular and tubular epithelial cells is suspected. Ibandronate is a newer agent that appears to be significantly less nephrotoxic.³¹² Dose modifications are recommended for pamidronate and zoledronate in the setting of impaired kidney function. Pamidronate, 90 mg, administered over 4 to 6 hours (rather than over 2 hours), is recommended for patients with stage 4 CKD, although 60 mg might be a more appropriate dose.³¹² The dose of zoledronate should be reduced in patients with stage 3 CKD and should be avoided in those with eGFR <30 mL/min.³¹²

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KEY REFERENCES

1. Christiansen CF, Johansen MB, Langeberg WJ, Fryzek JP, Sørensen HT. Incidence of acute kidney injury in cancer patients: a Danish population-based cohort study. *Eur J Intern Med.* 2011;22(4):399–406.
2. Liborio AB, Abreu KL, Silva Jr GB, et al. Predicting hospital mortality in critically ill cancer patients according to acute kidney injury severity. *Oncology.* 2011;80(3–4):160–166.
13. Rosner MH, Perazella MA. Acute kidney injury in patients with cancer. *N Engl J Med.* 2017;376(18):1770–1781.
15. Cairo MS, Bishop M. Tumour lysis syndrome: new therapeutic strategies and classification. *Br J Haematol.* 2004;127(1):3–11.
23. Cortes J, Moore JO, Maziarz RT, et al. Control of plasma uric acid in adults at risk for tumor lysis syndrome: efficacy and safety of rasburicase alone and rasburicase followed by allopurinol compared with allopurinol alone—results of a multicenter phase III study. *J Clin Oncol.* 2010;28(27):4207–4213.
32. Izzedine H, Perazella MA. Thrombotic microangiopathy, cancer, and cancer drugs. *Am J Kidney Dis.* 2015;66(5):857–868.
34. Hutchison CA, Batuman V, Behrens J, et al. The pathogenesis and diagnosis of acute kidney injury in multiple myeloma. *Nat Rev Nephrol.* 2011;8(1):43–51.
35. Sanders PW, Booker BB. Pathobiology of cast nephropathy from human Bence Jones proteins. *J Clin Invest.* 1992;89(2):630–639.
39. Dimopoulos MA, Roussou M, Gavriatopoulou M, et al. Bortezomib-based triplets are associated with a high probability of dialysis independence and rapid renal recovery in newly diagnosed myeloma patients with severe renal failure or those requiring dialysis. *Am J Hematol.* 2016;91(5):499–502.
45. Launay-Vacher V, Oudard S, Janus N, et al. Prevalence of Renal Insufficiency in cancer patients and implications for anticancer drug management: the renal insufficiency and anticancer medications (IRMA) study. *Cancer.* 2007;110(6):1376–1384.
54. Lowrance WT, Ordóñez J, Udaltsova N, Russo P, Go AS. CKD and the risk of incident cancer. *J Am Soc Nephrol.* 2014;25(10):2327–2334.
58. Birkeland SA, Lokkegaard H, Storm HH. Cancer risk in patients on dialysis and after renal transplantation. *Lancet.* 2000;355(9218):1886–1887.
61. Yanik EL, Clarke CA, Snyder JJ, Pfeiffer RM, Engels EA. Variation in cancer incidence among patients with ESRD during kidney function and nonfunction intervals. *J Am Soc Nephrol.* 2016;27(5):1495–1504.
63. Acuna SA, Fernandes KA, Daly C, et al. Cancer mortality among recipients of solid-organ transplantation in Ontario, Canada. *JAMA Oncol.* 2016;2(4):463–469.
65. Aapro M, Launay-Vacher V. Importance of monitoring renal function in patients with cancer. *Cancer Treat Rev.* 2012;38(3):235–240.
75. Janus N, Thariat J, Boulanger H, Deray G, Launay-Vacher V. Proposal for dosage adjustment and timing of chemotherapy in hemodialyzed patients. *Ann Oncol.* 2010;21(7):1395–1403.
90. Rosner MH. Cancer screening in patients with end-stage renal disease: an individualized approach. *J Onco-Nephrol.* 2017;1(1):36–41.
95. LeBrun CJ, Diehl LF, Abbott KC, Welch PG, Yuan CM. Life expectancy benefits of cancer screening in the end-stage renal disease population. *Am J Kidney Dis.* 2000;35(2):237–243.
105. Pani A, Porta C, Cosmai L, et al. Glomerular diseases and cancer: evaluation of underlying malignancy. *J Nephrol.* 2016;29(2):143–152.
109. Lefaucheur C, Stengel B, Nochy D, et al. Membranous nephropathy and cancer: epidemiologic evidence and determinants of high-risk cancer association. *Kidney Int.* 2006;70(8):1510–1517.
113. Beck Jr LH. Membranous nephropathy and malignancy. *Semin Nephrol.* 2010;30(6):635–644.
114. Qin W, Beck Jr LH, Zeng C, et al. Anti-phospholipase A2 receptor antibody in membranous nephropathy. *J Am Soc Nephrol.* 2011;22(6):1137–1143.
120. Audard V, Zhang SY, Copie-Bergman C, et al. Occurrence of minimal change nephrotic syndrome in classical Hodgkin lymphoma is closely related to the induction of c-mip in Hodgkin-Reed Sternberg cells and podocytes. *Blood.* 2010;115(18):3756–3762.
131. Rosner MH, Dalkin AC. Electrolyte disorders associated with cancer. *Adv Chron Kidney Dis.* 2014;21(1):7–17.
145. Rosner MH, Dalkin AC. Onco-nephrology: the pathophysiology and treatment of malignancy-associated hypercalcemia. *Clin J Am Soc Nephrol.* 2012;7(10):1722–1729.
162. Hingorani SR, Guthrie K, Batchelder A, et al. Acute renal failure after myeloablative hematopoietic cell transplant: incidence and risk factors. *Kidney Int.* 2005;67(1):272–277.
179. Keating GM. Defibrotide: a review of its use in severe hepatic veno-occlusive disease following haematopoietic stem cell transplantation. *Clin Drug Invest.* 2014;34(12):895–904.
209. Zuber J, Fakhouri F, Roumenina LT, Loirat C, Fremaux-Bacchi V, French Study Group for a HCG. Use of eculizumab for atypical haemolytic uraemic syndrome and C3 glomerulopathies. *Nat Rev Nephrol.* 2012;8(11):643–657.
229. Kogon A, Hingorani S. Acute kidney injury in hematopoietic cell transplantation. *Semin Nephrol.* 2010;30(6):615–626.
231. Ellis MJ, Parikh CR, Inrig JK, Kanbay M, Patel UD. Chronic kidney disease after hematopoietic cell transplantation: a systematic review. *Am J Transplant.* 2008;8(11):2378–2390.
241. Porta C, Cosmai L, Gallieni M, Pedrazzoli P, Malberti F. Renal effects of targeted anticancer therapies. *Nat Rev Nephrol.* 2015;11(6):354–370.
242. Perazella MA, Moeckel GW. Nephrotoxicity from chemotherapeutic agents: clinical manifestations, pathobiology, and prevention/therapy. *Semin Nephrol.* 2010;30(6):570–581.
246. Perazella MA, Izzedine H. New drug toxicities in the onco-nephrology world. *Kidney Int.* 2015;87(5):909–917.
249. Shirali AC, Perazella MA, Gettinger S. Association of acute interstitial nephritis with programmed cell death 1 inhibitor therapy in lung cancer patients. *Am J Kidney Dis.* 2016;68(2):287–291.
255. Arany I, Safirstein RL. Cisplatin nephrotoxicity. *Semin Nephrol.* 2003;23(5):460–464.
268. Perazella MA. Crystal-induced acute renal failure. *Am J Med.* 1999;106(4):459–465.
272. Wall SM, Johansen MJ, Molony DA, DuBose Jr TD, Jaffe N, Madden T. Effective clearance of methotrexate using high-flux hemodialysis membranes. *Am J Kidney Dis.* 1996;28(6):846–854.
273. Widemann BC, Schwartz S, Jayaprakash N, et al. Efficacy of glucarpidase (carboxypeptidase g2) in patients with acute kidney injury after high-dose methotrexate therapy. *Pharmacotherapy.* 2014;34(5):427–439.
275. Markowitz GS, Bombardier AS, Perazella MA. Drug-induced glomerular disease: direct cellular injury. *Clin J Am Soc Nephrol.* 2015;10(7):1291–1299.
279. Postow MA, Callahan MK, Wolchok JD. Immune checkpoint blockade in cancer therapy. *J Clin Oncol.* 2015;33(17):1974–1982.
286. Gurevich F, Perazella MA. Renal effects of anti-angiogenesis therapy: update for the internist. *Am J Med.* 2009;122(4):322–328.
287. Zhu X, Wu S, Dahut WL, Parikh CR. Risks of proteinuria and hypertension with bevacizumab, an antibody against vascular endothelial growth factor: systematic review and meta-analysis. *Am J Kidney Dis.* 2007;49(2):186–193.
313. Markowitz GS, Appel GB, Fine PL, et al. Collapsing focal segmental glomerulosclerosis following treatment with high-dose pamidronate. *J Am Soc Nephrol.* 2001;12(6):1164–1172.

REFERENCES

- Christiansen CF, Johansen MB, Langeberg WJ, Fryzek JP, Sørensen HT. Incidence of acute kidney injury in cancer patients: a Danish population-based cohort study. *Eur J Intern Med*. 2011;22(4):399–406.
- Liborio AB, Abreu KL, Silva Jr GB, et al. Predicting hospital mortality in critically ill cancer patients according to acute kidney injury severity. *Oncology*. 2011;80(3–4):160–166.
- Darmon M, Cioldi M, Thiery G, Schlemmer B, Azoulay E. Clinical review: specific aspects of acute renal failure in cancer patients. *Crit Care*. 2006;10(2):211.
- Lahoti A, Kantarjian H, Salahudeen AK, et al. Predictors and outcome of acute kidney injury in patients with acute myelogenous leukemia or high-risk myelodysplastic syndrome. *Cancer*. 2010;116(17):4063–4068.
- Soares M, Salluh JI, Carvalho MS, Darmon M, Rocco JR, Spector N. Prognosis of critically ill patients with cancer and acute renal dysfunction. *J Clin Oncol*. 2006;24(24):4003–4010.
- Gruss E, Bernis C, Tomas JF, et al. Acute renal failure in patients following bone marrow transplantation: prevalence, risk factors and outcome. *Am J Nephrol*. 1995;15(6):473–479.
- Parikh CR, McSweeney PA, Korular D, et al. Renal dysfunction in allogeneic hematopoietic cell transplantation. *Kidney Int*. 2002;62(2):566–573.
- Krishnappa V, Gupta M, Manu G, Kwatra S, Owusu OT, Raina R. Acute kidney injury in hematopoietic stem cell transplantation: a review. *Internet J Nephrol*. 2016;2016:5163789.
- Cho A, Lee JE, Kwon GY, et al. Post-operative acute kidney injury in patients with renal cell carcinoma is a potent risk factor for new-onset chronic kidney disease after radical nephrectomy. *Nephrol Dial Transplant*. 2011;26(11):3496–3501.
- Hu SL, Chang A, Perazella MA, et al. The nephrologist's tumor: basic biology and management of renal cell carcinoma. *J Am Soc Nephrol*. 2016;27(8):2227–2237.
- Campbell GA, Hu D, Okusa MD. Acute kidney injury in the cancer patient. *Adv Chron Kidney Dis*. 2014;21(1):64–71.
- Lameire N, Vanholder R, Van Biesen W, Benoit D. Acute kidney injury in critically ill cancer patients: an update. *Crit Care*. 2016;20(1):209.
- Rosner MH, Perazella MA. Acute kidney injury in patients with cancer. *N Engl J Med*. 2017;376(18):1770–1781.
- Mohty M, Malard F, Abecassis M, et al. Sinusoidal obstruction syndrome/veno-occlusive disease: current situation and perspectives—a position statement from the European Society for Blood and Marrow Transplantation (EBMT). *Bone Marrow Transplant*. 2015;50(6):781–789.
- Cairo MS, Bishop M. Tumor lysis syndrome: new therapeutic strategies and classification. *Br J Haematol*. 2004;127(1):3–11.
- Howard SC, Trifilio S, Gregory TK, Baxter N, McBride A. Tumor lysis syndrome in the era of novel and targeted agents in patients with hematologic malignancies: a systematic review. *Ann Hematol*. 2016;95(4):563–573.
- Wilson FP, Berns JS. Tumor lysis syndrome: new challenges and recent advances. *Adv Chron Kidney Dis*. 2014;21(1):18–26.
- Shimada M, Johnson RJ, May Jr WS, et al. A novel role for uric acid in acute kidney injury associated with tumor lysis syndrome. *Nephrol Dial Transplant*. 2009;24(10):2960–2964.
- Arnaud M, Loiseleur M, Vaganay C, et al. Tumor lysis syndrome and AKI: beyond crystal mechanisms. *J Am Soc Nephrol*. 2022;33(6):1154–1171.
- Uppal NN, Workeneh BT, Rondon-Berrios H, Jhaveri KD. Electrolyte and acid-base disorders associated with cancer immunotherapy. *Clin J Am Soc Nephrol*.
- Wanchoo R, Bernabe RC, Barrientos J, Jhaveri KD. Renal involvement in chronic lymphocytic leukemia. *Clin Kidney J*.
- Abernathy KM, Perciavalle MA, Gatwood KS, Chen H, Zakhari MM, Byrne M. Real-world analysis of tumor lysis syndrome in patients started on venetoclax combination for acute myeloid leukemia. *J Oncol Pharm Pract*.
- Cortes J, Moore JO, Maziarz RT, et al. Control of plasma uric acid in adults at risk for tumor lysis syndrome: efficacy and safety of rasburicase alone and rasburicase followed by allopurinol compared with allopurinol alone—results of a multicenter phase III study. *J Clin Oncol*. 2010;28(27):4207–4213.
- Sherwood GB, Paschal RD, Adamski J. Rasburicase-induced methemoglobinemia: case report, literature review, and proposed treatment algorithm. *Clin Case Rep*. 2016;4(4):315–319.
- Mir MA, Delamore IW. Metabolic disorders in acute myeloid leukemia. *Br J Haematol*. 1978;40(1):79–92.
- Kudose S, Cossey LN, Canetta PA, et al. Clinicopathologic spectrum of lysozyme-associated nephropathy. *Kidney Int Rep*. 2023;8(8):1585–1595. Published 2023 May 15.
- Gipe N, Leung N, Lasho T, et al. Spectrum of renal pathological findings in patients with chronic myelomonocytic leukemia and kidney injury. *Am J Hematol*. 2023;98(6):E148–E153.
- Lommatzsch SE, Bellizzi AM, Cathro HP, Rosner MH. Acute renal failure caused by renal infiltration by hematolymphoid malignancy. *Ann Diagn Pathol*. 2006;10(4):230–234.
- Lundberg WB, Cadman ED, Finch SC, Capizzi RL. Renal failure secondary to leukemic infiltration of the kidneys. *Am J Med*. 1977;62(4):636–642.
- Lohrmann HP, Adam W, Heymer B, Kubanek B. Microangiopathic hemolytic anemia in metastatic carcinoma. Report of eight cases. *Ann Intern Med*. 1973;79(3):368–375.
- Antman KH, Skarin AT, Mayer RJ, Hargreaves HK, Canellos GP. Microangiopathic hemolytic anemia and cancer: a review. *Medicine (Baltim)*. 1979;58(5):377–384.
- Izzedine H, Perazella MA. Thrombotic microangiopathy, cancer, and cancer drugs. *Am J Kidney Dis*. 2015;66(5):857–868.
- Chapin J, Shore T, Forsberg P, Desman G, Van Besien K, Laurence J. Hematopoietic transplant-associated thrombotic microangiopathy: case report and review of diagnosis and treatments. *Clin Adv Hematol Oncol*. 2014;12(9):565–573.
- Hutchison CA, Batuman V, Behrens J, et al. The pathogenesis and diagnosis of acute kidney injury in multiple myeloma. *Nat Rev Nephrol*. 2011;8(1):43–51.
- Sanders PW, Booker BB. Pathobiology of cast nephropathy from human Bence Jones proteins. *J Clin Invest*. 1992;89(2):630–639.
- Basnayake K, Ying WZ, Wang PX, Sanders PW. Immunoglobulin light chains activate tubular epithelial cells through redox signaling. *J Am Soc Nephrol*. 2010;21(7):1165–1173.
- Hutchison CA, Basnayake K, Cockwell P. Serum free light chain assessment in monoclonal gammopathy and kidney disease. *Nat Rev Nephrol*. 2009;5(11):621–628.
- Nasr SH, Valeri AM, Sethi S, et al. Clinicopathologic correlations in multiple myeloma: a case series of 190 patients with kidney biopsies. *Am J Kidney Dis*. 2012;59(6):786–794.
- Dimopoulos MA, Roussou M, Gavriatopoulou M, et al. Bortezomib-based triplets are associated with a high probability of dialysis independence and rapid renal recovery in newly diagnosed myeloma patients with severe renal failure or those requiring dialysis. *Am J Hematol*. 2016;91(5):499–502.
- Neri P, Bahlis NJ, Paba-Prada C, Richardson P. Treatment of relapsed/refractory multiple myeloma. *Cancer Treat Res*. 2016;169:169–194.
- Perazella MA, Finkel KW, American Society of Nephrology Oncology F. Paraprotein-related kidney disease: attack of the killer M proteins. *Clin J Am Soc Nephrol*. 2016;11(12):2256–2259.
- Luciano RL, Perazella MA. Crystalline-induced kidney disease: a case for urine microscopy. *Clin Kidney J*. 2015;8(2):131–136.
- Wong LM, Cleeve LK, Milner AD, Pitman AG. Malignant ureteral obstruction: outcomes after intervention. Have things changed? *J Urol*. 2007;178(1):178–183; discussion 183.
- van Bommel EFH, Jansen I, Hendriks TR, Aarnoudse A. Idiopathic retroperitoneal fibrosis: prospective evaluation of incidence and clinicoradiologic presentation. *Medicine (Baltim)*. 2009;88(4):193–201.
- Launay-Vacher V, Oudard S, Janus N, et al. Prevalence of Renal Insufficiency in cancer patients and implications for anticancer drug management: the renal insufficiency and anticancer medications (IRMA) study. *Cancer*. 2007;110(6):1376–1384.
- Launay-Vacher V. Epidemiology of chronic kidney disease in cancer patients: lessons from the IRMA study group. *Semin Nephrol*. 2010;30(6):548–556.
- Iff S, Craig JC, Turner R, et al. Reduced estimated GFR and cancer mortality. *Am J Kidney Dis*. 2014;63(1):23–30.
- Na SY, Sung JY, Chang JH, et al. Chronic kidney disease in cancer patients: an independent predictor of cancer-specific mortality. *Am J Nephrol*. 2011;33(2):121–130.

49. Yang Y, Li HY, Zhou Q, et al. Renal function and all-cause mortality risk among cancer patients. *Medicine (Baltim)*. 2016;95(20):e3728.
50. Sato M, Tateishi R, Yasunaga H, et al. Mortality and hemorrhagic complications associated with radiofrequency ablation for treatment of hepatocellular carcinoma in patients on hemodialysis for end-stage renal disease: a nationwide survey. *J Gastroenterol Hepatol*. 2017;32(11):1873–1878.
51. Kim MJ, Kang YU, Kim CS, et al. Proteinuria as a risk factor for mortality in patients with colorectal cancer. *Yonsei Med J*. 2013;54(5):1194–1201.
52. Konigsbrugge O, Lotsch F, Zielinski C, Pabinger I, Ay C. Chronic kidney disease in patients with cancer and its association with occurrence of venous thromboembolism and mortality. *Thromb Res*. 2014;134(1):44–49.
53. Cosmai L, Porta C, Gallieni M, Perazella MA. Onco-nephrology: a decalogue. *Nephrol Dial Transplant*. 2016;31(4):515–519.
54. Lowrance WT, Ordóñez J, Udaltsova N, Russo P, Go AS. CKD and the risk of incident cancer. *J Am Soc Nephrol*. 2014;25(10):2327–2334.
55. Stewart JH, Buccianti G, Agodoa L, et al. Cancers of the kidney and urinary tract in patients on dialysis for end-stage renal disease: analysis of data from the United States, Europe, and Australia and New Zealand. *J Am Soc Nephrol*. 2003;14(1):197–207.
56. Maisonneuve P, Agodoa L, Gellert R, et al. Cancer in patients on dialysis for end-stage renal disease: an international collaborative study. *Lancet*. 1999;354(9173):93–99.
57. Vajdic CM, McDonald SP, McCredie MR, et al. Cancer incidence before and after kidney transplantation. *JAMA*. 2006;296(23):2823–2831.
58. Birkeland SA, Lokkegaard H, Storm HH. Cancer risk in patients on dialysis and after renal transplantation. *Lancet*. 2000;355(9218):1886–1887.
59. Lin MY, Kuo MC, Hung CC, et al. Association of dialysis with the risks of cancers. *PLoS One*. 2015;10(4):e0122856.
60. Butler AM, Olshan AF, Kshirsagar AV, et al. Cancer incidence among US Medicare ESRD patients receiving hemodialysis, 1996–2009. *Am J Kidney Dis*. 2015;65(5):763–772.
61. Yanik EL, Clarke CA, Snyder JJ, Pfeiffer RM, Engels EA. Variation in cancer incidence among patients with ESRD during kidney function and nonfunction intervals. *J Am Soc Nephrol*. 2016;27(5):1495–1504.
62. Izzedine H, Perazella MA. Onco-nephrology: an appraisal of the cancer and chronic kidney disease links. *Nephrol Dial Transplant*. 2015;30(12):1979–1988.
63. Acuna SA, Fernandes KA, Daly C, et al. Cancer mortality among recipients of solid-organ transplantation in Ontario, Canada. *JAMA Oncol*. 2016;2(4):463–469.
64. Asch WS, Perazella MA. Cancer and mortality in solid-organ transplantation: preventable or inevitable? *Am J Kidney Dis*. 2016;68(6):839–842.
65. Aapro M, Launay-Vacher V. Importance of monitoring renal function in patients with cancer. *Cancer Treat Rev*. 2012;38(3):235–240.
66. Matzke GR, Aronoff GR, Atkinson Jr AJ, et al. Drug dosing consideration in patients with acute and chronic kidney disease—a clinical update from Kidney Disease: improving Global Outcomes (KDIGO). *Kidney Int*. 2011;80(11):1122–1137.
67. Wright JG, Boddy AV, Highley M, Fenwick J, McGill A, Calvert AH. Estimation of glomerular filtration rate in cancer patients. *Br J Cancer*. 2001;84(4):452–459.
68. Martin L, Chatelut E, Boneu A, et al. Improvement of the Cockcroft-Gault equation for predicting glomerular filtration in cancer patients. *Bull Cancer*. 1998;85(7):631–636.
69. Dooley MJ, Poole SG, Rischin D. Dosing of cytotoxic chemotherapy: impact of renal function estimates on dose. *Ann Oncol*. 2013;24(11):2746–2752.
70. Costa ESVT, et al. A prospective cross-sectional study estimated glomerular filtration rate from creatinine and cystatin C in adults with solid tumors. *Kidney Int*. 2022;101:607–614.
71. Titan SM, Lieske JC, Meeusen JW, et al. Performance of creatinine and cystatin-based equations on estimating measured GFR in people with hematological and solid cancers. *Clin J Am Soc Nephrol*. 2025;20(3):358–366.
72. Nolin TD. A synopsis of clinical pharmacokinetic alterations in advanced CKD. *Semin Dial*. 2015;28(4):325–329.
73. Naud J, Nolin TD, Leblond FA, Pichette V. Current understanding of drug disposition in kidney disease. *J Clin Pharmacol*. 2012;52(1 suppl):10S–22S.
74. Momper JD, Venkataramanan R, Nolin TD. Nonrenal drug clearance in CKD: searching for the path less traveled. *Adv Chron Kidney Dis*. 2010;17(5):384–391.
75. Janus N, Thariat J, Boulanger H, Deray G, Launay-Vacher V. Proposal for dosage adjustment and timing of chemotherapy in hemodialyzed patients. *Ann Oncol*. 2010;21(7):1395–1403.
76. Kintzel PE, Dorr RT. Anticancer drug renal toxicity and elimination: dosing guidelines for altered renal function. *Cancer Treat Rev*. 1995;21(1):33–64.
77. Yeung CK, Shen DD, Thummel KE, Himmelfarb J. Effects of chronic kidney disease and uremia on hepatic drug metabolism and transport. *Kidney Int*. 2014;85(3):522–528.
78. Perazella MA, Parikh C. Pharmacology. *Am J Kidney Dis*. 2005;46(6):1129–1139.
79. Perazella MA. Onco-nephrology: renal toxicities of chemotherapeutic agents. *Clin J Am Soc Nephrol*. 2012;7(10):1713–1721.
80. Salahudeen AK, Bonventre JV. Onconephrology: the latest frontier in the war against kidney disease. *J Am Soc Nephrol*. 2013;24(1):26–30.
81. Salahudeen AK, Doshi SM, Pawar T, Nowshad G, Lahoti A, Shah P. Incidence rate, clinical correlates, and outcomes of AKI in patients admitted to a comprehensive cancer center. *Clin J Am Soc Nephrol*. 2013;8(3):347–354.
82. Rowland M, Tozer TN. *Clinical Pharmacokinetics: Concepts and Applications*; 1989.
83. Reetze-Bonorden P, Bohler J, Keller E. Drug dosage in patients during continuous renal replacement therapy. Pharmacokinetic and therapeutic considerations. *Clin Pharmacokinet*. 1993;24(5):362–379.
84. Launay-Vacher V, Izzedine H, Baumelou A, Deray G. FHD: an index to evaluate drug elimination by hemodialysis. *Am J Nephrol*. 2005;25(4):342–351.
85. Wilson JMG, Jungner G, Organization WH. *Principles and Practice of Screening for Disease*; 1968.
86. Wong G, Howard K, Tong A, Craig JC. Cancer screening in people who have chronic disease: the example of kidney disease. *Semin Dial*. 2011;24(1):72–78.
87. *System USRD. Chapter 6: Mortality*. Accessed March 13, 2016. https://www.usrds.org/2015/view/v2_06.aspx.
88. Tang V, Boscardin WJ, Stijacic-Cenzer I, Lee SJ. Time to benefit for colorectal cancer screening: survival meta-analysis of flexible sigmoidoscopy trials. *BMJ*. 2015;350:h1662.
89. Holley JL. The importance of prognosis in cancer screening in patients with chronic kidney disease. *Semin Dial*. 2011;24(1):16–17.
90. Rosner MH. Cancer screening in patients with end-stage renal disease: an individualized approach. *J Onco-Nephrol*. 2017;1(1):36–41.
91. Holley JL. Screening, diagnosis, and treatment of cancer in long-term dialysis patients. *Clin J Am Soc Nephrol*. 2007;2(3):604–610.
92. Hu WH, Cajas-Monson LC, Eisenstein S, Parry L, Ramamoorthy S. Association of dialysis with adverse postoperative outcomes in colorectal cancer—an analysis of ACS-NSQIP. *Int J Colorectal Dis*. 2015;30(11):1557–1562.
93. Patel P, Henry LL, Ganti AK, Potti A. Clinical course of lung cancer in patients with chronic kidney disease. *Lung Cancer*. 2004;43(3):297–300.
94. Ciriaco P, Casiraghi M, Melloni G, et al. Pulmonary resection for non-small-cell lung cancer in patients on hemodialysis: clinical outcome and long-term results. *World J Surg*. 2005;29(11):1516–1519.
95. LeBrun CJ, Diehl LF, Abbott KC, Welch PG, Yuan CM. Life expectancy benefits of cancer screening in the end-stage renal disease population. *Am J Kidney Dis*. 2000;35(2):237–243.
96. Kajbaf S, Nichol G, Zimmerman D. Cancer screening and life expectancy of Canadian patients with kidney failure. *Nephrol Dial Transplant*. 2002;17(10):1786–1789.
97. Chertow GM, Paltiel AD, Owen Jr WF, Lazarus JM. Cost-effectiveness of cancer screening in end-stage renal disease. *Arch Intern Med*. 1996;156(12):1345–1350.
98. Wong G, Howard K, Chapman JR, Craig JC. Cost-effectiveness of breast cancer screening in women on dialysis. *Am J Kidney Dis*. 2008;52(5):916–929.

99. Carlos CA, McCulloch CE, Hsu CY, et al. Colon cancer screening among patients receiving dialysis in the United States: are we choosing wisely? *J Am Soc Nephrol*. 2017;28(8):2521–2528.
100. Rockville M. *Shared Decision-Making in the Appropriate Initiation of and Withdrawal from Dialysis*. Rockville, MD: The Renal Physicians Association; 2010.
101. Cohen LM, Ruthazer R, Moss AH, Germain MJ. Predicting six-month mortality for patients who are on maintenance hemodialysis. *Clin J Am Soc Nephrol*. 2010;5(1):72–79.
102. Thamer M, Kaufman JS, Zhang Y, Zhang Q, Cotter DJ, Bang H. Predicting early death among elderly dialysis patients: development and validation of a risk score to assist shared decision making for dialysis initiation. *Am J Kidney Dis*. 2015;66(6):1024–1032.
103. Wagner M, Ansell D, Kent DM, et al. Predicting mortality in incident dialysis patients: an analysis of the United Kingdom Renal Registry. *Am J Kidney Dis*. 2011;57(6):894–902.
104. Lee JC, Yamauchi H, Hopper Jr J. The association of cancer and the nephrotic syndrome. *Ann Intern Med*. 1966;64(1):41–51.
105. Pani A, Porta C, Cosmai L, et al. Glomerular diseases and cancer: evaluation of underlying malignancy. *J Nephrol*. 2016;29(2):143–152.
106. Pai P, Bone J, McDicken I, Bell G. Solid tumour and glomerulopathy. *QJM*. 1996;89(5):361–368.
107. Ronco PM. Paraneoplastic glomerulopathies: new insights into an old entity. *Kidney Int*. 1999;56(1):355–377.
108. Bacchetta J, Juillard L, Cochat P, Droz JP. Paraneoplastic glomerular diseases and malignancies. *Crit Rev Oncol Hematol*. 2009;70(1):39–58.
109. Lefaucheur C, Stengel B, Nochy D, et al. Membranous nephropathy and cancer: epidemiologic evidence and determinants of high-risk cancer association. *Kidney Int*. 2006;70(8):1510–1517.
110. Rihova Z, Honsova E, Merta M, et al. Secondary membranous nephropathy—one center experience. *Ren Fail*. 2005;27(4):397–402.
111. Cambier JF, Ronco P. Onco-nephrology: glomerular diseases with cancer. *Clin J Am Soc Nephrol*. 2012;7(10):1701–1712.
112. Beck Jr LH, Bonegio RG, Lambeau G, et al. M-type phospholipase A2 receptor as target antigen in idiopathic membranous nephropathy. *N Engl J Med*. 2009;361(1):11–21.
113. Beck Jr LH. Membranous nephropathy and malignancy. *Semin Nephrol*. 2010;30(6):635–644.
114. Qin W, Beck Jr LH, Zeng C, et al. Anti-phospholipase A2 receptor antibody in membranous nephropathy. *J Am Soc Nephrol*. 2011;22(6):1137–1143.
115. Group KGW. KDIGO clinical practice guideline for glomerulonephritis. *Kidney Int Suppl*. 2012;2(2):186–197.
116. Bjornekleit R, Vikse BE, Svarstad E, et al. Long-term risk of cancer in membranous nephropathy patients. *Am J Kidney Dis*. 2007;50(3):396–403.
117. Jhaveri KD, Shah HH, Calderon K, Campenot ES, Radhakrishnan J. Glomerular diseases seen with cancer and chemotherapy: a narrative review. *Kidney Int*. 2013;84(1):34–44.
118. Plager J, Stutzman L. Acute nephrotic syndrome as a manifestation of active Hodgkin's Disease. Report of four cases and review of the literature. *Am J Med*. 1971;50(1):56–66.
119. Audard V, Larousserie F, Grimbert P, et al. Minimal change nephrotic syndrome and classical Hodgkin's lymphoma: report of 21 cases and review of the literature. *Kidney Int*. 2006;69(12):2251–2260.
120. Audard V, Zhang SY, Copie-Bergman C, et al. Occurrence of minimal change nephrotic syndrome in classical Hodgkin lymphoma is closely related to the induction of c-mip in Hodgkin-Reed Sternberg cells and podocytes. *Blood*. 2010;115(18):3756–3762.
121. Seney Jr FD, Federgreen WR, Stein H, Kashgarian M. A review of nephrotic syndrome associated with chronic lymphocytic leukemia. *Arch Intern Med*. 1986;146(1):137–141.
122. Moulin B, Ronco PM, Mougnot B, Francois A, Fillastre JP, Mignon F. Glomerulonephritis in chronic lymphocytic leukemia and related B-cell lymphomas. *Kidney Int*. 1992;42(1):127–135.
123. Ivanyi B. Frequency of light chain deposition nephropathy relative to renal amyloidosis and Bence Jones cast nephropathy in a necropsy study of patients with myeloma. *Arch Pathol Lab Med*. 1990;114(9):986–987.
124. Ronco P, Plaisier E, Mougnot B, Aucouturier P. Immunoglobulin light (heavy)-chain deposition disease: from molecular medicine to pathophysiology-driven therapy. *Clin J Am Soc Nephrol*. 2006;1(6):1342–1350.
125. Doshi SM, Shah P, Lei X, Lahoti A, Salahudeen AK. Hyponatremia in hospitalized cancer patients and its impact on clinical outcomes. *Am J Kidney Dis*. 2012;59(2):222–228.
126. Berghmans T, Paesmans M, Body JJ. A prospective study on hyponatraemia in medical cancer patients: epidemiology, aetiology and differential diagnosis. *Support Care Cancer*. 2000;8(3):192–197.
127. Gill G, Huda B, Boyd A, et al. Characteristics and mortality of severe hyponatraemia—a hospital-based study. *Clin Endocrinol*. 2006;65(2):246–249.
128. Dhaliwal HS, Rohatiner AZ, Gregory W, et al. Combination chemotherapy for intermediate and high grade non-Hodgkin's lymphoma. *Br J Cancer*. 1993;68(4):767–774.
129. Osterlind K. Factors confounding evaluation of treatment effect in lung cancer. *Lung Cancer*. 1994;10(suppl 1):S97–S103.
130. Gross AJ, Steinberg SM, Reilly JG, et al. Atrial natriuretic factor and arginine vasopressin production in tumor cell lines from patients with lung cancer and their relationship to serum sodium. *Cancer Res*. 1993;53(1):67–74.
131. Rosner MH, Dalkin AC. Electrolyte disorders associated with cancer. *Adv Chron Kidney Dis*. 2014;21(1):7–17.
132. List AF, Hainsworth JD, Davis BW, Hande KR, Greco FA, Johnson DH. The syndrome of inappropriate secretion of antidiuretic hormone (SIADH) in small-cell lung cancer. *J Clin Oncol*. 1986;4(8):1191–1198.
133. Ferlito A, Rinaldo A, Devaney KO. Syndrome of inappropriate antidiuretic hormone secretion associated with head neck cancers: review of the literature. *Ann Otol Rhinol Laryngol*. 1997;106(10 Pt 1):878–883.
134. Comis RL, Miller M, Ginsberg SJ. Abnormalities in water homeostasis in small cell anaplastic lung cancer. *Cancer*. 1980;45(9):2414–2421.
135. Padfield PL, Morton JJ, Brown JJ, et al. Plasma arginine vasopressin in the syndrome of antidiuretic hormone excess associated with bronchogenic carcinoma. *Am J Med*. 1976;61(6):825–831.
136. Berghmans T. Hyponatremia related to medical anticancer treatment. *Support Care Cancer*. 1996;4(5):341–350.
137. Verbalis JG, Goldsmith SR, Greenberg A, Schrier RW, Sterns RH. Hyponatremia treatment guidelines 2007: expert panel recommendations. *Am J Med*. 2007;120(11 suppl 1):S1–S21.
138. Verbalis JG, Adler S, Schrier RW, et al. Efficacy and safety of oral tolvaptan therapy in patients with the syndrome of inappropriate antidiuretic hormone secretion. *Eur J Endocrinol*. 2011;164(5):725–732.
139. O'Regan S, Carson S, Chesney RW, Drummond KN. Electrolyte and acid-base disturbances in the management of leukemia. *Blood*. 1977;49(3):345–353.
140. Naparstek Y, Gutman A. Case report: spurious hypokalemia in myeloproliferative disorders. *Am J Med Sci*. 1984;288(4):175–177.
141. Isidori AM, Kaltsas GA, Grossman AB. Ectopic ACTH syndrome. *Front Horm Res*. 2006;35:143–156.
142. Milonis HJ, Bourantas CL, Siamopoulos KC, Elisaf MS. Acid-base and electrolyte abnormalities in patients with acute leukemia. *Am J Hematol*. 1999;62(4):201–207.
143. Mason DY, Howes DT, Taylor CR, Ross BD. Effect of human lysozyme (muramidase) on potassium handling by the perfused rat kidney. A mechanism for renal damage in human monocytic leukaemia. *J Clin Pathol*. 1975;28(9):722–727.
144. Sevastos N, Theodossiadis G, Archimandritis AJ. Pseudohyperkalemia in serum: a new insight into an old phenomenon. *Clin Med Res*. 2008;6(1):30–32.
145. Rosner MH, Dalkin AC. Onco-nephrology: the pathophysiology and treatment of malignancy-associated hypercalcemia. *Clin J Am Soc Nephrol*. 2012;7(10):1722–1729.
146. Vassilopoulou-Sellin R, Newman BM, Taylor SH, Guinee VF. Incidence of hypercalcemia in patients with malignancy referred to a comprehensive cancer center. *Cancer*. 1993;71(4):1309–1312.
147. Soki FN, Park SI, McCauley LK. The multifaceted actions of PTHrP in skeletal metastasis. *Future Oncol*. 2012;8(7):803–817.
148. Abildgaard N, Brixen K, Eriksen EF, Kristensen JE, Nielsen JL, Heickendorff L. Sequential analysis of biochemical markers of bone resorption and bone densitometry in multiple myeloma. *Haematologica*. 2004;89(5):567–577.

149. Seymour JF, Gagel RF. Calcitriol: the major humoral mediator of hypercalcemia in Hodgkin's disease and non-Hodgkin's lymphomas. *Blood*. 1993;82(5):1383–1394.
150. Reagan P, Pani A, Rosner MH. Approach to diagnosis and treatment of hypercalcemia in a patient with malignancy. *Am J Kidney Dis*. 2014;63(1):141–147.
151. Toka HR, Al-Romaih K, Koshy JM, et al. Deficiency of the calcium-sensing receptor in the kidney causes parathyroid hormone-independent hypocalciuria. *J Am Soc Nephrol*. 2012;23(11):1879–1890.
152. LeGrand SB, Leskusi D, Zama I. Narrative review: furosemide for hypercalcemia: an unproven yet common practice. *Ann Intern Med*. 2008;149(4):259–263.
153. Kyle RA, Yee GC, Somerfield MR, et al. American Society of Clinical Oncology 2007 clinical practice guideline update on the role of bisphosphonates in multiple myeloma. *J Clin Oncol*. 2007;25(17):2464–2472.
154. Castellano D, Sepulveda JM, Garcia-Escobar I, Rodriguez-Antolin A, Sundlov A, Cortes-Funes H. The role of RANK-ligand inhibition in cancer: the story of denosumab. *Oncologist*. 2011;16(2):136–145.
155. Brown-Glaberman U, Stopeck AT. Role of denosumab in the management of skeletal complications in patients with bone metastases from solid tumors. *Biologics*. 2012;6:89–99.
156. Bilezikian JP. Management of acute hypercalcemia. *N Engl J Med*. 1992;326(18):1196–1203.
157. Chong WH, Molinolo AA, Chen CC, Collins MT. Tumor-induced osteomalacia. *Endocr Relat Cancer*. 2011;18(3):R53–R77.
158. Andreopoulou PMC, Reynolds J. Multimodality diagnosis and treatment of tumor induced osteomalacia. *Endocr Rev*. 2010;31(suppl 1):0R08–6S49.
159. Fairley J, Glassford NJ, Zhang L, Bellomo R. Magnesium status and magnesium therapy in critically ill patients: a systematic review. *J Crit Care*. 2015;30(6):1349–1358.
160. Lajer H, Daugaard G. Cisplatin and hypomagnesemia. *Cancer Treat Rev*. 1999;25(1):47–58.
161. Fakih M, Vincent M. Adverse events associated with anti-EGFR therapies for the treatment of metastatic colorectal cancer. *Curr Oncol*. 2010;17(suppl 1):S18–S30.
162. Hingorani SR, Guthrie K, Batchelder A, et al. Acute renal failure after myeloablative hematopoietic cell transplant: incidence and risk factors. *Kidney Int*. 2005;67(1):272–277.
163. Parikh CR, Coca SG. Acute renal failure in hematopoietic cell transplantation. *Kidney Int*. 2006;69(3):430–435.
164. Go AS, Chertow GM, Fan D, McCulloch CE, Hsu CY. Chronic kidney disease and the risks of death, cardiovascular events, and hospitalization. *N Engl J Med*. 2004;351(13):1296–1305.
165. Zager RA, O'Quigley J, Zager BK, et al. Acute renal failure following bone marrow transplantation: a retrospective study of 272 patients. *Am J Kidney Dis*. 1989;13(3):210–216.
166. Hahn T, Rondeau C, Shaikat A, et al. Acute renal failure requiring dialysis after allogeneic blood and marrow transplantation identifies very poor prognosis patients. *Bone Marrow Transplant*. 2003;32(4):405–410.
167. Merouani A, Shpall EJ, Jones RB, Archer PG, Schrier RW. Renal function in high dose chemotherapy and autologous hematopoietic cell support treatment for breast cancer. *Kidney Int*. 1996;50(3):1026–1031.
168. Parikh CR, Sandmaier BM, Storb RF, et al. Acute renal failure after nonmyeloablative hematopoietic cell transplantation. *J Am Soc Nephrol*. 2004;15(7):1868–1876.
169. Parikh CR, Yarlagadda SG, Storer B, Sorror M, Storb R, Sandmaier B. Impact of acute kidney injury on long-term mortality after nonmyeloablative hematopoietic cell transplantation. *Biol Blood Marrow Transplant*. 2008;14(3):309–315.
170. Kersting S, Koomans HA, Hene RJ, Verdonck LF. Acute renal failure after allogeneic myeloablative stem cell transplantation: retrospective analysis of incidence, risk factors and survival. *Bone Marrow Transplant*. 2007;39(6):359–365.
171. Lopes JA, Jorge S, Silva S, et al. Acute renal failure following myeloablative autologous and allogeneic hematopoietic cell transplantation. *Bone Marrow Transplant*. 2006;38(10):707.
172. Michael M, Kuehnle I, Goldstein SL. Fluid overload and acute renal failure in pediatric stem cell transplant patients. *Pediatr Nephrol*. Jan 2004;19(1):91–95.
173. Hazar V, Gungor O, Guven AG, et al. Renal function after hematopoietic stem cell transplantation in children. *Pediatr Blood Cancer*. 2009;53(2):197–202.
174. Satwani P, Bavishi S, Jin Z, et al. Risk factors associated with kidney injury and the impact of kidney injury on overall survival in pediatric recipients following allogeneic stem cell transplant. *Biol Blood Marrow Transplant*. 2011;17(10):1472–1480.
175. Parikh CR, Schrier RW, Storer B, et al. Comparison of ARF after myeloablative and nonmyeloablative hematopoietic cell transplantation. *Am J Kidney Dis*. 2005;45(3):502–509.
176. Ando M, Mori J, Ohashi K, et al. A comparative assessment of the RIFLE, AKIN and conventional criteria for acute kidney injury after hematopoietic SCT. *Bone Marrow Transplant*. 2010;45(9):1427–1434.
177. Beyzadeoglu M, Arpacı F, Surenkok S, et al. Acute renal toxicity of 2 conditioning regimens in patients undergoing autologous peripheral blood stem-cell transplantation. Total body irradiation-cyclophosphamide versus ifosfamide, carboplatin, etoposide. *Saudi Med J*. 2008;29(6):832–836.
178. Wan L, Bagshaw SM, Langenberg C, Saotome T, May C, Bellomo R. Pathophysiology of septic acute kidney injury: what do we really know? *Crit Care Med*. 2008;36(4 suppl):S198–S203.
179. Keating GM. Defibrotide: a review of its use in severe hepatic veno-occlusive disease following haematopoietic stem cell transplantation. *Clin Drug Invest*. 2014;34(12):895–904.
180. Zager RA. Acute renal failure in the setting of bone marrow transplantation. *Kidney Int*. 1994;46(5):1443–1458.
181. Zager RA. Studies of mechanisms and protective maneuvers in myoglobinuric acute renal injury. *Lab Invest*. 1989;60(5):619–629.
182. Zager RA, Foerder C, Bredl C. The influence of mannitol on myoglobinuric acute renal failure: functional, biochemical, and morphological assessments. *J Am Soc Nephrol*. 1991;2(4):848–855.
183. Coppel JA, Richardson PG, Soiffer R, et al. Hepatic veno-occlusive disease following stem cell transplantation: incidence, clinical course, and outcome. *Biol Blood Marrow Transplant*. 2010;16(2):157–168.
184. Bearman SI. The syndrome of hepatic veno-occlusive disease after marrow transplantation. *Blood*. 1995;85(11):3005–3020.
185. McDonald GB, Hinds MS, Fisher LD, et al. Venous-occlusive disease of the liver and multiorgan failure after bone marrow transplantation: a cohort study of 355 patients. *Ann Intern Med*. 1993;118(4):255–267.
186. Jones RJ, Lee KS, Beschoner WE, et al. Venous-occlusive disease of the liver following bone marrow transplantation. *Transplantation*. 1987;44(6):778–783.
187. Attal M, Huguet F, Rubie H, et al. Prevention of hepatic veno-occlusive disease after bone marrow transplantation by continuous infusion of low-dose heparin: a prospective, randomized trial. *Blood*. 1992;79(11):2834–2840.
188. Gluckman E, Jolivet I, Scrobocaci ML, et al. Use of prostaglandin E1 for prevention of liver veno-occlusive disease in leukaemic patients treated by allogeneic bone marrow transplantation. *Br J Haematol*. 1990;74(3):277–281.
189. Bianco JA, Appelbaum FR, Nemunaitis J, et al. Phase II trial of pentoxifylline for the prevention of transplant-related toxicities following bone marrow transplantation. *Blood*. 1991;78(5):1205–1211.
190. Richardson PG, Murakami C, Jin Z, et al. Multi-institutional use of defibrotide in 88 patients after stem cell transplantation with severe veno-occlusive disease and multisystem organ failure: response without significant toxicity in a high-risk population and factors predictive of outcome. *Blood*. 2002;100(13):4337–4343.
191. Chopra R, Eaton JD, Grassi A, et al. Defibrotide for the treatment of hepatic veno-occlusive disease: results of the European compassionate-use study. *Br J Haematol*. 2000;111(4):1122–1129.
192. Batts ED, Lazarus HM. Diagnosis and treatment of transplantation-associated thrombotic microangiopathy: real progress or are we still waiting? *Bone Marrow Transplant*. 2007;40(8):709–719.
193. Siami K, Kojouri K, Swisher KK, Selby GB, George JN, Laszik ZG. Thrombotic microangiopathy after allogeneic hematopoietic stem cell transplantation: an autopsy study. *Transplantation*. 2008;85(1):22–28.
194. Laskin BL, Goebel J, Davies SM, Jodele S. Small vessels, big trouble in the kidneys and beyond: hematopoietic stem cell

- transplantation-associated thrombotic microangiopathy. *Blood*. 2011;118(6):1452–1462.
195. Laskin BL, Goebel J, Davies SM, et al. Early clinical indicators of transplant-associated thrombotic microangiopathy in pediatric neuroblastoma patients undergoing auto-SCT. *Bone Marrow Transplant*. 2011;46(5):682–689.
 196. Ho VT, Cutler C, Carter S, et al. Blood and marrow transplant clinical trials network toxicity committee consensus summary: thrombotic microangiopathy after hematopoietic stem cell transplantation. *Biol Blood Marrow Transplant*. 2005;11(8):571–575.
 197. Ruutu T, Barosi G, Benjamin RJ, et al. Diagnostic criteria for hematopoietic stem cell transplant-associated microangiopathy: results of a consensus process by an International Working Group. *Haematologica*. 2007;92(1):95–100.
 198. Cho BS, Yahng SA, Lee SE, et al. Validation of recently proposed consensus criteria for thrombotic microangiopathy after allogeneic hematopoietic stem-cell transplantation. *Transplantation*. 2010;90(8):918–926.
 199. Changsirikulchai S, Myerson D, Guthrie KA, McDonald GB, Alpers CE, Hingorani SR. Renal thrombotic microangiopathy after hematopoietic cell transplant: role of GVHD in pathogenesis. *Clin J Am Soc Nephrol*. 2009;4(2):345–353.
 200. George JN. Hematopoietic stem cell transplantation-associated thrombotic microangiopathy: defining a disorder. *Bone Marrow Transplant*. 2008;41(11):917–918.
 201. Kojouri K, George JN. Thrombotic microangiopathy following allogeneic hematopoietic stem cell transplantation. *Curr Opin Oncol*. 2007;19(2):148–154.
 202. Tichelli A, Passweg J, Wojcik D, et al. Late cardiovascular events after allogeneic hematopoietic stem cell transplantation: a retrospective multicenter study of the Late Effects Working Party of the European Group for Blood and Marrow Transplantation. *Haematologica*. 2008;93(8):1203–1210.
 203. Takatsuka H, Nakajima T, Nomura K, et al. Changes of clotting factors (7, 9 and 10) and hepatocyte growth factor in patients with thrombotic microangiopathy after bone marrow transplantation. *Clin Transplant*. 2006;20(5):640–643.
 204. Takatsuka H, Nakajima T, Nomura K, et al. Heparin cofactor II as a predictor of thrombotic microangiopathy after bone marrow transplantation. *Hematology*. 2006;11(2):101–103.
 205. Jodele S, Bleesing JJ, Mehta PA, et al. Successful early intervention for hyperacute transplant-associated thrombotic microangiopathy following pediatric hematopoietic stem cell transplantation. *Pediatr Transplant*. 2012;16(2):E39–E42.
 206. Mii A, Shimizu A, Kaneko T, et al. Renal thrombotic microangiopathy associated with chronic graft-versus-host disease after allogeneic hematopoietic stem cell transplantation. *Pathol Int*. 2011;61(9):518–527.
 207. Mii A, Shimizu A, Masuda Y, et al. Renal thrombotic microangiopathy associated with chronic humoral graft versus host disease after hematopoietic stem cell transplantation. *Pathol Int*. 2011;61(1):34–41.
 208. Caprioli J, Noris M, Brioschi S, et al. Genetics of HUS: the impact of MCP, CFH, and IF mutations on clinical presentation, response to treatment, and outcome. *Blood*. 2006;108(4):1267–1279.
 209. Zuber J, Fakhouri F, Roumenina LT, Loirat C, Fremeaux-Bacchi V, French Study Group for a HCG. Use of eculizumab for atypical haemolytic uraemic syndrome and C3 glomerulopathies. *Nat Rev Nephrol*. 2012;8(11):643–657.
 210. Kanamori H, Maruta A, Sasaki S, et al. Diagnostic value of hemostatic parameters in bone marrow transplant-associated thrombotic microangiopathy. *Bone Marrow Transplant*. 1998;21(7):705–709.
 211. Matsuda Y, Hara J, Osugi Y, et al. Serum levels of soluble adhesion molecules in stem cell transplantation-related complications. *Bone Marrow Transplant*. 2001;27(9):977–982.
 212. Choi CM, Schmaier AH, Snell MR, Lazarus HM. Thrombotic microangiopathy in haematopoietic stem cell transplantation: diagnosis and treatment. *Drugs*. 2009;69(2):183–198.
 213. Kennedy GA, Kearey N, Bleakley S, Butler J, Mudie K, Durrant S. Transplantation-associated thrombotic microangiopathy: effect of concomitant GVHD on efficacy of therapeutic plasma exchange. *Bone Marrow Transplant*. 2010;45(4):699–704.
 214. Cutler C, Henry NL, Magee C, et al. Sirolimus and thrombotic microangiopathy after allogeneic hematopoietic stem cell transplantation. *Biol Blood Marrow Transplant*. 2005;11(7):551–557.
 215. Hahn T, Alam AR, Lawrence D, et al. Thrombotic microangiopathy after allogeneic blood and marrow transplantation is associated with dose-intensive myeloablative conditioning regimens, unrelated donor, and methylprednisolone T-cell depletion. *Transplantation*. 2004;78(10):1515–1522.
 216. Uderzo C, Bonanomi S, Busca A, et al. Risk factors and severe outcome in thrombotic microangiopathy after allogeneic hematopoietic stem cell transplantation. *Transplantation*. 2006;82(5):638–644.
 217. Worel N, Greinix HT, Leitner G, et al. ABO-incompatible allogeneic hematopoietic stem cell transplantation following reduced-intensity conditioning: close association with transplant-associated microangiopathy. *Transfus Apher Sci*. 2007;36(3):297–304.
 218. Willems E, Baron F, Seidel L, Frere P, Fillet G, Beguin Y. Comparison of thrombotic microangiopathy after allogeneic hematopoietic cell transplantation with high-dose or nonmyeloablative conditioning. *Bone Marrow Transplant*. 2010;45(4):689–693.
 219. Erdbruegger U, Woywodt A, Kirsch T, Haller H, Haubitz M. Circulating endothelial cells as a prognostic marker in thrombotic microangiopathy. *Am J Kidney Dis*. 2006;48(4):564–570.
 220. Cho BS, Min CK, Eom KS, et al. Clinical impact of thrombotic microangiopathy on the outcome of patients with acute graft-versus-host disease after allogeneic hematopoietic stem cell transplantation. *Bone Marrow Transplant*. 2008;41(9):813–820.
 221. Oran B, Donato M, Aleman A, et al. Transplant-associated microangiopathy in patients receiving tacrolimus following allogeneic stem cell transplantation: risk factors and response to treatment. *Biol Blood Marrow Transplant*. 2007;13(4):469–477.
 222. Marr H, McDonald EJ, Merriman E, et al. Successful treatment of transplant-associated microangiopathy with rituximab. *N Z Med J*. 2009;122(1292):72–74.
 223. Naina HV, Gertz MA, Elliott MA. Thrombotic microangiopathy during peripheral blood stem cell mobilization. *J Clin Apher*. 2009;24(6):259–261.
 224. Carella AM, D'Arena G, Greco MM, Nobile M, Cascavilla N. Rituximab for allo-SCT-associated thrombotic thrombocytopenic purpura. *Bone Marrow Transplant*. 2008;41(12):1063–1065.
 225. Au WY, Ma ES, Lee TL, et al. Successful treatment of thrombotic microangiopathy after haematopoietic stem cell transplantation with rituximab. *Br J Haematol*. 2007;137(5):475–478.
 226. Hingorani S, Finn LS, Pao E, et al. Urinary elafin and kidney injury in hematopoietic cell transplant recipients. *Clin J Am Soc Nephrol*. 2015;10(1):12–20.
 227. Mori J, Ohashi K, Yamaguchi T, et al. Risk assessment for acute kidney injury after allogeneic hematopoietic stem cell transplantation based on Acute Kidney Injury Network criteria. *Intern Med*. 2012;51(16):2105–2110.
 228. Lopes JA, Jorge S. Acute kidney injury following HCT: incidence, risk factors and outcome. *Bone Marrow Transplant*. 2011;46(11):1399–1408.
 229. Kogon A, Hingorani S. Acute kidney injury in hematopoietic cell transplantation. *Semin Nephrol*. 2010;30(6):615–626.
 230. Weiss AS, Sandmaier BM, Storer B, Storb R, McSweeney PA, Parikh CR. Chronic kidney disease following non-myeloablative hematopoietic cell transplantation. *Am J Transplant*. 2006;6(1):89–94.
 231. Ellis MJ, Parikh CR, Inrig JK, Kanbay M, Patel UD. Chronic kidney disease after hematopoietic cell transplantation: a systematic review. *Am J Transplant*. 2008;8(11):2378–2390.
 232. Choi M, Sun CL, Kurian S, et al. Incidence and predictors of delayed chronic kidney disease in long-term survivors of hematopoietic cell transplantation. *Cancer*. 2008;113(7):1580–1587.
 233. Cohen EP, Piering WF, Kabler-Babbitt C, Moulder JE. End-stage renal disease (ESRD) after bone marrow transplantation: poor survival compared to other causes of ESRD. *Nephron*. 1998;79(4):408–412.
 234. Van Why SK, Friedman AL, Wei LJ, Hong R. Renal insufficiency after bone marrow transplantation in children. *Bone Marrow Transplant*. 1991;7(5):383–388.
 235. Hingorani S. Chronic kidney disease in long-term survivors of hematopoietic cell transplantation: epidemiology, pathogenesis, and treatment. *J Am Soc Nephrol*. 2006;17(7):1995–2005.

236. Schriber JR, Herzig GP. Transplantation-associated thrombotic thrombocytopenic purpura and hemolytic uremic syndrome. *Semin Hematol*. 1997;34(2):126–133.
237. Hingorani S, Guthrie KA, Schoch G, Weiss NS, McDonald GB. Chronic kidney disease in long-term survivors of hematopoietic cell transplant. *Bone Marrow Transplant*. 2007;39(4):223–229.
238. Chanswangphuwana C, Townamchai N, Intragumtornchai T, Bunworasate U. Glomerular diseases associated with chronic graft-versus-host disease after allogeneic peripheral blood stem cell transplantation: case reports. *Transplant Proc*. 2014;46(10):3616–3619.
239. Hu SL. The role of graft-versus-host disease in haematopoietic cell transplantation-associated glomerular disease. *Nephrol Dial Transplant*. 2011;26(6):2025–2031.
240. O'Donnell PH, Swanson K, Josephson MA, et al. BK virus infection is associated with hematuria and renal impairment in recipients of allogeneic hematopoietic stem cell transplants. *Biol Blood Marrow Transplant*. 2009;15(9):1038–1048 e1.
241. Porta C, Cosmai L, Gallieni M, Pedrazzoli P, Malberti F. Renal effects of targeted anticancer therapies. *Nat Rev Nephrol*. 2015;11(6):354–370.
242. Perazella MA, Moeckel GW. Nephrotoxicity from chemotherapeutic agents: clinical manifestations, pathobiology, and prevention/therapy. *Semin Nephrol*. 2010;30(6):570–581.
243. Shirali AC, Perazella MA. Tubulointerstitial injury associated with chemotherapeutic agents. *Adv Chron Kidney Dis*. 2014;21(1):56–63.
244. Eremina V, Jefferson JA, Kowalewska J, et al. VEGF inhibition and renal thrombotic microangiopathy. *N Engl J Med*. 2008;358(11):1129–1136.
245. Teuma C, Perier-Muzet M, Pelletier S, et al. New insights into renal toxicity of the B-Raf inhibitor, vemurafenib, in patients with metastatic melanoma. *Cancer Chemother Pharmacol*. 2016;78(2):419–426.
246. Perazella MA, Izzedine H. New drug toxicities in the onconephrology world. *Kidney Int*. 2015;87(5):909–917.
247. Izzedine H, El-Fekih RK, Perazella MA. The renal effects of ALK inhibitors. *Invest N Drugs*. 2016;34(5):643–649.
248. Cortazar FB, Marrone KA, Troxell ML, et al. Clinicopathological features of acute kidney injury associated with immune checkpoint inhibitors. *Kidney Int*. 2016;90(3):638–647.
249. Shirali AC, Perazella MA, Gettinger S. Association of acute interstitial nephritis with programmed cell death 1 inhibitor therapy in lung cancer patients. *Am J Kidney Dis*. 2016;68(2):287–291.
250. Izzedine H, Isnard-Bagnis C, Launay-Vacher V, et al. Gemcitabine-induced thrombotic microangiopathy: a systematic review. *Nephrol Dial Transplant*. 2006;21(11):3038–3045.
251. Zakarija A, Bennett C. Drug-induced thrombotic microangiopathy. *Semin Thromb Hemost*. 2005;31(6):681–690.
252. Humphreys BD, Sharman JP, Henderson JM, et al. Gemcitabine-associated thrombotic microangiopathy. *Cancer*. 2004;100(12):2664–2670.
253. Glezerman I, Kris MG, Miller V, Seshan S, Flombaum CD. Gemcitabine nephrotoxicity and hemolytic uremic syndrome: report of 29 cases from a single institution. *Clin Nephrol*. 2009;71(2):130–139.
254. Leibbrandt ME, Wolfgang GH, Metz AL, Ozobia AA, Haskins JR. Critical subcellular targets of cisplatin and related platinum analogs in rat renal proximal tubule cells. *Kidney Int*. 1995;48(3):761–770.
255. Arany I, Safirstein RL. Cisplatin nephrotoxicity. *Semin Nephrol*. 2003;23(5):460–464.
256. Santini V, Giles FJ. The potential of amifostine: from cytoprotectant to therapeutic agent. *Haematologica (Roma)*. 1999;84(11):1035–1042.
257. Skinner R. Strategies to prevent nephrotoxicity of anticancer drugs. *Curr Opin Oncol*. 1995;7(4):310–315.
258. McDonald BR, Kirmani S, Vasquez M, Mehta RL. Acute renal failure associated with the use of intraperitoneal carboplatin: a report of two cases and review of the literature. *Am J Med*. 1991;90(3):386–391.
259. Uehara T, Watanabe H, Itoh F, et al. Nephrotoxicity of a novel antineoplastic platinum complex, nedaplatin: a comparative study with cisplatin in rats. *Arch Toxicol*. 2005;79(8):451–460.
260. Gupta S, Glezerman IG, Hirsch JS, et al. Derivation and external validation of a simple risk score for predicting severe acute kidney injury after intravenous cisplatin: cohort study. [published correction appears in *BMJ*. 2024;385:e077169. <https://doi.org/10.1136/bmj.q1207>. 2024;384:e077169.
261. Gupta S, Glezerman IG, Hirsch JS, et al. Effect of mannitol on cisplatin-associated acute kidney injury: a target trial emulation. *Nephrol Dial Transplant*. 2025;40(2):405–408.
262. Gupta S, Glezerman IG, Hirsch JS, et al. Intravenous magnesium and cisplatin-associated acute kidney injury. *JAMA Oncol*. 2025;11(6):636–643. <https://doi.org/10.1001/jamaoncol.2025.0756>.
263. Lee BS, Lee JH, Kang HG, et al. Ifosfamide nephrotoxicity in pediatric cancer patients. *Pediatr Nephrol*. 2001;16(10):796–799.
264. Skinner R, Cotterill SJ, Stevens MC. Risk factors for nephrotoxicity after ifosfamide treatment in children: a UKCCSG late effects group study. United Kingdom children's cancer study group. *Br J Cancer*. 2000;82(10):1636–1645.
265. Aleksa K, Woodland C, Koren G. Young age and the risk for ifosfamide-induced nephrotoxicity: a critical review of two opposing studies. *Pediatr Nephrol*. 2001;16(12):1153–1158.
266. Friedlaender MM, Haviv YS, Rosenmann E, Peylan-Ramu N. End-stage renal interstitial fibrosis in an adult ten years after ifosfamide therapy. *Am J Nephrol*. 1998;18(2):131–133.
267. Sahni V, Choudhury D, Ahmed Z. Chemotherapy-associated renal dysfunction. *Nat Rev Nephrol*. 2009;5(8):450–462.
268. Perazella MA. Crystal-induced acute renal failure. *Am J Med*. 1999;106(4):459–465.
269. Pinheiro FV, Pimentel VC, De Bona KS, et al. Decrease of adenosine deaminase activity and increase of the lipid peroxidation after acute methotrexate treatment in young rats: protective effects of grape seed extract. *Cell Biochem Funct*. 2010;28(1):89–94.
270. de Miguel D, Garcia-Suarez J, Martin Y, Gil-Fernandez JJ, Burgaleta C. Severe acute renal failure following high-dose methotrexate therapy in adults with haematological malignancies: a significant number result from unrecognized co-administration of several drugs. *Nephrol Dial Transplant*. 2008;23(12):3762–3766.
271. Celticki B, Leclerc D, Lawrance AK, et al. Altered expression of methylenetetrahydrofolate reductase modifies response to methotrexate in mice. *Pharmacogenet Genom*. 2008;18(7):577–589.
272. Wall SM, Johansen MJ, Molony DA, DuBose Jr TD, Jaffe N, Madden T. Effective clearance of methotrexate using high-flux hemodialysis membranes. *Am J Kidney Dis*. 1996;28(6):846–854.
273. Widemann BC, Schwartz S, Jayaprakash N, et al. Efficacy of glucarpidase (carboxypeptidase g2) in patients with acute kidney injury after high-dose methotrexate therapy. *Pharmacotherapy*. 2014;34(5):427–439.
274. Gupta S, Kaunfer SA, Chen KL, et al. Glucarpidase for treatment of high-dose methotrexate toxicity. *Blood*. 2025. Published online January 6.
275. Markowitz GS, Bombardier AS, Perazella MA. Drug-induced glomerular disease: direct cellular injury. *Clin J Am Soc Nephrol*. 2015;10(7):1291–1299.
276. Markowitz GS, Nasr SH, Stokes MB, D'Agati VD. Treatment with IFN-alpha, -beta, or -gamma is associated with collapsing focal segmental glomerulosclerosis. *Clin J Am Soc Nephrol*. 2010;5(4):607–615.
277. Schwartz RN, Stover L, Dutcher JP. Managing toxicities of high-dose interleukin-2. *Oncology (Williston Park)*. 2002;16(11 suppl 13):11–20.
278. Memoli B, De Nicola L, Libetta C, et al. Interleukin-2-induced renal dysfunction in cancer patients is reversed by low-dose dopamine infusion. *Am J Kidney Dis*. 1995;26(1):27–33.
279. Postow MA, Callahan MK, Wolchok JD. Immune checkpoint blockade in cancer therapy. *J Clin Oncol*. 2015;33(17):1974–1982.
280. Gupta S, Short SAP, Sise ME, et al. Acute kidney injury in patients treated with immune checkpoint inhibitors. [published correction appears in *J Immunother Cancer*. 2021;9(10):e003467. <https://doi.org/10.1136/jitc-2021-003467corr1>.
- 280a. Isik B, Alexander MP, Manohar S, et al. Biomarkers, clinical features, and rechallenge for immune checkpoint inhibitor renal immune-related adverse events. *Kidney Int Rep*. 2021;6(4):1022–1031. <https://doi.org/10.1016/j.ekir.2021.01.013>.

281. Long JP, Singh S, Dong Y, Yee C, Lin JS. Urine proteomics defines an immune checkpoint-associated nephritis signature. *J Immunother Cancer*. 2025;13(1):e010680.
282. Jhaveri KD, Rosner MH. Chimeric antigen receptor T cell therapy and the kidney: what the nephrologist needs to know. *Clin J Am Soc Nephrol*. 2018;13:796.
283. Gupta S, Seethapathy H, Stroehbehn IA, et al. Acute kidney injury and electrolyte abnormalities after chimeric antigen receptor T-cell (CAR-T) therapy for diffuse large B-cell lymphoma. *Am J Kidney Dis*. 2020;76:63.
284. Kanduri SR, Cheungpasitporn W, Thongprayoon C, et al. Systematic review of risk factors and incidence of acute kidney injury among patients treated with CAR-T cell therapies. *Kidney Int Rep*. 2021;6:1416.
285. Lee MD, Stroehbehn IA, Seethapathy HS, et al. Acute kidney injury after the CAR-T therapy tisagenlecleucel. *Am J Kidney Dis*. 2021;77:990.
286. Gurevich F, Perazella MA. Renal effects of anti-angiogenesis therapy: update for the internist. *Am J Med*. 2009;122(4):322–328.
287. Zhu X, Wu S, Dahut WL, Parikh CR. Risks of proteinuria and hypertension with bevacizumab, an antibody against vascular endothelial growth factor: systematic review and meta-analysis. *Am J Kidney Dis*. 2007;49(2):186–193.
288. Muallem S, Moe OW. When EGF is offside, magnesium is wasted. *J Clin Invest*. 2007;117(8):2086–2089.
289. Fakih M. Anti-EGFR monoclonal antibody-induced hypomagnesaemia. *Lancet Oncol*. 2007;8(5):366–367.
290. Degirolamo C, Sabbà C, Moschetta A. Therapeutic potential of the endocrine fibroblast growth factors FGF19, FGF21 and FGF23. *Nat Rev Drug Discov*. 2016;15(1):51–69. Epub 2015 Nov 16. PMID: 26567701.
291. Attieh RM, Upadrista P, Levy AT, Jhaveri KD. Fibroblast growth factor receptor inhibitor-induced hyperphosphatemia: lessons for the nephrologist. *Clin Nephrol*. 2025;103(1):12–18.
292. Jhaveri KD, Sakhiya V, Wanchoo R, Ross D, Fishbane S. Renal effects of novel anticancer targeted therapies: a review of the food and drug administration adverse event reporting system. *Kidney Int*. 2016;90(3):706–707. PMID: 27521117.
293. Jhaveri KD, Sakhiya V, Fishbane S. Nephrotoxicity of the BRAF inhibitors vemurafenib and dabrafenib. *JAMA Oncol*. 2015;1(8):1133–1134.
294. Perico L, Mandalà M, Schieppati A, et al. BRAF signaling pathway inhibition, podocyte injury, and nephrotic syndrome. *Am J Kidney Dis*. 2017;70(1):145–150. Epub 2017 Feb 24. PMID: 28242136.
295. Perazella MA, Izzedine H. Crizotinib: renal safety evaluation. *J Onco-Nephrol*. 2017;1(1):49–56.
296. Hida T, Nokihara H, Kondo M, et al. Alectinib versus crizotinib in patients with ALK-positive non-small-cell lung cancer (J-ALEX): an open-label, randomised phase 3 trial. *Lancet*. 2017;390:29.
297. Lee CS, Wanchoo R, Seetharamu N. Lorlatinib induced proteinuria: a case report. *J Oncol Pharm Pract*. 2021;27:1037.
298. Fischer K, Al-Sawaf O, Hallek M. Preventing and monitoring for tumor lysis syndrome and other toxicities of venetoclax during treatment of chronic lymphocytic leukemia. *Hematology Am Soc Hematol Educ Program*. 2020;2020(1):357–362. PMID: 33275717; PMCID: PMC7727567.
299. Yui JC, Van Keer J, Weiss BM, et al. Proteasome inhibitor associated thrombotic microangiopathy. *Am J Hematol*. 2016;91(9):E348–E352. Epub 2016 Jul 4. PMID: 27286661.
300. Cortes JE, Gambacorti-Passerini C, Kim DW, et al. Effects of bosutinib treatment on renal function in patients with Philadelphia chromosome-positive leukemias. *Clin Lymphoma Myeloma Leuk*. 2017;17:684.
301. Ozkurt S, Temiz G, Acikalin MF, Soydan M. Acute renal failure under dasatinib therapy. *Ren Fail*. 2010;32:147.
302. Kaiafa G, Kakaletsis N, Savopoulos C, et al. Simultaneous manifestation of pleural effusion and acute renal failure associated with dasatinib: a case report. *J Clin Pharm Therapeut*. 2014;39:102.
303. Holstein SA, Stokes JB, Hohl RJ. Renal failure and recovery associated with second-generation Bcr-Abl kinase inhibitors in imatinib-resistant chronic myelogenous leukemia. *Leuk Res*. 2009;33:344.
304. Martino S, Daguindau E, Ferrand C, et al. A successful renal transplantation for renal failure after dasatinib-induced thrombotic thrombocytopenic purpura in a patient with imatinib-resistant chronic myelogenous leukaemia on nilotinib. *Leuk Res Rep*. 2013;2:29.
305. Uz B, Dolasik I. An unexpected and devastating adverse event of dasatinib: rhabdomyolysis. *Leuk Res Rep*. 2016;5:1.
306. Wallace E, Lyndon W, Chumley P, et al. Dasatinib-induced nephrotic-range proteinuria. *Am J Kidney Dis*. 2013;61:1026.
307. Hirano T, Hashimoto M, Korogi Y, et al. Dasatinib-induced nephrotic syndrome. *Leuk Lymphoma*. 2016;57:726.
308. Koshida T, Wu S, Suzuki H, et al. Thrombotic microangiopathy in dasatinib-treated patients with chronic myeloid leukemia. *J Onco-Nephrol*. 2019;4:41.
309. Abbas A, Mirza MM, Ganti AK, Tendulkar K. Renal toxicities of targeted therapies. *Targeted Oncol*. 2015;10:487.
310. Calizo RC, Bhattacharya S, van Hasselt JGC, et al. Disruption of podocyte cytoskeletal biomechanics by dasatinib leads to nephrotoxicity. *Nat Commun*. 2019;10:2061.
311. Wang ML, Rule S, Martin P, et al. Targeting BTK with ibrutinib in relapsed or refractory mantle-cell lymphoma. *N Engl J Med*. 2013;369(6):507–516. Epub 2013 Jun 19. PMID: 23782157; PMCID: PMC4513941.
312. Perazella MA, Markowitz GS. Bisphosphonate nephrotoxicity. *Kidney Int*. 2008;74(11):1385–1393.
313. Markowitz GS, Appel GB, Fine PL, et al. Collapsing focal segmental glomerulosclerosis following treatment with high-dose pamidronate. *J Am Soc Nephrol*. 2001;12(6):1164–1172.

QUESTIONS

1. A 66-year-old man with underlying stage 3b CKD, hypertension, type 2 diabetes mellitus, BPH, and gout undergoes partial nephrectomy for a 3-cm renal cell carcinoma in the left kidney. The operative procedure is associated with a longer than usual warm ischemia time due to complex renal anatomy. Which of the following are associated with the development of progressive CKD in this patient?
 - a. Partial nephrectomy
 - b. Stage 3b CKD
 - c. Type 2 diabetes mellitus
 - d. Prolonged intraoperative ischemia time
 - e. All of the above

Answer: e

Rationale: Patients undergoing nephrectomy for renal cell cancer may develop AKI, CKD, or progressive worsening of underlying CKD. Total (radical) nephrectomy and partial (nephron-sparing surgery nephrectomy) differ in terms of the incidence of postoperative CKD, which is lower with partial nephrectomy due to the reduced nephron loss. However, patients with various medical risk factors for CKD are at increased risk for progressive kidney function loss (potentially progressing to ESKD), irrespective of the use of partial nephrectomy, highlighting the key role played by comorbidities underlying CKD, hypertension, diabetes mellitus) in the development of postnephrectomy CKD. The duration of intraoperative ischemia is another important predictor of postoperative kidney function. Prolonged warm ischemia time is significantly associated with adverse postoperative kidney function including both AKI and worsening of CKD.

2. A 42-year-old woman undergoes hematopoietic stem cell transplant (HSCT) for multiple myeloma. Over the next several weeks, the patient develops a progressive rise in serum creatinine concentration from 1.2 mg/dL to 4.1 mg/dL. A complete blood count demonstrates hemoglobin 6.8 g/dL and platelet count 38. Review of the peripheral blood smear shows a reduced number of platelets and the presence of schistocytes. Concern for the possibility of transplant-associated thrombotic microangiopathy is raised. Laboratory tests show an elevated lactate dehydrogenase level, an undetectable haptoglobin level, with negative Coombs test. In addition to lowering the calcineurin inhibitor dose, all of the following are appropriate in the treatment of this patient except?
 - a. Administration of rituximab
 - b. Therapeutic plasma exchange
 - c. Red blood cell and platelet transfusion
 - d. Administration of Eculizumab
 - e. Defibrotide

Answer: b

Rationale: The patient has developed HSCT-associated thrombotic microangiopathy (TMA). Calcineurin inhibitor therapy, used for GVHD prophylaxis, has been associated with TMA in the HSCT population. First-line therapy includes lowering of calcineurin inhibitor dose (or discontinuing the medication). Supportive measures include red blood cell and platelet transfusion. Rituximab has been used in a small number of patients with favorable response. Eculizumab, a monoclonal antibody directed against C5, has also been used in patients with transplant-associated TMA and shown beneficial results. Defibrotide has also been shown to be efficacious. Evidence on the use of plasma exchange in this setting is of low quality and no longer recommended as the results have been disappointing.

3. A 49-year-old female patient has been given a clinical diagnosis of non-small cell lung cancer. The patient is started on pembrolizumab, a PD-1 antagonist. The patient develops acute kidney injury within 4 weeks of therapy. The kidney biopsy reveals acute interstitial nephritis. The patient is started on steroids and immunotherapy is held. Which one of the following are known risk factors for immune checkpoint inhibitor-related acute kidney injury?
 - a. Chronic Kidney Disease
 - b. Use of PD1 inhibitor and CTLA-4 inhibitor in combination
 - c. Use of Proton pump inhibitor
 - d. Presence of another immune related adverse event in another organ
 - e. All of the above

Answer: e

Rationale: Few retrospective studies of patients with AKI from immune checkpoint inhibitors have been done. The use of proton pump inhibitor therapy, underlying low eGFR, combined immune checkpoint inhibitor use and presence of extra renal immune-related adverse events were risk factors. AIN was the most common kidney biopsy finding in 83% of patients.

4. Which of the following is not a risk factors for the development of Methotrexate (MTX) associated nephrotoxicity?
 - a. Volume depletion
 - b. Acidic urine
 - c. Dosing of MTX (high versus low)
 - d. Alkaline urine

Answer: d

Rationale: Methotrexate-associated nephrotoxicity is a crystalline nephropathy occurring as a result of precipitation of methotrexate and its metabolites (hydroxymethotrexate) within the tubular lumen. Prevention of AKI can be achieved by maintaining high flow rates within the tubular lumen. Reduced flow rates in the setting of volume depletion increase the concentration of MTX within the tubular lumen. Acidic urine is an additional risk factor as it results in decreased solubility of MTX, as well as its metabolite. Alkaline urine enhances its solubility and increasing the urine pH from 5 to 7 increases methotrexate solubility by 10-fold. High doses of MTX are associated with AKI in close to 1.8% of treated patients despite preventive measures.